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# Dual-View Native Time-Series Prompting for Few-Shot Time Series Classification

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## Abstract

Few-shot time series classification remains challenging because extremely limited supervision is often insufficient to learn robust class-discriminative patterns from raw sequences. While large language models (LLMs) offer strong pretrained priors, directly applying them to time series is nontrivial: continuous temporal signals must first be converted into representations that are both compatible with a pretrained language backbone and stable under scarce labels. In this work, we propose **DuoTSP**, a **Duo**-view native **Time-Series Prompting** framework for few-shot time series classification. Instead of relying on a single time-series-to-language pathway, DuoTSP encodes each sequence from two complementary views: a temporal branch captures local patch relations and long-range dynamics, while a visual branch transforms the sequence into a pseudo-image and leverages a pretrained vision transformer to model global morphology and cross-local structures. The resulting token-level representations are fused into soft time-series prompts and inserted into a pretrained causal LLM. During downstream few-shot adaptation, the language side is lightly updated with LoRA rather than an additional classifier head. To bridge the gap between time-series representations and language modeling, we study a three-stage curriculum in which dual-view self-supervision and time-series question answering provide the default downstream checkpoint, while an optional third stage adds semantic bridging through captioning and synthetic attribute supervision. Finally, at inference time we score each candidate class-token-plus-eos sequence in the dataset-specific label space, yielding a parameter-efficient and LLM-consistent decision rule. Experiments on few-shot UCR benchmarks show that DuoTSP provides a strong and stable transfer paradigm for time series classification under limited supervision.

## 1 Introduction

Time series classification (TSC) is a fundamental problem in machine learning, with applications spanning healthcare monitoring, industrial diagnosis, sensor analytics, and scientific discovery Ismail Fawaz et al. [2019]. In many real-world scenarios, however, labeled time series are scarce: obtaining expert annotations can be expensive, class distributions are often imbalanced, and newly introduced categories may have only a handful of examples. These challenges make few-shot time series classification particularly difficult, as the model must generalize from very limited supervision while still capturing subtle temporal patterns that distinguish classes Tang et al. [2020]. Although recent deep architectures have substantially improved TSC performance, their effectiveness often degrades in the low-data regime, where robust inductive biases and transferable representations become especially important Pan et al. [2024a].

Existing few-shot TSC methods have mainly explored meta-learning, prototype- or shapelet-based reasoning, and spectral or contrastive representation learning Barreda et al. [2024], Tang et al. [2020],

Yang et al. [2022]. In parallel, a growing line of work has attempted to exploit pretrained foundation models for time-series understanding, especially through large language models (LLMs), which offer strong sequence modeling priors and a unified generative interface Zhang et al. [2024], Jin et al. [2024], Cheng et al. [2024]. However, applying LLMs to time series is far from straightforward. Unlike text, time series are continuous numerical signals whose discriminative cues may appear simultaneously in local temporal dependencies, long-range dynamics, and global waveform morphology. Existing approaches often address only part of this mismatch. Some methods focus on a single temporal encoding pathway and project time-series features into the language space, while others discretize time series into textual prototypes or convert them into images for visual encoders Zhang et al. [2024], Jin et al. [2024], Cheng et al. [2024]. These strategies are useful, but they leave three key issues insufficiently resolved for few-shot classification: incomplete representation, unstable cross-modal transfer, and a decision mechanism that is not fully aligned with the generative nature of LLMs.

First, representation is often incomplete when time series are modeled from only one perspective. The class identity of a sequence may depend not only on local segments and temporal evolution, but also on holistic shape, texture-like patterns, and cross-local structural cues. Recent work increasingly suggests that visualized time-series signals can expose morphology-aware information that is complementary to sequence-space modeling Ni et al. [2025], Liu et al. [2025]. Under few-shot supervision, missing any of these signals can be especially harmful, because the model has little data to compensate for a weak inductive bias. Second, transferring time-series features into a pretrained causal LLM is itself a difficult optimization problem. A small labeled support set is rarely sufficient to simultaneously learn task-specific classification, bridge the distribution gap between continuous time-series representations and the hidden space of the language model, and stably adapt even a lightweight language-side interface such as LoRA Chang et al. [2025], Cheng et al. [2024]. Third, even when time-series features are successfully injected into an LLM, many approaches still rely on a conventional classifier head for downstream prediction or on unconstrained open-vocabulary generation. This makes the final decision process only weakly aligned with the native autoregressive mechanism of the pretrained LLM. Recent evidence that prompt outputs can underestimate the time-series classification capability already contained in LLM representations further reinforces this concern Schumacher et al. [2026].

In this paper, we argue that effective few-shot TSC with pretrained causal LLMs requires a complete transfer pipeline that jointly addresses these three challenges. Based on this view, we propose **DuoTSP**, a **Duo**-view native **Time-Series Prompting** framework that turns time series into native soft prompts for LLMs. Instead of mapping a sequence to language through a single pathway, DuoTSP builds token-level time-series prompts from two complementary views. A temporal branch models patch-level relations and long-range dependencies directly on the raw sequence, preserving the native sequential structure of time series. In parallel, a visual branch converts the sequence into a pseudo-image and uses a pretrained vision transformer to capture global morphology, texture-like patterns, and cross-local structures that are difficult to express through a purely temporal pathway. The temporal encoder is inspired by patchified time-series Transformers Nie et al. [2022]. The two branches are fused at the token level, producing soft time-series tokens that are inserted together with textual tokens into the input sequence of a pretrained causal LLM. In this way, the model conditions generation on a richer and more complete time-series representation before any downstream supervision is applied.

To make this dual-view prompting mechanism transferable under limited labels, we further study a three-stage curriculum. Stage I performs branch-faithful self-supervised representation learning, Stage II establishes the time-series-to-language generation pathway through time-series question answering, and Stage III optionally adds semantic bridging through captioning and synthetic attribute supervision. In our main UCR benchmark, the default downstream initialization is the Stage II checkpoint; Stage III is reserved for additional semantic enrichment rather than being required for the default few-shot results.

Finally, for downstream few-shot classification, we avoid introducing an extra classifier head and instead formulate prediction as candidate label-sequence scoring. For each dataset, we dynamically add class-specific tokens to the vocabulary and only expose the corresponding embedding and output rows for learning, thereby confining task adaptation to a compact label subspace. At inference time, we score each candidate class token followed by the end-of-sequence token, rather than allowing open-vocabulary generation. This design preserves consistency with the autoregressive objective of the pretrained LLM, reduces unnecessary output freedom under scarce labels, and provides a parameter-efficient adaptation mechanism well suited to few-shot classification.

Our work makes three main contributions. First, we propose a dual-view native time-series prompting framework that jointly preserves temporal dependency patterns and morphology-level structures before conditioning a pretrained causal LLM. Second, we study a three-stage transfer pipeline in which Stage I and Stage II provide the default downstream checkpoint and Stage III offers optional semantic enrichment through captioning and synthetic attributes. Third, we cast few-shot time series classification as candidate label-sequence scoring in a dataset-specific label space, avoiding an additional discriminative head and yielding a stable parameter-efficient adaptation paradigm. Extensive experiments on few-shot UCR classification benchmarks demonstrate the effectiveness of the proposed framework.

## 2 Related Work

**Few-shot time series classification.** Few-shot time series classification (TSC) has mainly been studied through meta-learning, metric learning, and prototype-based inductive biases. Narwariya et al. [2020] introduced a meta-learning framework that trains residual networks across heterogeneous UCR tasks for rapid adaptation under scarce supervision. Tang et al. [2020] proposed Dual Prototypical Shapelet Networks (DPSN), which combine representative class prototypes with discriminative local shapelets to improve both interpretability and data efficiency. Yang et al. [2022] further explored spectral relations and introduced the Spectral Propagation Graph Network (SPGN) to propagate class information through spectrum-wise comparisons. More recently, COSCO Barreda et al. [2024] improved few-shot multivariate TSC from the optimization perspective by combining sharpness-aware minimization with a prototypical loss. These methods demonstrate the importance of transferable priors and structured inductive biases, but they remain essentially discriminative pipelines and mostly operate within a single temporal representation space.

**LLMs and foundation models for time series.** Recent surveys show that foundation-model research for time series has rapidly expanded from adapting pretrained language models to building native temporal foundation models Jiang et al. [2024], Ye et al. [2024], Shi et al. [2025]. Representative adaptation-based approaches include TEST Sun et al. [2023], which aligns time-series embeddings with the LLM embedding space through text-prototype-aware contrastive learning; TimeLLM Jin et al. [2024], which reprograms time-series patches into a frozen LLM via text prototypes and prompt-as-prefix; and S<sup>2</sup>IP-LLM Pan et al. [2024b], which aligns time-series embeddings with semantic anchors derived from pretrained word embeddings. LLM4TS Chang et al. [2025] further adopts a two-stage alignment-and-task fine-tuning strategy, while OpenTSLM Langer et al. [2025] moves toward native time-series language models with soft-prompt and cross-attention interfaces. A recent line of work has also begun to study few-shot classification with LLM-based architectures. The closest to our setting is LLMFew Chen et al. [2025], which uses a patch-wise temporal encoder, LoRA-tuned decoder, and a conventional classification head for few-shot multivariate TSC. Compared with this line, we focus on dual-view temporal evidence, explicit stage-wise alignment before few-shot adaptation, and generative label-token prediction rather than an external classifier. We are also motivated by recent ablation studies suggesting that naively attaching time-series inputs to pretrained LLMs is insufficient unless the temporal interface is meaningfully designed and aligned Tan et al. [2024].

**Vision-based representations for time series.** Beyond sequence-space modeling, a growing body of work represents time series as images so that waveform morphology, texture, periodicity, and holistic structural patterns can be captured by vision backbones. Recent surveys identify this as an increasingly important branch of time-series foundation modeling Ni et al. [2025]. In the context of multimodal reasoning, Liu et al. [2025] show that visualization can substantially improve LLM reasoning over time series. For classification, TiViT Roschmann et al. [2025] demonstrates that pretrained vision transformers applied to imaged time series can yield highly competitive representations and are complementary to time-series foundation models. These results suggest that visualized signals expose discriminative cues that are not fully captured by sequence-only modeling. However, existing visual approaches are typically used as alternatives to temporal encoders or as interfaces for reasoning, rather than being fused at the token level with a temporal branch for few-shot generative classification.

**Generative classification and constrained decoding.** Casting classification as generation has been widely explored in NLP, most notably in the text-to-text paradigm of T5 Raffel et al. [2020]. Rather than relying on an external discriminative head, this paradigm formulates prediction as conditional token generation, making the decision process more naturally aligned with the pretrained objective of language models. In parallel, constrained decoding has been studied as a general mechanism for restricting generation to valid outputs under lexical or structural constraints Hokamp and Liu [2017], Koo et al. [2024]. Inspired by these ideas, we formulate few-shot time series classification as generation of class-specific label tokens conditioned on textual instructions and time-series soft prompts. Instead of attaching a separate classifier, our model performs prediction in the native autoregressive manner of a pretrained causal LLM, while restricting the final decision to the candidate label space of the current episode further reduces irrelevant output freedom and improves decision stability under low-data uncertainty.

In summary, our method lies at the intersection of these four lines of work: few-shot TSC, LLM-based time-series adaptation, vision-based time-series representation learning, and generative classification with constrained decoding. It differs from prior work by jointly combining dual-view token-level encoding, stage-wise time-series-to-LLM alignment, and episode-wise constrained label-token generation in a unified framework for few-shot TSC.

### 3 Method

#### 3.1 Problem Setup and Unified Time-Series-to-Language Formulation

We study few-shot time series classification with a pretrained causal large language model (LLM). During curriculum pretraining, the LLM backbone remains fixed. During downstream few-shot adaptation, we optionally enable lightweight language-side adaptation through LoRA. Given a time series  $x \in \mathbb{R}^T$ , we construct each training example as a unified sequence-to-sequence instance consisting of a textual prefix  $p^{\text{pre}}$ , an optional textual description  $p^{\text{ts}}$ , the time series itself, a textual suffix  $p^{\text{post}}$ , and a target answer  $y$ . The answer can be either a natural-language response (e.g., for question answering or captioning) or a dataset-specific class token (e.g., for classification).

Our key design principle is to treat time series as a native conditioning modality for language modeling, rather than serializing raw values into text. Concretely, the model converts the input series into a sequence of continuous soft tokens, projects them into the hidden space of the LLM, and inserts them into the token stream together with the textual prompt. The resulting mixed sequence is processed by the pretrained LLM under the standard autoregressive objective.

Let  $\text{Tok}(\cdot)$  denote the tokenizer,  $\text{Emb}(\cdot)$  the input embedding layer of the LLM, and  $P(\cdot)$  the projector from time-series features to the LLM hidden space. If  $Z_x$  denotes the time-series soft prompt extracted from  $x$ , the final input sequence to the LLM is

$$H = [\text{Emb}(\text{Tok}(p^{\text{pre}})); \text{Emb}(\text{Tok}(p^{\text{ts}})); P(Z_x); \text{Emb}(\text{Tok}(p^{\text{post}}))] . \quad (1)$$

During training, the answer tokens are appended to the end of this mixed sequence, and the causal language modeling loss is computed only on answer positions. This formulation provides a single interface for all stages of training and all downstream tasks. In particular, few-shot classification is not treated as a separate discriminative problem with an additional classifier head, but as label-space answer generation within the same autoregressive framework. Figure 1 provides an overview of the proposed pipeline.

#### 3.2 Dual-View Native Time-Series Prompting

As illustrated in Figure 1, our method consists of three tightly coupled components: dual-view native time-series prompting, curriculum bridging from time series to language, and candidate label-sequence scoring for few-shot classification. A central challenge in few-shot time series learning is that class-discriminative cues are not concentrated in a single representational view. Some cues arise from local temporal relations, long-range dependencies, phase shifts, and trend evolution, while others are more naturally expressed as global waveform shapes, texture-like patterns, or cross-local morphological structures. Relying on a single encoding pathway therefore risks discarding useful inductive bias precisely when supervision is scarce.

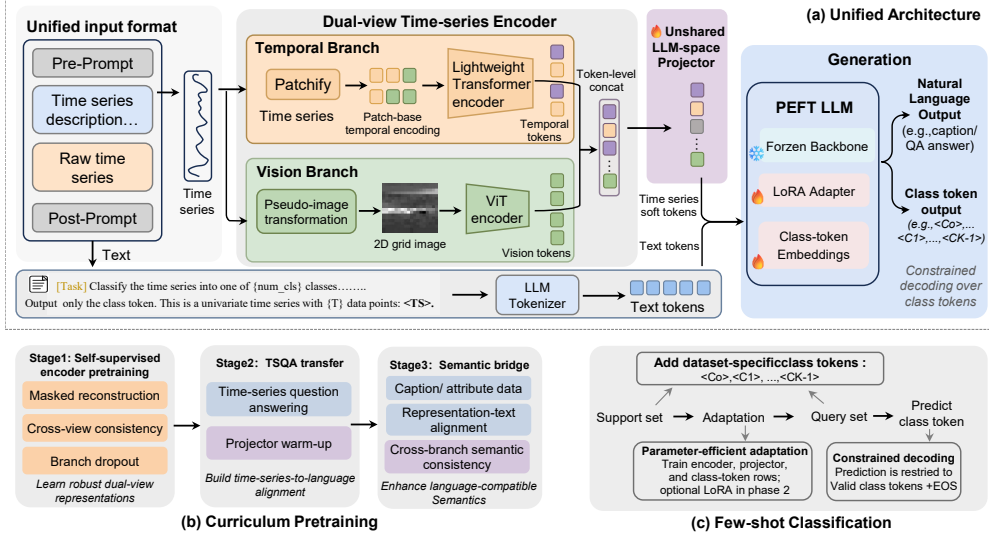


Figure 1: Overview of the proposed framework. Given a raw time series, we construct dual-view native time-series prompts using a temporal branch and a visual branch. The fused soft tokens are projected into the hidden space of a pretrained causal LLM and combined with textual prompts for autoregressive generation; during few-shot adaptation, the language side can be lightly updated with LoRA adapters. Training proceeds through Stage I dual-view self-supervised representation learning and Stage II time-series question answering transfer, with an optional Stage III semantic-bridging and caption-specialization continuation. For downstream few-shot classification, we introduce dataset-specific class tokens and score candidate label sequences in the label subspace.

To address this issue, we introduce a *dual-view native time-series prompting* mechanism that generates soft time-series tokens from two complementary branches under a unified interface.

**Temporal branch.** The temporal branch preserves the sequential nature of the signal. We divide the input series into overlapping patches and encode them with a Transformer-style time-series backbone, producing a sequence of temporally contextualized tokens,

$$Z^t = E_t(x), \quad (2)$$

where  $E_t$  denotes the temporal encoder. This branch is designed to capture local patch interactions as well as longer-range temporal dependencies.

**Visual branch.** The visual branch captures morphology-level patterns that may be difficult to express through a purely sequential pathway. In our main implementation, we construct  $I(x)$  with the `tivit_sqrt_overlap` transform. We first robustly normalize the 1D series by its median and interquartile range, set the effective patch size to  $\lfloor \sqrt{T} \rfloor$ , and unfold the sequence into an overlapping 2D grid. The grid is then min-max normalized, lightly contrast-adjusted by a power transform, resized to the vision input resolution, and replicated across RGB channels to form a pseudo-image  $I(x)$ . We then feed  $I(x)$  into a pretrained vision transformer to obtain visual tokens,

$$Z^v = E_v(I(x)), \quad (3)$$

where  $E_v$  denotes the visual encoder. This branch emphasizes global waveform geometry, texture-like regularities, and structural patterns across distant parts of the sequence.

After branch-specific projection into a shared encoder space, we fuse the two token streams at the token level to form the final soft time-series prompt,

$$Z_x = F(Z^t, Z^v). \quad (4)$$

In our main instantiation,  $F(\cdot)$  is implemented by direct token concatenation, which preserves branch-specific information without prematurely collapsing the two views into a single vector. We further maintain pooled representations  $\bar{z}^t$ ,  $\bar{z}^v$ , and  $\bar{z}^f$  for branch-level and fused-level auxiliary objectives.

This design differs from standard “encoder + projector + LLM” pipelines in an important way. We do not map time series to language through a single pathway. Instead, we construct native time-series prompts from dual-view inductive biases, so that both temporal dependency patterns and morphology-level structures are preserved before language conditioning. It also differs from pure image-based approaches, where the visual view replaces time-series modeling. In our framework, the visual pathway complements rather than substitutes temporal encoding.

### 3.3 Curriculum Bridging from Time Series to Language

Simply attaching a time-series encoder to a pretrained LLM is often unstable, especially in the few-shot regime. The difficulty is not only representational, but also architectural and semantic: the model must first learn robust dual-view features, then learn how these features should condition language generation, and finally organize the internal representation into a more language-compatible semantic space. We therefore adopt a three-stage curriculum that progressively bridges time-series representation and language modeling.

#### 3.3.1 Stage I: Dual-view Self-Supervised Representation Learning

The first stage trains the dual-branch encoder before interfacing it with the LLM. The goal is to obtain stable and complementary representations from both branches while preventing the fused representation from collapsing onto a single dominant view.

We use a combination of masked reconstruction and multi-view self-supervision. Specifically, the temporal branch is trained with a masked reconstruction objective  $L_{\text{rec}}$ , which encourages recovery of masked temporal content from partially observed inputs. In addition, we apply branch-specific and fused self-supervised regularization across augmented views of the same series, yielding losses  $L_{\text{ssl}}^t$ ,  $L_{\text{ssl}}^v$ , and  $L_{\text{ssl}}^f$  for the temporal, visual, and fused representations, respectively. The overall objective of Stage I is

$$L_I = \lambda_{\text{rec}} L_{\text{rec}} + \lambda_t L_{\text{ssl}}^t + \lambda_v L_{\text{ssl}}^v + \lambda_f L_{\text{ssl}}^f. \quad (5)$$

To reduce the risk that training overfits to a single branch, we further introduce *branch dropout*: during pretraining, each forward pass randomly uses the temporal branch only, the visual branch only, or both branches together. This simple strategy encourages each branch to remain individually informative and leads to a more balanced fused representation. In our implementation, Stage I is trained on raw time-series inputs from the TSQA train split, raw time-series inputs from the M4 caption train split, and additional unlabeled official TRAIN-split sequences from 98 UCR datasets listed in `ucr_train_98_datasets.txt`. Validation and test monitoring for this stage use only the corresponding TSQA and M4 validation/test raw series. No UCR official TEST split is used in Stage I.

#### 3.3.2 Stage II: Generative Transfer via Time-Series Question Answering

The second stage establishes the functional path from time-series soft prompts to language outputs. We initialize the dual-branch encoder from Stage I, attach the projector and the pretrained LLM, and train the model on time-series question answering data. Given the mixed prompt sequence defined in Section 3.1, the model is optimized with the standard causal language modeling objective,

$$L_{II} = L_{\text{LM}}^{\text{TSQA}}. \quad (6)$$

This stage is not merely another pretraining phase. It is the first stage that explicitly trains the model to generate text conditioned on time-series soft prompts, thereby converting a time-series encoder into a time-series-conditioned language model. To mitigate optimization instability when moving from the encoder feature space to the LLM hidden space, we first warm up the projector while keeping the encoder fixed, and then jointly optimize the encoder and projector. This warm-start procedure stabilizes cross-space interfacing and reduces alignment shock at the beginning of language-conditioned training. Stage II uses only the TSQA train/validation/test splits for training and monitoring, without introducing UCR classification labels into the transfer stage.

### 3.3.3 Stage III: Optional Semantic Bridging and Caption Specialization

Although Stage II teaches the model to answer language queries from time-series prompts, the learned internal representation may still be overly task-specific. For downstream few-shot classification, we want the representation to be more explicitly organized around semantically expressible properties of time series.

We therefore define an optional third stage that trains on a mixture of time-series question answering, M4 captioning, and automatically constructed attribute-oriented supervision. The synthetic attribute samples are derived from TSQA and M4 raw series and describe interpretable properties such as trend, periodicity, spikes, abrupt changes, volatility, or stationarity in textual form. These tasks encourage the model to connect latent time-series structure with language-compatible semantics.

In addition to the standard language modeling loss, Stage III introduces two auxiliary objectives. First, we apply a representation-text alignment loss  $L_{\text{align}}$  that pulls time-series representations closer to their matched textual semantics in a shared alignment space. Second, we use a branch-consistency loss  $L_{\text{cons}}$  that encourages the temporal and visual branches to produce semantically consistent representations for the same input. The Stage III objective is

$$L_{\text{III}} = L_{\text{LM}}^{\text{sem}} + \lambda_{\text{align}} L_{\text{align}} + \lambda_{\text{cons}} L_{\text{cons}}. \quad (7)$$

In our main UCR benchmark, the default downstream checkpoint is still the Stage II model. Stage III is reserved for additional semantic enrichment and caption specialization rather than being required for the default few-shot results. The three stages therefore address three distinct obstacles in time-series-to-language transfer: representation learning, generative interfacing, and optional semantic bridging.

### 3.4 Few-Shot Classification as Candidate Label-Sequence Scoring

Most existing methods that combine time-series features with LLMs still revert to a conventional classifier head for final decision making. While effective in some settings, this design breaks the generative interface of the pretrained LLM and introduces a separate discriminative output space that must be learned from very limited supervision.

Instead, we reformulate few-shot classification as candidate label-sequence scoring in a compact dataset-specific label space. For each dataset  $D$  with  $K$  classes, we dynamically introduce a set of class tokens

$$\mathcal{C}_D = \{\langle c_0 \rangle, \langle c_1 \rangle, \dots, \langle c_{K-1} \rangle\}. \quad (8)$$

The input prompt ends with a task-specific classification suffix (e.g., “Class:”), and the target answer is the corresponding class token. Therefore, downstream classification remains a next-token prediction problem under the same autoregressive objective used throughout training.

To keep adaptation compact and stable in the few-shot regime, we do not open the full vocabulary for task-specific learning. Instead, only the embedding vectors and output-layer rows corresponding to the newly introduced class tokens are made trainable, while the rest of the vocabulary remains fixed. This creates a small dataset-specific label subspace in which task adaptation is concentrated.

At inference time, we score each candidate label sequence consisting of a class token followed by the end-of-sequence token. The predicted label is

$$\hat{y} = \arg \max_{c \in \mathcal{C}_D} [\log p(c \mid H) + \log p(\text{eos} \mid H, c)]. \quad (9)$$

This procedure preserves the generative interface of the LLM while replacing open-vocabulary decoding with direct scoring over a compact candidate set. As a result, few-shot adaptation becomes both more parameter-efficient and more decision-stable.

### 3.5 Downstream Adaptation

For downstream few-shot adaptation, we use a two-phase training schedule. In the first phase, we warm up the task-critical components, namely the time-series encoder, the projector, and the dataset-specific class-token parameters. In the second phase, we jointly tune the trainable components together with lightweight language-side adaptation through LoRA. In the main benchmark configuration,

LoRA is enabled with rank 16 and alpha 32. This staged adaptation avoids introducing excessive optimization freedom at the beginning of few-shot learning, where the number of labeled examples is extremely limited.

Overall, our method is built around a simple but crucial claim: few-shot time series classification with pretrained causal LLMs becomes effective only when representation completeness, cross-modal transfer, and label-space decision are co-designed as a single pipeline. Dual-view native prompting allows the model to see more complete evidence, curriculum bridging allows the LLM to consume time-series information more reliably, and candidate label-sequence scoring allows the final decision to remain stable under scarce supervision.

## 4 Experiments

### 4.1 Experimental Setup

**Protocol.** We evaluate DuoTSP on all 128 datasets in the UCR Archive 2018 under a fixed full-label-space few-shot classification protocol. For each dataset  $D$ , we preserve its native label space rather than sampling episode-specific class subsets. Let  $C_D$  denote the number of classes in dataset  $D$ . For each support size  $K \in \{1, 2, 5, 10\}$ , we randomly sample up to  $K$  labeled examples per class from the official training split. If class  $c$  contains fewer than  $K$  training examples, we use all available training examples from that class, so the effective support size becomes  $k_{\text{eff}}(c) = \min(K, n_c^{\text{train}})$ . Only these support examples are used for downstream few-shot adaptation. All remaining examples in the original training split are excluded from downstream adaptation, and evaluation is performed on the official test split.

Each method is evaluated over five independent runs with different random support-set samples. For fairness, within each run and shot setting, all compared methods use the same sampled support set. This ensures that performance differences are attributable to the methods rather than to different few-shot data selections.

**Pretraining data and default initialization.** The default DuoTSP initialization used in the main few-shot benchmark is the Stage II checkpoint. This checkpoint is obtained after Stage I dual-view self-supervised representation learning on raw TSQA-train series, raw M4-train series, and additional unlabeled official TRAIN-split sequences from 98 UCR datasets, followed by Stage II transfer on the TSQA train/validation/test splits. Stage III is an optional continuation that mixes TSQA, M4, and synthetic attribute supervision for additional semantic enrichment and caption specialization, but it is not used for the default results in Table 1 unless otherwise stated. No pretraining stage uses any official UCR TEST split. At the same time, Stage I does access raw TRAIN-split sequences from 98 UCR datasets, so the pretraining setup is not benchmark-disjoint with respect to UCR TRAIN data. Downstream few-shot adaptation nevertheless uses only the sampled support set of the target dataset.

**Baselines and metrics.** We compare DuoTSP with three groups of baselines. The first group contains classical supervised or few-shot time-series classification architectures, including RESNET, TAPNET, and COSCO-RESNET. The second group includes Transformer-style or decomposition-based time-series backbones, namely PATCHTST, DLINEAR, TIMESNET, AUTOFORMER, CROSSFORMER, FEDFORMER, and INFORMER. The third group contains a foundation-style baseline, GPT4TS.

We report classification accuracy in percentage form. For each dataset, shot setting, and method, accuracy is first averaged over five runs. Benchmark-level mean accuracy is then computed as the macro-average over all 128 datasets. In the main summary table, Avg denotes the arithmetic mean of the four benchmark-level mean accuracies under the 1-, 2-, 5-, and 10-shot settings.

To evaluate benchmark-wide robustness, we also report average rank. For each dataset and shot setting, methods are ranked in descending order according to their averaged accuracy, with ties assigned the average of the corresponding ranks. For a fixed shot setting, the shot-wise average rank is computed by averaging these ranks over all 128 datasets. The reported AvgRank is then obtained by averaging the four shot-wise average ranks across the 1-, 2-, 5-, and 10-shot settings. Lower AvgRank indicates stronger and more consistent performance across datasets. Standard deviations and complete per-dataset results are provided in the appendix.



Table 1: Few-shot classification accuracy (%) on 128 shared UCR datasets. We report the mean over 5 runs. Avg and AvgRank average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

| Model        | 1-shot       | 2-shot       | 5-shot       | 10-shot      | Avg          | AvgRank     |
|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
| ResNet       | 47.79        | 53.98        | 61.27        | 66.24        | 57.32        | 4.89        |
| TapNet       | 45.40        | 50.79        | 59.47        | 64.93        | 55.15        | 5.45        |
| COSCO-ResNet | <u>47.88</u> | <u>54.95</u> | <u>63.55</u> | <u>67.18</u> | <u>58.39</u> | <u>4.70</u> |
| DLinear      | 44.21        | 48.01        | 52.30        | 55.43        | 49.99        | 7.24        |
| PatchTST     | 42.39        | 48.43        | 56.94        | 62.72        | 52.62        | 6.54        |
| TimesNet     | 45.16        | 49.95        | 57.86        | 62.79        | 53.94        | 5.90        |
| Autoformer   | 36.78        | 39.45        | 43.20        | 46.04        | 41.37        | 9.47        |
| Crossformer  | 44.31        | 49.45        | 57.20        | 61.94        | 53.23        | 6.04        |
| FEDformer    | 37.71        | 41.71        | 47.30        | 51.50        | 44.56        | 8.68        |
| Informer     | 46.42        | 51.14        | 58.35        | 63.13        | 54.76        | 5.59        |
| GPT4TS       | 35.61        | 38.30        | 42.94        | 46.88        | 40.93        | 9.26        |
| DuoTSP       | <b>48.31</b> | <b>55.55</b> | <b>64.14</b> | <b>69.16</b> | <b>59.29</b> | <b>4.24</b> |

**Fairness and training details.** All methods share the same official TRAIN support pool, the same official TEST query split, and, within each run, the same sampled support indices. Under this few-shot protocol, we do not introduce an additional validation split by default. Instead, each method follows the checkpoint-selection rule implemented in its released few-shot runner. ResNet and TapNet use support-set model selection; COSCO-ResNet trains for a fixed number of epochs and evaluates with support centroids; PatchTST uses a two-phase head-warm-up/joint-fine-tuning schedule and reports the last Phase II checkpoint; TSLib models and GPT4TS use fixed single-stage schedules without a separate validation split or early stopping. Appendix B.2 summarizes the corresponding training recipes and hyperparameters.

**Implementation details.** Our main instantiation of DuoTSP uses meta-llama/Llama-3.2-1B as the pretrained causal LLM and facebook/dinov2-base as the visual backbone. The temporal branch uses patch length 16, stride 8, and hidden size 128. The visual branch adopts the tivit\_sqrt\_overlap pseudo-image construction. The main benchmark uses the Stage II checkpoint as the default downstream initialization. During downstream few-shot adaptation, we use a two-phase schedule consisting of 5 warm-up epochs followed by 55 joint-training epochs. In the reported main runs, language-side adaptation is enabled via LoRA with rank 16 and alpha 32. Additional stage-wise data sources, hyperparameters, and baseline training details are summarized in Appendix B.

## 4.2 Main Results

**Benchmark-level comparison.** Table 1 reports the shot-specific benchmark-level mean accuracy on all 128 UCR datasets, together with Avg and AvgRank. DuoTSP achieves the highest mean accuracy under all four support sizes, reaching 48.31, 55.55, 64.14, and 69.16 in the 1-, 2-, 5-, and 10-shot settings, respectively. The strongest non-DuoTSP baseline is COSCO-RESNET, which obtains 47.88, 54.95, 63.55, and 67.18 under the same settings. DuoTSP therefore improves over the strongest baseline by 0.43, 0.60, 0.59, and 1.98 percentage points from 1-shot to 10-shot.

The improvements in the extremely low-shot settings are relatively modest, but they are consistent across all support sizes. This consistency is meaningful because the protocol preserves the full native label space of each dataset and restricts adaptation to only  $C_D \times K$  labeled support examples. DuoTSP also achieves the best overall average accuracy, with an Avg score of 59.29, compared with 58.39 for COSCO-RESNET and 57.32 for RESNET. Moreover, DuoTSP obtains the lowest AvgRank of 4.24, outperforming COSCO-RESNET and RESNET, whose average ranks are 4.70 and 4.89, respectively. These results indicate that DuoTSP provides not only higher average accuracy but also stronger benchmark-wide ranking stability.

**Dataset-level robustness.** Mean accuracy alone may be influenced by a small number of datasets with large absolute gains. To further examine whether the improvement is broadly distributed,

Table 2: Summary of top-level ablations on the 128 shared UCR datasets under the current few-shot protocol. Each value is mean accuracy (%) over five runs. Best is bold, and the best ablated variant in each column is underlined.

| Variant                     | 1-shot       | 2-shot       | 5-shot       | 10-shot      | Avg          |
|-----------------------------|--------------|--------------|--------------|--------------|--------------|
| DuoTSP                      | <b>48.31</b> | <b>55.55</b> | <b>64.14</b> | <b>69.16</b> | <b>59.29</b> |
| w/o Curriculum Pretraining  | 44.70        | <u>51.81</u> | 60.65        | <u>66.60</u> | <u>55.94</u> |
| Temporal only               | 43.51        | 50.55        | 58.40        | 64.39        | 54.21        |
| Visual only                 | <u>45.88</u> | 50.02        | <u>60.86</u> | 63.58        | 55.09        |
| w/o Candidate Label Scoring | 44.46        | 50.44        | 60.08        | 65.59        | 55.14        |

Appendix D reports head-to-head Win/Tie/Loss comparisons against representative baselines. Across the four shot settings, DuoTSP wins more often than it loses against Transformer-style and foundation-style baselines such as PATCHTST, TIMESNET, and GPT4TS. It also remains competitive with strong CNN-based baselines such as COSCO-RESNET and RESNET. Together with the average-rank results in Table 1, this suggests that the gains of DuoTSP are not driven only by a few favorable datasets, but are distributed across the benchmark.

### 4.3 Ablation Studies

We conduct ablation studies to examine three top-level design choices of DuoTSP: aggregate curriculum pretraining, dual-view native prompting, and candidate label scoring in the dataset label space. Table 2 summarizes the benchmark-level results. Unless otherwise specified, each ablation follows the same full-label-space few-shot protocol, the same support-set sampling procedure, and the same evaluation metrics as the main experiment, while modifying only the component under study. These ablations are intended to assess whether the overall design choices are helpful under the current protocol, rather than to separately validate every pretraining substage.

**Effect of curriculum pretraining.** Removing curriculum pretraining decreases the average accuracy from 59.29 to 55.94, corresponding to an overall drop of 3.35 percentage points. The degradation is observed under all support sizes, with drops of 3.61, 3.74, 3.49, and 2.56 points in the 1-, 2-, 5-, and 10-shot settings, respectively. Under the current protocol, this comparison suggests that the aggregate Stage I+II initialization is helpful when downstream supervision is too limited to learn robust time-series representations and the time-series-to-LLM interface from scratch. Because this ablation removes the full pretraining pipeline at once, it should not be interpreted as separately validating each pretraining stage.

**Effect of dual-view native prompting.** We next compare the full dual-view model with temporal-only and visual-only variants. The temporal-only and visual-only variants obtain average accuracies of 54.21 and 55.09, respectively, whereas the full DuoTSP model reaches 59.29. This corresponds to drops of 5.08 and 4.20 percentage points. Neither single-view variant consistently dominates the other across all support sizes: the visual-only variant performs better in the 1-shot and 5-shot settings, while the temporal-only variant is slightly stronger in the 2-shot and 10-shot settings. This pattern is consistent with the two branches capturing complementary information. The temporal branch contributes sequential dependency and local patch-level dynamics, whereas the visual branch provides morphology-level and cross-local structural cues. Their fusion therefore appears more effective than relying on either view alone under the current protocol.

**Effect of candidate label scoring.** Finally, we evaluate the importance of restricting prediction to candidate label sequences. When this label-space scoring constraint is disabled, the average accuracy decreases from 59.29 to 55.14, yielding a drop of 4.15 percentage points. The performance decline is consistent across all support sizes, with drops of 3.85, 5.11, 4.06, and 3.57 points from 1-shot to 10-shot. This result suggests that generative adaptation alone is insufficient for stable few-shot classification. Restricting prediction to the dataset-specific label-token set reduces unnecessary open-vocabulary uncertainty and better aligns the prediction process with the target classification space.

Overall, the ablation results are consistent with the central design motivation of DuoTSP. Aggregate curriculum pretraining appears to improve the transferability of time-series prompts, dual-view prompting provides more complete time-series evidence, and candidate label scoring stabilizes the final label decision. Their combination yields the strongest few-shot performance under the current benchmark protocol.

## References

- J. Barreda, A. Gomez, R. Puga, K. Zhou, and L. Zhang. Cosco: a sharpness-aware training framework for few-shot multivariate time series classification. In *Proceedings of the 33rd ACM international conference on information and knowledge management*, pages 3622–3626, 2024.
- C. Chang, W.-Y. Wang, W.-C. Peng, and T.-F. Chen. Llm4ts: Aligning pre-trained llms as data-efficient time-series forecasters. *ACM Transactions on Intelligent Systems and Technology*, 16(3): 1–20, 2025.
- Y. Chen, Z. Li, C. Yang, X. Wang, and G. Xu. Large language models are few-shot multivariate time series classifiers. *Data Mining and Knowledge Discovery*, 39(5):66, 2025.
- M. Cheng, Y. Chen, Q. Liu, Z. Liu, and Y. Luo. Advancing time series classification with multimodal language modeling. *arXiv preprint arXiv:2403.12371*, 2024.
- C. Hokamp and Q. Liu. Lexically constrained decoding for sequence generation using grid beam search. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1535–1546, 2017.
- H. Ismail Fawaz, G. Forestier, J. Weber, L. Idoumghar, and P.-A. Muller. Deep learning for time series classification: a review. *Data mining and knowledge discovery*, 33(4):917–963, 2019.
- Y. Jiang, Z. Pan, X. Zhang, S. Garg, A. Schneider, Y. Nevmyvaka, and D. Song. Empowering time series analysis with large language models: A survey. *arXiv preprint arXiv:2402.03182*, 2024.
- M. Jin, S. Wang, L. Ma, Z. Chu, J. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-f. Li, S. Pan, et al. Time-llm: Time series forecasting by reprogramming large language models. In *International Conference on Learning Representations*, 2024.
- T. Koo, F. Liu, and L. He. Automata-based constraints for language model decoding. *arXiv preprint arXiv:2407.08103*, 2024.
- P. Langer, T. Kaar, M. Rosenblattl, M. A. Xu, W. Chow, M. Maritsch, R. Jakob, N. Wang, J. Liu, A. Verma, et al. Opentslm: Time-series language models for reasoning over multivariate medical text-and time-series data. *arXiv preprint arXiv:2510.02410*, 2025.
- H. Liu, C. Liu, and B. A. Prakash. A picture is worth a thousand numbers: Enabling llms reason about time series via visualization. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7486–7518, 2025.
- J. Narwariya, P. Malhotra, L. Vig, G. Shroff, and T. Vishnu. Meta-learning for few-shot time series classification. In *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, pages 28–36. 2020.
- J. Ni, Z. Zhao, C. Shen, H. Tong, D. Song, W. Cheng, D. Luo, and H. Chen. Harnessing vision models for time series analysis: a survey. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 10612–10620, 2025.
- Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *The Eleventh International Conference on Learning Representations*, 2022.
- S. Pan, H. Luo, M. C. Chua, K. Pugalenth, and E. K. Yapp. A review of similarity-based few-shot learning methods for time series classification in manufacturing. In *2024 6th International Conference on Industrial Artificial Intelligence (IAI)*, pages 1–6. IEEE, 2024a.

477 Z. Pan, Y. Jiang, S. Garg, A. Schneider, Y. Nevmyvaka, and D. Song. S<sup>2</sup>IP-LLM: Semantic space  
478 informed prompt learning with LLM for time series forecasting. In *International Conference on*  
479 *Machine Learning*, pages 39135–39153. PMLR, 2024b.

480 C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu.  
481 Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine*  
482 *learning research*, 21(140):1–67, 2020.

483 S. Roschmann, Q. Bouniot, V. Feofanov, I. Redko, and Z. Akata. Time series representations for  
484 classification lie hidden in pretrained vision transformers. *arXiv preprint arXiv:2506.08641*, 2025.

485 D. Schumacher, E. Nourbakhsh, R. Slavin, and A. Rios. Prompting underestimates llm capability for  
486 time series classification. *arXiv preprint arXiv:2601.03464*, 2026.

487 F. Shi, X. Yin, K. Wang, W. Tu, Q. Sun, and H. Ning. Large language models for time series analysis:  
488 Techniques, applications, and challenges. *arXiv preprint arXiv:2506.11040*, 2025.

489 C. Sun, H. Li, Y. Li, and S. Hong. Test: Text prototype aligned embedding to activate llm’s ability for  
490 time series. *arXiv preprint arXiv:2308.08241*, 2023.

491 M. Tan, M. A. Merrill, V. Gupta, T. Althoff, and T. Hartvigsen. Are language models actually useful  
492 for time series forecasting? *Advances in Neural Information Processing Systems*, 37:60162–60191,  
493 2024.

494 W. Tang, L. Liu, and G. Long. Interpretable time-series classification on few-shot samples. In *2020*  
495 *International joint conference on neural networks (IJCNN)*, pages 1–8. IEEE, 2020.

496 L. Yang, S. Hong, and L. Zhang. Spectral propagation graph network for few-shot time series  
497 classification. *arXiv preprint arXiv:2202.04769*, 2022.

498 J. Ye, W. Zhang, K. Yi, Y. Yu, Z. Li, J. Li, and F. Tsung. A survey of time series foundation  
499 models: Generalizing time series representation with large language model. *arXiv e-prints*, pages  
500 arXiv–2405, 2024.

501 X. Zhang, R. R. Chowdhury, R. K. Gupta, and J. Shang. Large language models for time series: A  
502 survey. *arXiv preprint arXiv:2402.01801*, 2024.

## 503 A Full UCR Benchmark Results

504 This appendix reports the complete per-dataset few-shot benchmark results for all 128 shared UCR  
505 datasets. Each cell is shown as mean  $\pm$  standard deviation over five runs, and the summary rows at  
506 the end of each table report average accuracy, average rank, and the number of datasets attaining the  
507 best mean accuracy (counting ties).

Table 3: Per-dataset 1-shot classification results on 128 shared UCR datasets. Each value is reported as mean  $\pm$  standard deviation over 5 runs.

| Dataset                      | ResNet                             | TapNet                              | COSCO-ResNet                        | DLinear                             | PatchTST                           | TimesNet                            | Autoformer        | Crossformer                        | FEDformer         | Informer                            | GPT4TS                              | DuoTSP                              |
|------------------------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|-------------------|------------------------------------|-------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| ACSF1                        | <b>53.80 <math>\pm</math> 4.97</b> | 42.00 $\pm$ 4.95                    | 48.40 $\pm$ 7.30                    | 24.00 $\pm$ 4.24                    | 35.60 $\pm$ 3.21                   | 27.67 $\pm$ 7.02                    | 24.33 $\pm$ 9.71  | 28.33 $\pm$ 2.52                   | 27.80 $\pm$ 7.09  | 31.67 $\pm$ 9.81                    | 18.60 $\pm$ 4.51                    | 44.00 $\pm$ 5.20                    |
| Adiac                        | 33.81 $\pm$ 1.23                   | 12.84 $\pm$ 3.09                    | 33.81 $\pm$ 1.23                    | 8.85 $\pm$ 2.36                     | 11.97 $\pm$ 1.69                   | 21.40 $\pm$ 4.24                    | 34.53 $\pm$ 1.77  | 8.10 $\pm$ 1.92                    | 40.82 $\pm$ 3.61  | 24.21 $\pm$ 3.24                    | 3.02 $\pm$ 1.05                     | <b>45.17 <math>\pm</math> 5.28</b>  |
| AllGestureWiimoteX           | 25.71 $\pm$ 4.01                   | <b>26.86 <math>\pm</math> 2.52</b>  | 24.43 $\pm$ 2.55                    | 15.57 $\pm$ 2.95                    | 21.86 $\pm$ 4.59                   | 20.71 $\pm$ 2.87                    | 11.33 $\pm$ 1.19  | 16.76 $\pm$ 2.52                   | 13.43 $\pm$ 1.74  | 18.90 $\pm$ 1.08                    | 21.86 $\pm$ 4.59                    | 25.11 $\pm$ 4.65                    |
| AllGestureWiimoteY           | <b>26.37 <math>\pm</math> 2.89</b> | 26.17 $\pm$ 2.88                    | 22.71 $\pm$ 4.62                    | 17.49 $\pm$ 2.02                    | 25.17 $\pm$ 5.50                   | 21.81 $\pm$ 2.78                    | 12.29 $\pm$ 1.27  | 20.29 $\pm$ 1.98                   | 13.06 $\pm$ 1.34  | 21.33 $\pm$ 2.38                    | 25.17 $\pm$ 5.50                    | 25.54 $\pm$ 4.42                    |
| AllGestureWiimoteZ           | <b>27.11 <math>\pm</math> 1.95</b> | 22.66 $\pm$ 4.13                    | <u>25.49 <math>\pm</math> 2.57</u>  | 18.26 $\pm$ 2.01                    | 19.71 $\pm$ 2.87                   | 19.24 $\pm$ 1.57                    | 13.29 $\pm$ 0.43  | 18.76 $\pm$ 2.79                   | 14.14 $\pm$ 1.72  | 19.71 $\pm$ 1.79                    | 19.71 $\pm$ 2.87                    | 22.26 $\pm$ 3.96                    |
| ArrowHead                    | 47.66 $\pm$ 8.94                   | 46.63 $\pm$ 9.45                    | 48.00 $\pm$ 7.74                    | 52.00 $\pm$ 6.38                    | 49.71 $\pm$ 1.40                   | 47.62 $\pm$ 9.46                    | 38.29 $\pm$ 4.00  | 50.86 $\pm$ 1.71                   | 35.89 $\pm$ 2.60  | 50.86 $\pm$ 6.05                    | 41.49 $\pm$ 12.34                   | <b>52.57 <math>\pm</math> 3.43</b>  |
| BME                          | 49.60 $\pm$ 2.85                   | 50.80 $\pm$ 2.13                    | 49.73 $\pm$ 3.42                    | 56.67 $\pm$ 3.43                    | 62.93 $\pm$ 5.35                   | 55.33 $\pm$ 7.86                    | 51.56 $\pm$ 3.15  | 61.33 $\pm$ 4.06                   | 55.87 $\pm$ 7.71  | 58.89 $\pm$ 6.19                    | 35.73 $\pm$ 3.22                    | <b>66.40 <math>\pm</math> 3.88</b>  |
| Beef                         | 36.00 $\pm$ 6.41                   | 31.33 $\pm$ 5.06                    | 27.33 $\pm$ 7.96                    | 38.67 $\pm$ 9.89                    | 32.00 $\pm$ 7.67                   | 40.00 $\pm$ 8.82                    | 34.44 $\pm$ 6.94  | <b>41.11 <math>\pm</math> 1.92</b> | 30.67 $\pm$ 6.83  | 40.00 $\pm$ 6.67                    | 31.33 $\pm$ 9.31                    | 34.00 $\pm$ 2.79                    |
| BeetleFly                    | 55.00 $\pm$ 24.24                  | 60.00 $\pm$ 25.00                   | 61.00 $\pm$ 24.85                   | 67.00 $\pm$ 4.47                    | 61.00 $\pm$ 9.62                   | 60.00 $\pm$ 13.23                   | 55.00 $\pm$ 5.00  | 61.67 $\pm$ 11.55                  | 48.00 $\pm$ 2.74  | 60.00 $\pm$ 13.23                   | <b>70.00 <math>\pm</math> 11.18</b> | <b>70.00 <math>\pm</math> 11.73</b> |
| BirdChicken                  | 62.00 $\pm$ 8.37                   | <b>71.00 <math>\pm</math> 9.62</b>  | <u>70.00 <math>\pm</math> 17.68</u> | 56.00 $\pm$ 4.18                    | 56.00 $\pm$ 4.18                   | 51.67 $\pm$ 2.89                    | 51.67 $\pm$ 12.58 | 53.33 $\pm$ 5.77                   | 55.00 $\pm$ 5.00  | 51.67 $\pm$ 7.64                    | 59.00 $\pm$ 9.62                    | 68.00 $\pm$ 14.83                   |
| CBF                          | 81.67 $\pm$ 6.47                   | <b>87.62 <math>\pm</math> 2.73</b>  | 71.71 $\pm$ 7.91                    | 54.02 $\pm$ 3.44                    | 66.38 $\pm$ 7.40                   | 55.85 $\pm$ 2.16                    | 33.63 $\pm$ 1.20  | 58.00 $\pm$ 5.61                   | 36.89 $\pm$ 2.37  | 54.78 $\pm$ 1.68                    | 35.20 $\pm$ 3.75                    | 72.48 $\pm$ 4.85                    |
| Car                          | 37.00 $\pm$ 4.62                   | 39.00 $\pm$ 4.80                    | 41.67 $\pm$ 9.28                    | 38.67 $\pm$ 11.63                   | 27.33 $\pm$ 9.90                   | 42.22 $\pm$ 5.09                    | 24.44 $\pm$ 3.47  | 43.89 $\pm$ 5.85                   | 27.00 $\pm$ 4.15  | 40.00 $\pm$ 7.26                    | 30.00 $\pm$ 9.65                    | <b>55.67 <math>\pm</math> 8.94</b>  |
| Chinatown                    | 66.53 $\pm$ 19.14                  | 63.73 $\pm$ 26.98                   | 62.51 $\pm$ 27.20                   | 67.41 $\pm$ 23.86                   | 67.35 $\pm$ 11.32                  | 77.07 $\pm$ 16.93                   | 81.83 $\pm$ 2.19  | 77.94 $\pm$ 9.20                   | 77.61 $\pm$ 8.88  | 75.41 $\pm$ 7.92                    | 56.68 $\pm$ 20.52                   | <b>93.59 <math>\pm</math> 3.92</b>  |
| ChlorineConcentration        | 34.61 $\pm$ 2.26                   | 35.41 $\pm$ 4.36                    | 34.04 $\pm$ 2.01                    | 33.26 $\pm$ 2.96                    | 35.54 $\pm$ 3.20                   | 33.64 $\pm$ 3.96                    | 32.92 $\pm$ 4.26  | 33.42 $\pm$ 5.09                   | 33.78 $\pm$ 2.74  | 34.11 $\pm$ 3.50                    | <b>39.93 <math>\pm</math> 15.64</b> | 32.61 $\pm$ 3.30                    |
| CinCECGTorso                 | 37.20 $\pm$ 9.73                   | 36.03 $\pm$ 6.56                    | 36.51 $\pm$ 8.82                    | 40.06 $\pm$ 3.86                    | 35.72 $\pm$ 7.09                   | 44.86 $\pm$ 6.03                    | 35.36 $\pm$ 5.77  | <b>50.53 <math>\pm</math> 4.66</b> | 34.97 $\pm$ 4.68  | 45.68 $\pm$ 5.50                    | 42.49 $\pm$ 6.76                    | 48.74 $\pm$ 6.47                    |
| Coffee                       | 78.57 $\pm$ 9.78                   | <b>92.86 <math>\pm</math> 8.38</b>  | 84.29 $\pm$ 13.02                   | 72.86 $\pm$ 14.42                   | 74.29 $\pm$ 18.80                  | 63.10 $\pm$ 10.91                   | 65.48 $\pm$ 7.43  | 71.43 $\pm$ 16.37                  | 76.43 $\pm$ 4.07  | 64.29 $\pm$ 7.14                    | 51.43 $\pm$ 4.07                    | 75.71 $\pm$ 13.69                   |
| Computers                    | 57.52 $\pm$ 7.64                   | <b>60.40 <math>\pm</math> 6.13</b>  | <u>59.12 <math>\pm</math> 4.47</u>  | 47.12 $\pm$ 5.87                    | 52.64 $\pm$ 10.26                  | 50.93 $\pm$ 1.29                    | 48.80 $\pm$ 0.69  | 51.87 $\pm$ 2.57                   | 48.96 $\pm$ 2.79  | 52.27 $\pm$ 3.23                    | 45.20 $\pm$ 5.79                    | 56.40 $\pm$ 5.77                    |
| CricketX                     | 26.15 $\pm$ 3.29                   | 26.10 $\pm$ 3.67                    | 20.56 $\pm$ 4.67                    | 17.28 $\pm$ 1.96                    | 18.15 $\pm$ 4.50                   | 20.34 $\pm$ 0.39                    | 8.89 $\pm$ 0.82   | 17.52 $\pm$ 1.26                   | 11.38 $\pm$ 2.35  | 20.68 $\pm$ 0.74                    | 7.28 $\pm$ 1.05                     | <b>26.26 <math>\pm</math> 5.29</b>  |
| CricketY                     | <b>26.97 <math>\pm</math> 3.03</b> | 24.46 $\pm$ 4.25                    | 22.72 $\pm$ 1.74                    | 18.56 $\pm$ 4.13                    | 17.90 $\pm$ 3.61                   | 22.82 $\pm$ 1.56                    | 10.34 $\pm$ 1.89  | 20.94 $\pm$ 3.72                   | 11.23 $\pm$ 0.80  | 21.20 $\pm$ 3.98                    | 9.54 $\pm$ 2.33                     | 23.33 $\pm$ 5.19                    |
| CricketZ                     | <b>32.51 <math>\pm</math> 2.32</b> | 28.72 $\pm$ 2.15                    | 26.56 $\pm$ 4.19                    | 19.64 $\pm$ 2.73                    | 22.26 $\pm$ 3.91                   | 20.34 $\pm$ 4.11                    | 11.03 $\pm$ 1.54  | 20.00 $\pm$ 2.31                   | 12.26 $\pm$ 1.25  | 20.34 $\pm$ 2.57                    | 7.64 $\pm$ 1.05                     | <u>29.28 <math>\pm</math> 4.82</u>  |
| Crop                         | 25.10 $\pm$ 4.94                   | 28.14 $\pm$ 2.90                    | 25.10 $\pm$ 4.94                    | 10.23 $\pm$ 2.66                    | 17.27 $\pm$ 1.43                   | 27.73 $\pm$ 2.89                    | 24.10 $\pm$ 4.30  | 26.19 $\pm$ 2.96                   | 27.27 $\pm$ 3.58  | 30.31 $\pm$ 5.00                    | 3.80 $\pm$ 0.58                     | <b>36.48 <math>\pm</math> 3.38</b>  |
| DiatomSizeReduction          | 79.67 $\pm$ 5.51                   | 55.36 $\pm$ 19.14                   | 84.12 $\pm$ 4.82                    | <b>89.22 <math>\pm</math> 8.91</b>  | 49.61 $\pm$ 24.86                  | 75.71 $\pm$ 7.74                    | 55.01 $\pm$ 11.77 | 56.43 $\pm$ 20.40                  | 50.07 $\pm$ 12.66 | 76.47 $\pm$ 11.70                   | 41.44 $\pm$ 12.35                   | 85.82 $\pm$ 8.55                    |
| DistalPhalanxOutlineAgeGroup | 43.02 $\pm$ 17.16                  | 43.02 $\pm$ 18.34                   | 43.60 $\pm$ 14.92                   | 42.73 $\pm$ 19.01                   | 46.76 $\pm$ 16.49                  | 45.80 $\pm$ 19.81                   | 46.28 $\pm$ 21.27 | <b>56.59 <math>\pm</math> 9.36</b> | 50.94 $\pm$ 19.66 | 46.28 $\pm$ 20.78                   | 35.68 $\pm$ 14.61                   | 42.01 $\pm$ 16.44                   |
| DistalPhalanxOutlineCorrect  | 62.10 $\pm$ 1.32                   | 57.32 $\pm$ 9.22                    | <b>63.04 <math>\pm</math> 2.44</b>  | 57.54 $\pm$ 11.13                   | 53.55 $\pm$ 11.63                  | 53.99 $\pm$ 14.78                   | 53.26 $\pm$ 12.27 | 53.62 $\pm$ 11.97                  | 55.14 $\pm$ 10.52 | 53.26 $\pm$ 12.97                   | 56.74 $\pm$ 7.30                    | 60.58 $\pm$ 2.98                    |
| DistalPhalanxTW              | 51.22 $\pm$ 5.15                   | 44.03 $\pm$ 11.70                   | 53.38 $\pm$ 4.02                    | 43.74 $\pm$ 5.17                    | 51.22 $\pm$ 6.58                   | 45.32 $\pm$ 9.89                    | 42.69 $\pm$ 4.40  | 38.13 $\pm$ 7.51                   | 48.20 $\pm$ 5.84  | 47.48 $\pm$ 5.62                    | 18.56 $\pm$ 14.51                   | <b>54.53 <math>\pm</math> 1.72</b>  |
| DodgerLoopDay                | 32.50 $\pm$ 7.34                   | 25.00 $\pm$ 4.05                    | 29.75 $\pm$ 7.68                    | <b>41.50 <math>\pm</math> 2.05</b>  | 26.00 $\pm$ 6.58                   | 40.42 $\pm$ 4.73                    | 20.42 $\pm$ 1.91  | 32.92 $\pm$ 3.15                   | 21.50 $\pm$ 4.45  | 40.42 $\pm$ 7.94                    | 26.00 $\pm$ 6.58                    | 34.50 $\pm$ 4.38                    |
| DodgerLoopGame               | 52.17 $\pm$ 6.30                   | 55.07 $\pm$ 17.38                   | 49.13 $\pm$ 8.90                    | <b>60.87 <math>\pm</math> 12.35</b> | 59.86 $\pm$ 22.05                  | 51.45 $\pm$ 9.96                    | 51.21 $\pm$ 5.14  | 56.52 $\pm$ 10.82                  | 50.43 $\pm$ 2.44  | 51.45 $\pm$ 9.59                    | 59.86 $\pm$ 22.05                   | 56.96 $\pm$ 11.69                   |
| DodgerLoopWeekend            | 84.49 $\pm$ 8.37                   | 74.78 $\pm$ 12.27                   | 83.19 $\pm$ 10.24                   | 93.62 $\pm$ 3.81                    | <b>96.23 <math>\pm</math> 1.19</b> | 95.65 $\pm$ 1.26                    | 63.77 $\pm$ 4.41  | 94.44 $\pm$ 0.42                   | 68.41 $\pm$ 4.57  | 93.72 $\pm$ 5.86                    | <b>96.23 <math>\pm</math> 1.19</b>  | 85.36 $\pm$ 4.30                    |
| ECG200                       | 63.00 $\pm$ 12.61                  | <b>72.00 <math>\pm</math> 6.75</b>  | 66.20 $\pm$ 8.32                    | 71.80 $\pm$ 4.44                    | 61.60 $\pm$ 12.82                  | 61.33 $\pm$ 10.97                   | 65.00 $\pm$ 11.79 | 63.00 $\pm$ 20.66                  | 64.20 $\pm$ 14.29 | 68.67 $\pm$ 7.09                    | 51.40 $\pm$ 12.93                   | 60.60 $\pm$ 6.91                    |
| ECG5000                      | 54.22 $\pm$ 21.00                  | 50.05 $\pm$ 20.59                   | 61.53 $\pm$ 13.66                   | 62.80 $\pm$ 19.19                   | 53.35 $\pm$ 20.14                  | <b>69.07 <math>\pm</math> 12.52</b> | 54.44 $\pm$ 9.79  | 57.20 $\pm$ 16.09                  | 62.35 $\pm$ 12.99 | <u>68.84 <math>\pm</math> 13.99</u> | 19.70 $\pm$ 25.30                   | 67.18 $\pm$ 10.57                   |
| ECGFiveDays                  | 58.65 $\pm$ 7.69                   | 57.24 $\pm$ 6.51                    | 54.80 $\pm$ 13.00                   | 57.63 $\pm$ 2.63                    | <b>62.21 <math>\pm</math> 8.35</b> | 57.99 $\pm$ 4.84                    | 54.86 $\pm$ 3.17  | 55.87 $\pm$ 1.56                   | 54.08 $\pm$ 6.73  | <u>58.46 <math>\pm</math> 5.08</u>  | 51.54 $\pm$ 6.38                    | 61.77 $\pm$ 6.61                    |
| EOGHorizontalSignal          | 20.11 $\pm$ 2.01                   | 22.38 $\pm$ 1.86                    | 22.60 $\pm$ 2.77                    | 22.43 $\pm$ 4.01                    | 21.22 $\pm$ 3.15                   | 20.63 $\pm$ 4.63                    | 15.29 $\pm$ 2.63  | 23.57 $\pm$ 5.45                   | 10.39 $\pm$ 2.26  | 22.01 $\pm$ 4.37                    | 27.24 $\pm$ 2.99                    | <b>29.06 <math>\pm</math> 4.16</b>  |
| EOGVerticalSignal            | 11.27 $\pm$ 3.67                   | 14.09 $\pm$ 3.23                    | 13.26 $\pm$ 3.09                    | 24.20 $\pm$ 3.27                    | 23.76 $\pm$ 3.73                   | 23.39 $\pm$ 2.79                    | 11.23 $\pm$ 0.70  | 24.86 $\pm$ 4.17                   | 11.27 $\pm$ 3.83  | 23.11 $\pm$ 4.07                    | 19.06 $\pm$ 6.08                    | <b>27.35 <math>\pm</math> 1.90</b>  |
| Earthquakes                  | 50.07 $\pm$ 8.55                   | 51.08 $\pm$ 6.33                    | 50.94 $\pm$ 6.46                    | 49.35 $\pm$ 5.05                    | 45.61 $\pm$ 20.94                  | 48.68 $\pm$ 19.16                   | 49.40 $\pm$ 1.10  | 49.40 $\pm$ 14.03                  | 47.05 $\pm$ 5.90  | 47.00 $\pm$ 24.65                   | 46.04 $\pm$ 13.71                   | <b>55.54 <math>\pm</math> 11.57</b> |
| ElectricDevices              | 28.83 $\pm$ 9.01                   | <u>32.82 <math>\pm</math> 12.31</u> | 28.35 $\pm$ 12.24                   | 17.59 $\pm$ 1.67                    | 27.21 $\pm$ 7.38                   | 21.93 $\pm$ 4.54                    | 17.11 $\pm$ 1.87  | 15.12 $\pm$ 2.81                   | 17.04 $\pm$ 2.05  | 15.61 $\pm$ 7.10                    | 17.06 $\pm$ 3.10                    | <b>33.14 <math>\pm</math> 10.96</b> |
| EthanolLevel                 | 24.44 $\pm$ 1.30                   | <u>25.40 <math>\pm</math> 0.85</u>  | 25.28 $\pm$ 1.57                    | 24.92 $\pm$ 1.50                    | 23.80 $\pm$ 1.65                   | 26.13 $\pm$ 0.58                    | 26.53 $\pm$ 1.55  | <u>26.67 <math>\pm</math> 1.29</u> | 25.32 $\pm$ 1.50  | 25.80 $\pm$ 1.39                    | 24.76 $\pm$ 1.40                    | <b>26.68 <math>\pm</math> 1.86</b>  |
| FaceAll                      | <b>42.17 <math>\pm</math> 4.79</b> | 32.36 $\pm$ 9.47                    | <b>42.17 <math>\pm</math> 4.79</b>  | 28.66 $\pm$ 9.09                    | 19.36 $\pm$ 3.07                   | 29.11 $\pm$ 4.19                    | 33.12 $\pm$ 5.46  | 25.94 $\pm$ 1.75                   | 34.05 $\pm$ 3.40  | 32.92 $\pm$ 2.10                    | 8.64 $\pm$ 3.46                     | 32.84 $\pm$ 6.86                    |
| FaceFour                     | 68.64 $\pm$ 10.56                  | 47.50 $\pm$ 11.12                   | 56.36 $\pm$ 4.30                    | 64.32 $\pm$ 4.73                    | 54.09 $\pm$ 8.34                   | 70.08 $\pm$ 4.30                    | 52.65 $\pm$ 3.99  | 67.05 $\pm$ 7.10                   | 51.36 $\pm$ 4.28  | <b>73.11 <math>\pm</math> 3.99</b>  | 42.95 $\pm$ 6.60                    | 54.77 $\pm$ 8.25                    |
| FacesUCR                     | <b>54.83 <math>\pm</math> 5.21</b> | 37.63 $\pm$ 5.89                    | 43.65 $\pm$ 3.88                    | 35.69 $\pm$ 7.47                    | 20.81 $\pm$ 1.95                   | 36.24 $\pm$ 3.26                    | 41.98 $\pm$ 2.95  | 40.05 $\pm$ 2.80                   | 41.87 $\pm$ 3.09  | 42.63 $\pm$ 2.58                    | 16.92 $\pm$ 7.54                    | 50.37 $\pm$ 2.98                    |
| FiftyWords                   | 13.27 $\pm$ 1.89                   | 12.79 $\pm$ 3.45                    | 13.27 $\pm$ 1.89                    | 32.70 $\pm$ 4.33                    | 19.60 $\pm$ 4.90                   | 35.90 $\pm$ 3.31                    | 3.44 $\pm$ 1.59   | <b>38.97 <math>\pm</math> 6.44</b> | 11.30 $\pm$ 2.86  | 36.70 $\pm$ 4.41                    | 3.74 $\pm$ 2.40                     | 36.48 $\pm$ 2.66                    |
| Fish                         | <b>49.03 <math>\pm</math> 5.96</b> | 21.26 $\pm$ 3.60                    | <u>45.71 <math>\pm</math> 1.71</u>  | 36.00 $\pm$ 3.98                    | 23.20 $\pm$ 3.80                   | 32.57 $\pm$ 4.57                    | 16.95 $\pm$ 2.93  | 30.29 $\pm$ 2.62                   | 21.71 $\pm$ 2.77  | 31.81 $\pm$ 4.76                    | 20.80 $\pm$ 5.10                    | 34.74 $\pm$ 4.91                    |
| FordA                        | 59.23 $\pm$ 9.68                   | 57.68 $\pm$ 10.00                   | <b>59.83 <math>\pm</math> 8.76</b>  | 50.73 $\pm$ 1.01                    | 56.70 $\pm$ 8.04                   | 50.30 $\pm$ 0.33                    | 49.57 $\pm$ 1.42  | 49.72 $\pm$ 0.63                   | 49.68 $\pm$ 1.11  | 49.92 $\pm$ 0.80                    | 50.61 $\pm$ 1.68                    | 50.71 $\pm$ 0.98                    |
| FordB                        | 52.54 $\pm$ 6.44                   | 52.64 $\pm$ 6.88                    | 52.40 $\pm$ 4.53                    | 50.12 $\pm$ 1.83                    | <b>53.60 <math>\pm</math> 4.09</b> | 49.96 $\pm$ 0.94                    | 5                 |                                    |                   |                                     |                                     |                                     |

Table 3: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

| Dataset                        | ResNet               | TapNet               | COSCO-ResNet         | DLinear              | PatchTST             | TimesNet             | Autoformer          | Crossformer          | FEDformer           | Informer             | GPT4TS               | DuoTSP               |
|--------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|---------------------|----------------------|----------------------|----------------------|
| GesturePebbleZ1                | 49.07 ± 5.03         | 38.37 ± 6.03         | <b>53.26 ± 6.97</b>  | 46.16 ± 13.08        | 23.60 ± 6.95         | 40.50 ± 15.60        | 20.93 ± 3.54        | 42.44 ± 7.05         | 18.95 ± 2.08        | 43.99 ± 8.75         | 23.60 ± 6.95         | 35.81 ± 4.48         |
| GesturePebbleZ2                | 55.19 ± 11.38        | 42.91 ± 15.92        | <b>57.47 ± 8.89</b>  | 48.35 ± 13.33        | 21.01 ± 3.48         | 47.68 ± 13.93        | 18.35 ± 4.94        | 43.88 ± 16.17        | 16.20 ± 5.17        | 55.06 ± 14.60        | 21.01 ± 3.48         | 33.29 ± 5.85         |
| GunPoint                       | 55.47 ± 4.79         | 58.13 ± 9.13         | 55.87 ± 5.32         | 53.20 ± 9.11         | 56.40 ± 7.58         | <b>64.00 ± 8.08</b>  | 58.89 ± 10.59       | 63.33 ± 4.81         | 58.93 ± 8.45        | 62.89 ± 3.01         | 52.40 ± 11.44        | 58.00 ± 7.79         |
| GunPointAgeSpan                | 49.94 ± 8.02         | 48.67 ± 3.53         | 49.11 ± 3.99         | 60.82 ± 22.09        | 58.73 ± 19.72        | 59.70 ± 21.56        | 52.11 ± 7.60        | 59.07 ± 20.81        | 52.41 ± 6.32        | <b>61.92 ± 19.97</b> | 48.10 ± 2.61         | 60.51 ± 10.32        |
| GunPointMaleVersusFemale       | 63.99 ± 11.97        | 63.29 ± 11.59        | 64.30 ± 12.20        | 54.56 ± 4.47         | 56.65 ± 3.27         | 55.91 ± 6.97         | 50.95 ± 7.17        | 55.59 ± 9.92         | 50.19 ± 6.98        | 56.75 ± 8.76         | 58.04 ± 11.38        | <b>81.01 ± 10.12</b> |
| GunPointOldVersusYoung         | <b>92.38 ± 10.44</b> | 89.97 ± 13.74        | 90.35 ± 13.23        | 53.90 ± 25.42        | 53.33 ± 25.91        | 58.31 ± 30.15        | 54.81 ± 4.33        | 57.46 ± 29.14        | 54.79 ± 8.98        | 58.10 ± 28.87        | 78.41 ± 20.62        | 57.97 ± 10.78        |
| Ham                            | 50.10 ± 9.32         | 51.24 ± 7.01         | 47.81 ± 11.28        | 55.24 ± 11.13        | 49.33 ± 2.81         | 48.89 ± 5.25         | 53.65 ± 3.06        | 51.43 ± 7.44         | <b>55.81 ± 6.79</b> | 50.16 ± 4.89         | 51.81 ± 10.55        | 52.76 ± 9.30         |
| HandOutlines                   | 46.81 ± 15.76        | 58.86 ± 12.82        | 55.84 ± 9.47         | 70.59 ± 7.20         | 75.68 ± 8.02         | 71.17 ± 7.55         | 51.08 ± 2.34        | <b>76.76 ± 10.41</b> | 52.81 ± 15.40       | 74.86 ± 11.22        | 61.19 ± 14.73        | 61.73 ± 6.31         |
| Haptics                        | 23.96 ± 5.22         | 22.66 ± 2.12         | 22.86 ± 4.94         | <b>26.49 ± 6.85</b>  | 22.66 ± 4.38         | 25.11 ± 9.19         | 19.16 ± 1.81        | 23.16 ± 7.32         | 23.44 ± 3.46        | 25.43 ± 9.51         | 25.65 ± 5.26         | 25.65 ± 4.99         |
| Herring                        | 49.69 ± 6.39         | 49.69 ± 10.50        | 50.94 ± 11.51        | 48.12 ± 8.44         | 49.69 ± 7.44         | 47.40 ± 10.40        | <b>51.04 ± 2.39</b> | 49.48 ± 9.55         | 48.75 ± 8.44        | 48.44 ± 9.50         | 48.12 ± 10.39        | 50.00 ± 10.00        |
| HouseTwenty                    | <b>80.84 ± 7.67</b>  | 74.62 ± 5.52         | 68.40 ± 16.13        | 50.25 ± 3.68         | 57.65 ± 8.25         | 68.63 ± 3.50         | 49.30 ± 7.15        | 61.62 ± 2.57         | 46.05 ± 6.81        | 66.67 ± 1.28         | 53.61 ± 8.52         | 62.18 ± 8.04         |
| InlineSkate                    | 16.40 ± 1.60         | 15.64 ± 0.76         | 18.00 ± 0.73         | 15.89 ± 3.32         | 14.69 ± 2.92         | 18.06 ± 2.11         | 15.15 ± 0.73        | 18.00 ± 2.36         | 15.02 ± 3.34        | <b>18.67 ± 3.56</b>  | 14.87 ± 3.22         | 17.13 ± 2.24         |
| InsectEPGRegularTrain          | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | 43.53 ± 11.83        | 50.44 ± 13.06        | 48.19 ± 5.92         | 42.03 ± 6.13        | 45.25 ± 3.22         | 46.27 ± 8.59        | 49.26 ± 6.69         | 87.15 ± 20.64        | 55.34 ± 14.47        |
| InsectEPGSmallTrain            | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | 47.95 ± 6.69         | 55.10 ± 5.49         | 51.14 ± 8.16         | 39.36 ± 1.75        | 52.88 ± 6.50         | 37.67 ± 14.87       | 52.48 ± 11.47        | <b>85.78 ± 19.79</b> | 50.76 ± 7.86         |
| InsectWingbeatSound            | 16.18 ± 3.00         | 14.23 ± 3.87         | 14.35 ± 2.82         | 39.88 ± 4.03         | 22.79 ± 2.65         | 43.42 ± 0.26         | 17.56 ± 4.92        | 43.16 ± 3.28         | 21.14 ± 2.99        | <b>45.61 ± 5.13</b>  | 12.52 ± 3.37         | 25.27 ± 2.20         |
| ItalyPowerDemand               | 61.83 ± 17.15        | 62.29 ± 17.84        | 64.74 ± 16.58        | 55.32 ± 7.10         | 60.35 ± 16.10        | 69.81 ± 17.24        | 68.35 ± 9.65        | <b>78.33 ± 24.61</b> | 68.98 ± 15.67       | 71.62 ± 21.13        | 47.31 ± 2.37         | 58.52 ± 15.27        |
| LargeKitchenAppliances         | 48.48 ± 12.88        | 52.85 ± 11.74        | 46.72 ± 15.63        | 33.39 ± 3.38         | <b>54.88 ± 8.28</b>  | 40.09 ± 11.55        | 34.93 ± 5.70        | 32.53 ± 3.23         | 31.47 ± 3.69        | 41.69 ± 12.56        | 33.81 ± 2.87         | 45.44 ± 5.60         |
| Lightning2                     | 50.49 ± 6.80         | 55.08 ± 10.15        | 51.80 ± 4.27         | 56.72 ± 11.39        | 51.80 ± 12.52        | 58.47 ± 11.16        | 48.63 ± 3.41        | 57.38 ± 11.36        | 51.48 ± 5.26        | <b>59.02 ± 8.67</b>  | 55.08 ± 13.25        | <b>59.67 ± 8.87</b>  |
| Lightning7                     | 44.93 ± 4.48         | 44.38 ± 4.60         | 40.82 ± 7.47         | 43.01 ± 4.18         | 42.19 ± 7.34         | 53.42 ± 10.70        | 15.07 ± 6.28        | 45.66 ± 6.76         | 18.63 ± 4.40        | <b>56.16 ± 8.33</b>  | 24.11 ± 7.72         | 41.64 ± 7.60         |
| Mallat                         | 64.03 ± 14.52        | 50.42 ± 9.84         | 64.49 ± 9.43         | <b>80.37 ± 4.76</b>  | 56.96 ± 10.09        | <b>82.32 ± 3.73</b>  | 54.67 ± 2.15        | 75.96 ± 4.82         | 64.55 ± 4.53        | 75.72 ± 2.60         | 37.30 ± 5.46         | 68.24 ± 7.54         |
| Meat                           | 42.00 ± 9.96         | 44.00 ± 10.18        | 58.33 ± 10.99        | <b>77.67 ± 17.26</b> | 52.67 ± 11.94        | 46.11 ± 9.18         | 38.33 ± 8.66        | 33.33 ± 0.00         | 36.00 ± 2.53        | 66.11 ± 20.97        | 33.67 ± 0.75         | 55.67 ± 9.62         |
| MedicalImages                  | 27.92 ± 5.84         | 28.45 ± 4.75         | 24.95 ± 5.95         | 26.34 ± 8.67         | 24.16 ± 4.26         | 22.32 ± 5.35         | 26.84 ± 12.59       | 23.86 ± 6.12         | <b>28.53 ± 8.23</b> | 23.16 ± 6.10         | 4.21 ± 2.27          | 27.66 ± 4.41         |
| MelbournePedestrian            | <b>51.29 ± 5.58</b>  | 44.27 ± 4.81         | 47.47 ± 6.50         | 9.85 ± 0.23          | 31.99 ± 4.95         | 21.74 ± 7.37         | 46.44 ± 8.26        | 40.62 ± 4.65         | 43.40 ± 7.11        | 43.73 ± 5.33         | 31.99 ± 4.95         | 39.16 ± 5.22         |
| MiddlePhalanxOutlineAgeGroup   | 38.83 ± 7.74         | 37.53 ± 10.57        | 38.05 ± 7.83         | 34.03 ± 9.03         | 35.97 ± 8.21         | 37.45 ± 11.78        | 29.87 ± 4.55        | 38.53 ± 15.64        | 33.90 ± 7.76        | 32.90 ± 7.86         | <b>41.69 ± 18.94</b> | 39.48 ± 7.40         |
| MiddlePhalanxOutlineCorrect    | 53.68 ± 13.63        | 52.10 ± 14.21        | 54.43 ± 13.33        | 53.75 ± 17.89        | 56.49 ± 9.58         | <b>58.42 ± 17.26</b> | 55.56 ± 15.16       | 54.07 ± 23.81        | 54.50 ± 10.22       | 57.50 ± 21.09        | 52.85 ± 12.08        | 51.68 ± 5.00         |
| MiddlePhalanxTW                | 34.03 ± 16.27        | 44.68 ± 7.76         | 32.47 ± 17.97        | 30.52 ± 10.75        | 32.34 ± 10.78        | 29.00 ± 15.96        | 38.96 ± 3.37        | <b>46.10 ± 3.62</b>  | 35.45 ± 13.57       | 42.21 ± 5.07         | 15.19 ± 11.64        | 37.14 ± 16.84        |
| MixedShapesRegularTrain        | 45.36 ± 5.29         | 40.59 ± 8.80         | 43.58 ± 10.39        | 47.60 ± 10.74        | 39.59 ± 5.62         | 43.05 ± 15.67        | 17.31 ± 1.15        | 45.53 ± 11.76        | 25.71 ± 3.77        | 42.76 ± 16.81        | 45.03 ± 12.47        | <b>56.06 ± 5.34</b>  |
| MixedShapesSmallTrain          | 52.78 ± 8.02         | 39.55 ± 6.53         | 48.84 ± 9.40         | 53.90 ± 7.66         | 45.57 ± 7.91         | 52.81 ± 6.78         | 17.32 ± 0.61        | 53.84 ± 4.25         | 24.17 ± 3.24        | 53.31 ± 5.69         | 48.73 ± 8.44         | <b>63.30 ± 3.34</b>  |
| MoteStrain                     | 62.99 ± 13.96        | 60.67 ± 12.58        | 64.35 ± 11.70        | 63.40 ± 15.50        | 63.27 ± 13.76        | 68.45 ± 11.94        | 56.26 ± 12.55       | 63.82 ± 16.98        | 61.65 ± 9.57        | <b>68.66 ± 11.89</b> | 57.99 ± 9.98         | 54.36 ± 12.77        |
| NonInvasiveFetalECGThorax1     | 19.05 ± 4.42         | 8.04 ± 2.52          | 19.05 ± 4.42         | 38.23 ± 1.81         | 12.07 ± 7.41         | 36.76 ± 3.30         | 12.57 ± 2.38        | 28.62 ± 3.97         | 14.44 ± 2.07        | <b>42.58 ± 4.62</b>  | 5.58 ± 2.48          | 39.16 ± 2.28         |
| NonInvasiveFetalECGThorax2     | 20.69 ± 5.30         | 9.71 ± 2.65          | 20.69 ± 5.30         | 47.48 ± 2.72         | 14.81 ± 5.33         | 46.67 ± 0.18         | 20.97 ± 3.22        | 34.59 ± 7.78         | 23.69 ± 1.73        | <b>49.89 ± 4.96</b>  | 8.30 ± 1.72          | 47.48 ± 2.06         |
| OSULeaf                        | <b>45.62 ± 7.45</b>  | 30.41 ± 8.35         | 38.35 ± 12.00        | 24.05 ± 2.14         | 22.15 ± 5.63         | 27.82 ± 3.13         | 17.77 ± 1.80        | 26.03 ± 1.89         | 20.00 ± 5.09        | 26.17 ± 1.56         | 17.60 ± 2.44         | 29.83 ± 5.82         |
| OliveOil                       | 28.00 ± 23.64        | 26.67 ± 11.06        | 47.33 ± 13.21        | <b>65.33 ± 15.20</b> | 36.00 ± 22.66        | 32.22 ± 16.44        | 44.44 ± 10.72       | 28.89 ± 11.71        | 32.00 ± 9.60        | 61.11 ± 15.40        | 27.33 ± 11.88        | 28.00 ± 14.26        |
| PLAID                          | 15.46 ± 6.16         | 13.97 ± 4.45         | 16.95 ± 4.17         | 17.77 ± 3.34         | 21.79 ± 0.97         | 21.54 ± 5.77         | 13.78 ± 3.08        | 19.93 ± 5.41         | 14.41 ± 2.63        | 20.30 ± 4.75         | 21.79 ± 0.97         | <b>32.10 ± 5.72</b>  |
| PhalangesOutlinesCorrect       | <b>54.90 ± 6.98</b>  | 51.28 ± 10.93        | 54.41 ± 7.38         | 53.40 ± 8.33         | 52.54 ± 10.38        | 48.87 ± 4.48         | 51.01 ± 10.34       | 53.38 ± 8.94         | 52.73 ± 9.70        | 52.49 ± 9.30         | <b>54.83 ± 6.32</b>  | 51.93 ± 9.54         |
| Phoneme                        | <b>10.43 ± 1.42</b>  | 8.46 ± 1.00          | <b>10.43 ± 1.42</b>  | 4.21 ± 0.54          | 5.61 ± 2.38          | 6.14 ± 1.81          | 2.39 ± 1.01         | 5.87 ± 1.54          | 3.20 ± 2.06         | 4.98 ± 3.20          | 3.42 ± 0.41          | 6.87 ± 1.15          |
| PickupGestureWiimoteZ          | 35.60 ± 7.80         | 44.80 ± 4.82         | 42.00 ± 5.10         | 40.80 ± 5.40         | 36.00 ± 3.16         | 46.67 ± 9.87         | 17.33 ± 5.03        | 39.33 ± 6.43         | 20.40 ± 4.98        | <b>48.67 ± 8.33</b>  | 36.00 ± 3.16         | 47.20 ± 7.29         |
| PigAirwayPressure              | 23.94 ± 3.76         | 9.13 ± 2.66          | <b>38.65 ± 2.92</b>  | 3.65 ± 1.47          | 3.75 ± 1.33          | 5.29 ± 0.83          | 4.01 ± 0.73         | 4.01 ± 1.11          | 3.27 ± 0.92         | 4.81 ± 0.96          | 6.15 ± 1.64          | 5.48 ± 2.19          |
| PigArtPressure                 | 40.58 ± 5.14         | 10.10 ± 2.36         | <b>60.10 ± 2.28</b>  | 9.90 ± 2.42          | 2.21 ± 0.80          | 10.58 ± 2.10         | 8.01 ± 2.00         | 10.58 ± 1.73         | 4.81 ± 0.76         | 9.94 ± 2.27          | 15.58 ± 3.05         | 15.38 ± 4.07         |
| PigCVP                         | 22.50 ± 3.47         | 14.33 ± 2.79         | <b>50.29 ± 2.90</b>  | 5.00 ± 1.25          | 3.27 ± 0.71          | 5.93 ± 0.73          | 3.53 ± 1.39         | 6.89 ± 0.56          | 2.02 ± 0.22         | 4.33 ± 0.83          | 8.94 ± 1.94          | 11.44 ± 2.08         |
| Plane                          | <b>98.86 ± 2.56</b>  | 91.62 ± 6.26         | 95.43 ± 3.65         | 87.24 ± 8.67         | 85.14 ± 9.49         | 82.86 ± 4.15         | 78.73 ± 5.25        | 88.57 ± 9.09         | 86.48 ± 4.39        | 88.57 ± 6.67         | 26.10 ± 13.02        | 93.90 ± 6.92         |
| PowerCons                      | 76.89 ± 9.68         | <b>80.22 ± 9.41</b>  | 78.11 ± 13.88        | 60.22 ± 11.82        | 61.00 ± 7.74         | 62.96 ± 3.70         | 49.44 ± 7.47        | 58.33 ± 2.78         | 51.33 ± 4.56        | 62.78 ± 6.41         | 55.22 ± 21.33        | 60.89 ± 11.09        |
| ProximalPhalanxOutlineAgeGroup | 48.49 ± 34.51        | 48.20 ± 31.13        | 48.68 ± 30.82        | 39.61 ± 19.82        | 39.12 ± 18.60        | 43.74 ± 32.89        | 49.59 ± 25.02       | <b>60.81 ± 40.85</b> | 38.15 ± 22.87       | 46.50 ± 27.14        | 32.68 ± 16.57        | 48.98 ± 23.27        |
| ProximalPhalanxOutlineCorrect  | 54.43 ± 13.05        | 53.40 ± 15.85        | 51.20 ± 16.57        | 56.22 ± 5.47         | 53.26 ± 10.68        | 44.67 ± 13.09        | 46.96 ± 12.63       | <b>58.76 ± 6.63</b>  | 52.78 ± 12.27       | 43.99 ± 6.30         | 57.59 ± 13.02        | 52.23 ± 12.03        |
| ProximalPhalanxTW              | 54.63 ± 10.31        | 57.56 ± 9.33         | 58.15 ± 6.64         | 54.34 ± 11.16        | 49.76 ± 11.67        | <b>60.81 ± 11.71</b> | 55.93 ± 9.23        | 38.37 ± 0.75         | 55.80 ± 7.24        | 57.07 ± 10.70        | 17.37 ± 16.23        | 55.22 ± 6.63         |
| RefrigerationDevices           | 36.05 ± 6.49         | 35.68 ± 5.16         | 37.44 ± 5.37         | 34.99 ± 2.77         | 33.97 ± 4.76         | <b>41.16 ± 5.10</b>  | 34.58 ± 1.47        | 34.93 ± 0.92         | 34.72 ± 0.99        | 38.84 ± 5.04         | 32.75 ± 2.24         | 29.76 ± 11.02        |
| Rock                           | 29.20 ± 8.07         | 34.40 ± 7.92         | 28.40 ± 6.54         | 41.60 ± 16.52        | 26.40 ± 8.53         | 38.67 ± 25.32        | <b>57.33 ± 5.03</b> | 44.67 ± 18.90        | 36.80 ± 8.07        | 40.00 ± 24.25        | 36.40 ± 14.93        | 39.20 ± 13.75        |
| ScreenType                     | 33.92 ± 4.73         | 34.19 ± 7.39         | <b>34.29 ± 6.61</b>  | 33.87 ± 2.72         | 33.87 ± 4.61         | 32.44 ± 5.27         | 32.18 ± 3.99        | 28.00 ± 3.40         | 30.99 ± 3.01        | 33.60 ± 4.19         | 30.61 ± 2.66         | 32.53 ± 4.84         |
| SemgHandGenderCh2              | 50.87 ± 2.35         | 55.37 ± 5.38         | 53.17 ± 4.35         | 50.17 ± 11.45        | <b>61.00 ± 15.57</b> | 42.78 ± 13.47        | 52.78 ± 1.84        | 39.89 ± 8.47         | 44.67 ± 12.14       | 35.61 ± 1.06         | 55.73 ± 4.74         | 51.90 ± 14.92        |
| SemgHandMovementCh2            | 28.89 ± 4.22         | <b>29.87 ± 4.86</b>  | 29.60 ± 5.24         | 25.29 ± 1.13         | 24.89 ± 3.57         | 20.37 ± 0.13         | 21.70 ± 2.31        | 22.52 ± 2.76         | 18.71 ± 1.91        | 20.07 ± 3.46         | 27.02 ± 5.10         | 28.13 ± 4.95         |

Continued on next page

Table 3: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

| Dataset                | ResNet                              | TapNet                              | COSCO-ResNet                        | DLinear                            | PatchTST                           | TimesNet                            | Autoformer        | Crossformer                        | FEDformer         | Informer                           | GPT4TS            | DuoTSP                              |
|------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|------------------------------------|-------------------------------------|-------------------|------------------------------------|-------------------|------------------------------------|-------------------|-------------------------------------|
| SemgHandSubjectCh2     | 28.89 $\pm$ 5.77                    | 29.07 $\pm$ 4.60                    | 28.00 $\pm$ 2.65                    | 35.60 $\pm$ 4.19                   | 29.91 $\pm$ 10.77                  | 26.37 $\pm$ 1.89                    | 23.11 $\pm$ 2.89  | 26.67 $\pm$ 4.87                   | 21.56 $\pm$ 1.94  | 26.96 $\pm$ 3.68                   | 27.91 $\pm$ 3.11  | <b>37.42 <math>\pm</math> 8.82</b>  |
| ShakeGestureWiimoteZ   | 63.20 $\pm$ 7.69                    | 58.00 $\pm$ 11.83                   | <b>67.60 <math>\pm</math> 11.61</b> | 40.40 $\pm$ 11.26                  | 43.20 $\pm$ 9.55                   | 45.33 $\pm$ 3.06                    | 16.00 $\pm$ 7.21  | 40.00 $\pm$ 3.46                   | 24.40 $\pm$ 4.98  | 46.00 $\pm$ 6.00                   | 43.20 $\pm$ 9.55  | 60.40 $\pm$ 7.92                    |
| ShapeletSim            | 66.33 $\pm$ 12.25                   | <b>87.00 <math>\pm</math> 16.91</b> | 73.89 $\pm$ 23.85                   | 50.11 $\pm$ 2.27                   | 75.22 $\pm$ 8.30                   | 53.89 $\pm$ 1.67                    | 52.96 $\pm$ 1.40  | 48.89 $\pm$ 5.64                   | 50.44 $\pm$ 1.82  | 51.11 $\pm$ 0.96                   | 47.78 $\pm$ 3.07  | 62.44 $\pm$ 7.44                    |
| ShapesAll              | 23.63 $\pm$ 4.60                    | 15.60 $\pm$ 1.98                    | 23.63 $\pm$ 4.60                    | 45.70 $\pm$ 2.59                   | 19.27 $\pm$ 4.58                   | 44.72 $\pm$ 2.58                    | 6.00 $\pm$ 0.58   | 43.94 $\pm$ 2.14                   | 8.07 $\pm$ 3.90   | 44.06 $\pm$ 1.11                   | 8.13 $\pm$ 2.75   | <b>50.20 <math>\pm</math> 3.03</b>  |
| SmallKitchenAppliances | 40.37 $\pm$ 15.51                   | <b>46.24 <math>\pm</math> 15.42</b> | 44.32 $\pm$ 15.50                   | 42.19 $\pm$ 3.16                   | 38.13 $\pm$ 22.42                  | 32.98 $\pm$ 5.23                    | 32.80 $\pm$ 3.67  | 35.82 $\pm$ 4.27                   | 35.41 $\pm$ 3.13  | 36.89 $\pm$ 7.74                   | 38.88 $\pm$ 7.80  | 42.88 $\pm$ 15.27                   |
| SmoothSubspace         | <b>74.93 <math>\pm</math> 5.59</b>  | 74.40 $\pm$ 5.13                    | <u>74.80 <math>\pm</math> 8.48</u>  | 31.73 $\pm$ 3.96                   | 50.00 $\pm$ 10.52                  | 46.67 $\pm$ 1.76                    | 53.33 $\pm$ 9.45  | 50.22 $\pm$ 10.96                  | 62.27 $\pm$ 6.56  | 60.67 $\pm$ 10.35                  | 56.67 $\pm$ 5.19  | 68.53 $\pm$ 8.95                    |
| SonyAIBORobotSurface1  | 77.57 $\pm$ 10.30                   | <b>80.80 <math>\pm</math> 8.99</b>  | <u>80.37 <math>\pm</math> 5.89</u>  | 59.80 $\pm$ 6.73                   | 64.53 $\pm$ 18.83                  | 52.86 $\pm$ 10.38                   | 56.96 $\pm$ 5.82  | 55.35 $\pm$ 9.37                   | 56.07 $\pm$ 4.43  | 57.68 $\pm$ 9.32                   | 55.61 $\pm$ 9.09  | 57.20 $\pm$ 8.62                    |
| SonyAIBORobotSurface2  | 80.29 $\pm$ 6.43                    | <b>84.09 <math>\pm</math> 5.18</b>  | <u>80.90 <math>\pm</math> 6.99</u>  | 67.39 $\pm$ 19.08                  | 73.03 $\pm$ 13.78                  | 66.18 $\pm$ 11.61                   | 60.37 $\pm$ 27.05 | 61.35 $\pm$ 27.10                  | 69.49 $\pm$ 15.13 | 68.07 $\pm$ 16.06                  | 57.31 $\pm$ 15.03 | 66.09 $\pm$ 10.78                   |
| StarLightCurves        | 63.75 $\pm$ 3.45                    | 62.30 $\pm$ 9.88                    | <u>57.59 <math>\pm</math> 9.08</u>  | 57.39 $\pm$ 13.40                  | 57.55 $\pm$ 15.28                  | <b>67.54 <math>\pm</math> 17.11</b> | 37.78 $\pm$ 3.53  | 57.92 $\pm$ 15.55                  | 27.99 $\pm$ 0.00  | 67.36 $\pm$ 17.40                  | 64.81 $\pm$ 14.61 | 63.03 $\pm$ 14.23                   |
| Strawberry             | 52.27 $\pm$ 4.26                    | 51.84 $\pm$ 3.86                    | 54.43 $\pm$ 4.33                    | 54.86 $\pm$ 4.34                   | 53.62 $\pm$ 3.45                   | <u>58.29 <math>\pm</math> 0.95</u>  | 54.86 $\pm$ 13.21 | <b>60.18 <math>\pm</math> 0.68</b> | 56.11 $\pm$ 9.26  | <u>57.93 <math>\pm</math> 0.83</u> | 45.89 $\pm$ 11.15 | 51.46 $\pm$ 5.21                    |
| SwedishLeaf            | <b>65.95 <math>\pm</math> 6.01</b>  | 46.08 $\pm$ 6.36                    | <b>65.95 <math>\pm</math> 6.01</b>  | 27.42 $\pm$ 3.44                   | 40.83 $\pm$ 4.40                   | 36.69 $\pm$ 1.99                    | 27.20 $\pm$ 2.42  | 24.53 $\pm$ 4.08                   | 31.65 $\pm$ 2.48  | 38.19 $\pm$ 0.81                   | 8.10 $\pm$ 1.23   | <u>57.98 <math>\pm</math> 2.72</u>  |
| Symbols                | <b>85.77 <math>\pm</math> 4.83</b>  | 70.17 $\pm$ 4.91                    | 76.68 $\pm$ 7.53                    | 79.60 $\pm$ 7.65                   | 72.50 $\pm$ 6.12                   | 81.54 $\pm$ 3.92                    | 37.15 $\pm$ 0.97  | 82.18 $\pm$ 0.42                   | 46.13 $\pm$ 9.06  | 80.57 $\pm$ 0.95                   | 51.54 $\pm$ 6.72  | 80.94 $\pm$ 7.54                    |
| SyntheticControl       | 72.53 $\pm$ 6.61                    | 76.40 $\pm$ 5.24                    | 68.67 $\pm$ 10.59                   | 38.47 $\pm$ 6.28                   | 80.33 $\pm$ 9.12                   | 47.78 $\pm$ 5.87                    | 44.89 $\pm$ 3.20  | <u>48.22 <math>\pm</math> 0.38</u> | 42.13 $\pm$ 2.87  | 50.00 $\pm$ 4.48                   | 24.53 $\pm$ 7.68  | <b>84.20 <math>\pm</math> 6.54</b>  |
| ToeSegmentation1       | <b>69.82 <math>\pm</math> 7.39</b>  | 69.65 $\pm$ 4.27                    | 65.09 $\pm$ 8.05                    | 51.67 $\pm$ 1.47                   | <u>55.79 <math>\pm</math> 6.85</u> | 49.27 $\pm$ 1.10                    | 49.71 $\pm$ 3.29  | 46.64 $\pm$ 4.73                   | 52.46 $\pm$ 2.82  | 48.39 $\pm$ 3.23                   | 49.47 $\pm$ 2.47  | 58.16 $\pm$ 3.70                    |
| ToeSegmentation2       | 68.92 $\pm$ 8.63                    | 70.31 $\pm$ 3.38                    | <u>70.92 <math>\pm</math> 10.76</u> | 50.31 $\pm$ 5.37                   | 65.54 $\pm$ 9.84                   | 50.26 $\pm$ 2.91                    | 35.90 $\pm$ 10.81 | 50.51 $\pm$ 4.24                   | 33.23 $\pm$ 16.47 | 52.05 $\pm$ 6.54                   | 41.54 $\pm$ 14.95 | <b>75.23 <math>\pm</math> 12.13</b> |
| Trace                  | 61.00 $\pm$ 4.64                    | 65.20 $\pm$ 9.09                    | <u>65.80 <math>\pm</math> 6.61</u>  | 49.60 $\pm$ 3.65                   | <u>74.40 <math>\pm</math> 6.15</u> | 51.33 $\pm$ 5.51                    | 42.33 $\pm$ 4.51  | 47.67 $\pm$ 0.58                   | 44.80 $\pm$ 7.69  | 51.33 $\pm$ 4.93                   | 37.80 $\pm$ 6.42  | <b>99.20 <math>\pm</math> 1.79</b>  |
| TwoLeadECG             | <b>69.48 <math>\pm</math> 12.74</b> | 64.97 $\pm$ 10.26                   | <u>66.51 <math>\pm</math> 13.23</u> | 54.03 $\pm$ 2.33                   | <u>57.82 <math>\pm</math> 5.04</u> | 59.32 $\pm$ 3.05                    | 53.32 $\pm$ 7.30  | 52.62 $\pm$ 3.32                   | 54.47 $\pm$ 5.41  | 60.32 $\pm$ 1.06                   | 50.73 $\pm$ 1.02  | 64.88 $\pm$ 11.28                   |
| TwoPatterns            | 48.77 $\pm$ 2.41                    | <b>61.51 <math>\pm</math> 6.85</b>  | <u>46.17 <math>\pm</math> 11.65</u> | 31.86 $\pm$ 3.24                   | 36.86 $\pm$ 6.10                   | 29.76 $\pm$ 2.56                    | 25.52 $\pm$ 1.93  | 28.23 $\pm$ 0.74                   | 26.06 $\pm$ 1.27  | 29.87 $\pm$ 1.68                   | 26.32 $\pm$ 2.53  | 45.74 $\pm$ 2.64                    |
| UMD                    | 43.06 $\pm$ 4.02                    | <b>54.58 <math>\pm</math> 9.53</b>  | 45.56 $\pm$ 6.14                    | 46.81 $\pm$ 7.70                   | 50.14 $\pm$ 6.28                   | 46.30 $\pm$ 5.02                    | 45.60 $\pm$ 3.21  | 47.69 $\pm$ 4.19                   | 48.61 $\pm$ 2.55  | 47.45 $\pm$ 6.56                   | 31.39 $\pm$ 4.35  | 45.47 $\pm$ 8.73                    |
| UWaveGestureLibraryAll | 29.36 $\pm$ 4.20                    | 24.46 $\pm$ 5.19                    | 28.34 $\pm$ 3.82                    | 66.77 $\pm$ 5.89                   | 41.63 $\pm$ 5.40                   | <u>69.19 <math>\pm</math> 5.01</u>  | 25.78 $\pm$ 1.68  | 67.23 $\pm$ 2.91                   | 26.67 $\pm$ 3.53  | <b>69.21 <math>\pm</math> 6.24</b> | 50.70 $\pm$ 5.42  | 33.37 $\pm$ 0.26                    |
| UWaveGestureLibraryX   | 26.45 $\pm$ 4.19                    | 26.60 $\pm$ 3.44                    | 24.71 $\pm$ 1.83                    | 47.27 $\pm$ 6.01                   | 35.36 $\pm$ 5.04                   | <b>49.30 <math>\pm</math> 7.17</b>  | 22.37 $\pm$ 0.14  | 47.22 $\pm$ 8.02                   | 24.34 $\pm$ 5.30  | 49.17 $\pm$ 7.61                   | 17.28 $\pm$ 8.20  | 30.33 $\pm$ 0.23                    |
| UWaveGestureLibraryY   | 25.37 $\pm$ 4.63                    | 26.13 $\pm$ 4.45                    | 24.62 $\pm$ 4.93                    | 43.95 $\pm$ 4.51                   | 38.35 $\pm$ 7.26                   | 47.05 $\pm$ 4.02                    | 24.58 $\pm$ 1.89  | 46.37 $\pm$ 4.93                   | 26.81 $\pm$ 2.91  | <b>47.18 <math>\pm</math> 5.20</b> | 20.37 $\pm$ 4.71  | 40.36 $\pm$ 0.25                    |
| UWaveGestureLibraryZ   | 24.77 $\pm$ 4.74                    | 27.03 $\pm$ 3.18                    | 24.45 $\pm$ 3.15                    | 42.90 $\pm$ 1.96                   | 36.37 $\pm$ 3.72                   | <u>44.61 <math>\pm</math> 1.78</u>  | 22.47 $\pm$ 2.82  | 44.70 $\pm$ 2.64                   | 25.10 $\pm$ 2.80  | <u>44.75 <math>\pm</math> 1.63</u> | 22.77 $\pm$ 3.47  | <b>47.44 <math>\pm</math> 2.61</b>  |
| Wafer                  | 50.39 $\pm$ 4.56                    | 44.94 $\pm$ 10.31                   | 49.91 $\pm$ 13.57                   | <b>67.87 <math>\pm</math> 4.81</b> | 51.56 $\pm$ 17.59                  | 63.90 $\pm$ 3.73                    | 64.15 $\pm$ 3.76  | 63.99 $\pm$ 9.60                   | 63.82 $\pm$ 9.82  | <u>65.48 <math>\pm</math> 1.03</u> | 43.02 $\pm$ 26.21 | 46.27 $\pm$ 7.11                    |
| Wine                   | <b>57.41 <math>\pm</math> 14.70</b> | 54.07 $\pm$ 4.61                    | <u>54.81 <math>\pm</math> 11.31</u> | 52.96 $\pm$ 8.34                   | 50.74 $\pm$ 6.23                   | 48.77 $\pm$ 3.85                    | 50.00 $\pm$ 0.00  | 50.00 $\pm$ 0.00                   | 52.22 $\pm$ 4.97  | 37.65 $\pm$ 19.80                  | 51.48 $\pm$ 3.31  | 51.11 $\pm$ 7.48                    |
| WordSynonyms           | 14.92 $\pm$ 3.03                    | 13.70 $\pm$ 5.79                    | <u>14.92 <math>\pm</math> 3.03</u>  | 19.78 $\pm$ 2.70                   | 16.96 $\pm$ 4.46                   | 19.75 $\pm$ 1.90                    | 6.79 $\pm$ 1.46   | <b>21.16 <math>\pm</math> 1.50</b> | 10.06 $\pm$ 3.54  | 20.53 $\pm$ 2.41                   | 4.98 $\pm$ 1.34   | 20.91 $\pm$ 2.73                    |
| Worms                  | <b>36.10 <math>\pm</math> 9.87</b>  | 33.25 $\pm$ 6.66                    | <u>35.06 <math>\pm</math> 11.43</u> | 17.14 $\pm$ 4.04                   | 23.90 $\pm$ 9.39                   | 19.91 $\pm$ 6.54                    | 23.38 $\pm$ 4.50  | 18.61 $\pm$ 5.86                   | 17.66 $\pm$ 7.03  | 19.05 $\pm$ 5.25                   | 17.66 $\pm$ 2.36  | 24.68 $\pm$ 8.95                    |
| WormsTwoClass          | 44.79 $\pm$ 3.48                    | 49.09 $\pm$ 6.96                    | <u>52.99 <math>\pm</math> 6.71</u>  | 49.09 $\pm$ 3.72                   | 50.39 $\pm$ 3.23                   | 49.35 $\pm$ 3.44                    | 48.92 $\pm$ 10.50 | 52.38 $\pm$ 1.98                   | 53.51 $\pm$ 4.81  | <b>53.68 <math>\pm</math> 3.75</b> | 52.99 $\pm$ 2.82  | 47.53 $\pm$ 10.69                   |
| Yoga                   | <b>55.60 <math>\pm</math> 4.57</b>  | 50.68 $\pm$ 5.18                    | <u>53.54 <math>\pm</math> 4.28</u>  | 49.99 $\pm$ 1.44                   | 49.75 $\pm$ 4.01                   | 49.48 $\pm$ 0.91                    | 51.11 $\pm$ 3.62  | 49.62 $\pm$ 0.72                   | 51.12 $\pm$ 3.30  | 48.62 $\pm$ 0.52                   | 49.35 $\pm$ 2.85  | 50.42 $\pm$ 4.51                    |
| Avg. Acc.              | 47.79                               | 45.40                               | <u>47.88</u>                        | 44.21                              | 42.39                              | 45.16                               | 36.78             | 44.31                              | 37.71             | 46.42                              | 35.61             | <b>48.31</b>                        |
| Avg. Rank              | <u>5.11</u>                         | 5.80                                | 5.19                                | 6.45                               | 6.92                               | 6.04                                | 9.11              | 6.17                               | 8.51              | 5.30                               | 8.82              | <b>4.58</b>                         |
| #Best                  | 25                                  | 16                                  | 14                                  | 7                                  | 5                                  | 11                                  | 2                 | 11                                 | 2                 | 15                                 | 4                 | <b>26</b>                           |

Table 4: Per-dataset 2-shot classification results on 128 shared UCR datasets. Each value is reported as mean  $\pm$  standard deviation over 5 runs.

| Dataset                      | ResNet                              | TapNet                             | COSCO-ResNet                        | DLinear                             | PatchTST                            | TimesNet                            | Autoformer                         | Crossformer                        | FEDformer         | Informer                           | GPT4TS                             | DuoTSP                              |
|------------------------------|-------------------------------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|------------------------------------|-------------------------------------|
| ACSF1                        | <b>61.60 <math>\pm</math> 7.86</b>  | 47.20 $\pm$ 6.30                   | 59.20 $\pm$ 5.26                    | 25.80 $\pm$ 4.38                    | 42.00 $\pm$ 3.16                    | 40.33 $\pm$ 2.31                    | 30.67 $\pm$ 2.52                   | 44.00 $\pm$ 6.24                   | 35.80 $\pm$ 3.19  | 40.67 $\pm$ 5.69                   | 24.80 $\pm$ 4.66                   | 54.33 $\pm$ 4.16                    |
| Adiac                        | 35.45 $\pm$ 2.60                    | 14.27 $\pm$ 3.74                   | 35.45 $\pm$ 2.60                    | 18.87 $\pm$ 2.86                    | 17.80 $\pm$ 5.08                    | 27.79 $\pm$ 2.14                    | 44.67 $\pm$ 0.39                   | 16.62 $\pm$ 1.77                   | 55.24 $\pm$ 4.67  | 32.91 $\pm$ 3.49                   | 3.48 $\pm$ 1.27                    | <b>56.01 <math>\pm</math> 1.25</b>  |
| AllGestureWiimoteX           | <b>32.43 <math>\pm</math> 2.94</b>  | 27.69 $\pm$ 3.48                   | 32.09 $\pm$ 3.58                    | 19.46 $\pm$ 2.47                    | 23.77 $\pm$ 3.63                    | 24.43 $\pm$ 4.71                    | 11.86 $\pm$ 1.00                   | 22.67 $\pm$ 2.29                   | 14.09 $\pm$ 1.28  | 21.10 $\pm$ 5.46                   | 23.77 $\pm$ 3.63                   | 26.51 $\pm$ 4.36                    |
| AllGestureWiimoteY           | <b>35.11 <math>\pm</math> 5.43</b>  | 28.57 $\pm$ 4.31                   | 28.91 $\pm$ 5.81                    | 20.54 $\pm$ 2.69                    | 33.31 $\pm$ 2.65                    | 24.33 $\pm$ 2.50                    | 12.00 $\pm$ 1.38                   | 21.90 $\pm$ 1.33                   | 16.14 $\pm$ 1.67  | 22.57 $\pm$ 1.00                   | 33.31 $\pm$ 2.65                   | 29.43 $\pm$ 1.69                    |
| AllGestureWiimoteZ           | <b>29.80 <math>\pm</math> 2.53</b>  | 26.94 $\pm$ 2.84                   | 29.31 $\pm$ 4.56                    | 16.77 $\pm$ 2.02                    | 22.97 $\pm$ 5.35                    | 19.86 $\pm$ 4.12                    | 12.76 $\pm$ 0.36                   | 19.57 $\pm$ 2.48                   | 14.97 $\pm$ 0.98  | 20.29 $\pm$ 3.48                   | 22.97 $\pm$ 5.35                   | 27.91 $\pm$ 5.21                    |
| ArrowHead                    | 50.17 $\pm$ 5.13                    | 44.80 $\pm$ 3.82                   | 45.14 $\pm$ 5.25                    | <b>57.03 <math>\pm</math> 4.92</b>  | 49.14 $\pm$ 13.02                   | 53.14 $\pm$ 6.02                    | 34.10 $\pm$ 6.27                   | 50.67 $\pm$ 2.01                   | 42.29 $\pm$ 4.92  | 55.43 $\pm$ 6.44                   | 43.77 $\pm$ 8.76                   | <b>57.03 <math>\pm</math> 2.81</b>  |
| BME                          | 48.40 $\pm$ 4.91                    | 55.33 $\pm$ 7.02                   | 54.40 $\pm$ 8.12                    | 50.80 $\pm$ 5.11                    | 67.33 $\pm$ 7.90                    | 48.89 $\pm$ 2.14                    | 64.00 $\pm$ 5.21                   | 58.89 $\pm$ 3.91                   | 67.33 $\pm$ 7.44  | 54.89 $\pm$ 9.20                   | 50.80 $\pm$ 8.05                   | <b>79.47 <math>\pm</math> 11.40</b> |
| Beef                         | 42.00 $\pm$ 2.98                    | 36.67 $\pm$ 8.50                   | 36.70 $\pm$ 9.60                    | 49.33 $\pm$ 7.23                    | 39.33 $\pm$ 6.41                    | 48.89 $\pm$ 5.09                    | 40.00 $\pm$ 3.33                   | 34.44 $\pm$ 5.09                   | 42.67 $\pm$ 4.94  | <b>51.11 <math>\pm</math> 7.70</b> | 32.67 $\pm$ 7.23                   | 43.33 $\pm$ 5.27                    |
| BeetleFly                    | 50.00 $\pm$ 9.35                    | 67.00 $\pm$ 18.23                  | 55.00 $\pm$ 12.75                   | 52.00 $\pm$ 11.51                   | 57.00 $\pm$ 16.43                   | 45.00 $\pm$ 10.00                   | 48.33 $\pm$ 12.58                  | 38.33 $\pm$ 7.64                   | 45.00 $\pm$ 9.35  | 56.67 $\pm$ 7.64                   | 57.00 $\pm$ 4.47                   | <b>75.00 <math>\pm</math> 18.03</b> |
| BirdChicken                  | 71.00 $\pm$ 6.52                    | <b>72.00 <math>\pm</math> 4.47</b> | <b>76.00 <math>\pm</math> 7.42</b>  | 52.00 $\pm$ 14.40                   | 65.00 $\pm$ 10.61                   | 45.00 $\pm$ 8.66                    | 51.67 $\pm$ 7.64                   | 46.67 $\pm$ 5.77                   | 53.00 $\pm$ 2.74  | 50.00 $\pm$ 13.23                  | 49.00 $\pm$ 2.24                   | 69.00 $\pm$ 22.19                   |
| CBF                          | 85.76 $\pm$ 5.83                    | <b>90.62 <math>\pm</math> 7.16</b> | 82.18 $\pm$ 10.67                   | 64.78 $\pm$ 6.67                    | 82.69 $\pm$ 10.89                   | 68.07 $\pm$ 6.00                    | 34.89 $\pm$ 2.41                   | 73.78 $\pm$ 6.88                   | 36.78 $\pm$ 1.75  | 70.00 $\pm$ 6.65                   | 42.84 $\pm$ 7.09                   | 85.91 $\pm$ 4.23                    |
| Car                          | 42.00 $\pm$ 15.43                   | 41.67 $\pm$ 13.49                  | 52.67 $\pm$ 7.87                    | 55.33 $\pm$ 10.63                   | 43.00 $\pm$ 11.45                   | 54.44 $\pm$ 6.74                    | 32.78 $\pm$ 0.96                   | 52.78 $\pm$ 10.84                  | 37.00 $\pm$ 7.76  | 55.56 $\pm$ 4.19                   | 31.00 $\pm$ 10.71                  | <b>63.33 <math>\pm</math> 7.45</b>  |
| Chinatown                    | 95.74 $\pm$ 1.26                    | <b>96.73 <math>\pm</math> 1.19</b> | <b>96.73 <math>\pm</math> 0.81</b>  | 54.52 $\pm$ 24.75                   | 91.08 $\pm$ 5.39                    | 85.33 $\pm$ 8.50                    | 86.10 $\pm$ 2.71                   | 84.55 $\pm$ 3.36                   | 80.70 $\pm$ 5.41  | 82.80 $\pm$ 4.37                   | 45.01 $\pm$ 24.65                  | 91.84 $\pm$ 4.85                    |
| ChlorineConcentration        | 34.61 $\pm$ 4.95                    | 35.24 $\pm$ 5.79                   | 35.19 $\pm$ 5.79                    | 34.54 $\pm$ 2.93                    | 34.28 $\pm$ 2.22                    | 31.22 $\pm$ 6.21                    | 33.24 $\pm$ 2.95                   | 33.84 $\pm$ 2.69                   | 33.91 $\pm$ 2.07  | 33.69 $\pm$ 2.59                   | <b>39.08 <math>\pm</math> 9.47</b> | 32.47 $\pm$ 3.03                    |
| CinCECGTorso                 | 46.52 $\pm$ 12.75                   | 48.35 $\pm$ 9.58                   | 51.06 $\pm$ 10.06                   | 39.77 $\pm$ 4.46                    | <b>54.91 <math>\pm</math> 12.73</b> | 48.14 $\pm$ 2.63                    | 36.18 $\pm$ 3.15                   | 53.94 $\pm$ 2.51                   | 50.30 $\pm$ 12.26 | 48.89 $\pm$ 2.81                   | 49.00 $\pm$ 5.40                   | <b>62.95 <math>\pm</math> 6.66</b>  |
| Coffee                       | <b>94.29 <math>\pm</math> 7.41</b>  | 88.57 $\pm$ 12.73                  | <b>94.29 <math>\pm</math> 3.19</b>  | 92.14 $\pm$ 3.91                    | 90.71 $\pm$ 10.89                   | 73.81 $\pm$ 8.99                    | 63.10 $\pm$ 7.43                   | 85.71 $\pm$ 3.57                   | 84.29 $\pm$ 10.29 | 77.38 $\pm$ 4.12                   | 65.71 $\pm$ 15.07                  | 88.57 $\pm$ 4.66                    |
| Computers                    | 52.24 $\pm$ 10.15                   | 54.40 $\pm$ 10.81                  | 52.72 $\pm$ 12.14                   | 48.08 $\pm$ 7.70                    | 53.28 $\pm$ 11.25                   | 48.00 $\pm$ 10.51                   | 48.67 $\pm$ 4.60                   | 48.00 $\pm$ 11.45                  | 50.40 $\pm$ 1.72  | 46.67 $\pm$ 11.22                  | 50.16 $\pm$ 7.53                   | <b>57.73 <math>\pm</math> 17.44</b> |
| CricketerX                   | <b>36.51 <math>\pm</math> 3.27</b>  | 34.10 $\pm$ 3.98                   | 33.03 $\pm$ 3.86                    | 20.62 $\pm$ 2.07                    | 30.56 $\pm$ 3.87                    | 27.35 $\pm$ 2.06                    | 8.12 $\pm$ 1.41                    | 26.15 $\pm$ 1.85                   | 12.15 $\pm$ 1.24  | 27.44 $\pm$ 2.96                   | 11.54 $\pm$ 1.86                   | 35.98 $\pm$ 3.13                    |
| CricketY                     | <b>37.54 <math>\pm</math> 5.48</b>  | 30.62 $\pm$ 2.69                   | 33.28 $\pm$ 3.86                    | 22.56 $\pm$ 4.44                    | 24.21 $\pm$ 2.69                    | 26.50 $\pm$ 7.27                    | 9.06 $\pm$ 0.39                    | 25.90 $\pm$ 6.29                   | 12.67 $\pm$ 0.94  | 27.52 $\pm$ 6.80                   | 10.00 $\pm$ 3.24                   | 30.43 $\pm$ 5.23                    |
| CricketZ                     | <b>40.41 <math>\pm</math> 4.15</b>  | <b>35.49 <math>\pm</math> 2.72</b> | 33.69 $\pm$ 5.21                    | 20.36 $\pm$ 1.13                    | 24.92 $\pm$ 3.09                    | 26.75 $\pm$ 2.72                    | 8.12 $\pm$ 1.50                    | 24.44 $\pm$ 0.78                   | 12.97 $\pm$ 2.47  | 26.50 $\pm$ 2.74                   | 11.18 $\pm$ 2.42                   | 33.33 $\pm$ 3.02                    |
| Crop                         | 31.31 $\pm$ 3.87                    | 37.82 $\pm$ 2.09                   | 31.31 $\pm$ 3.87                    | 14.67 $\pm$ 0.98                    | 24.66 $\pm$ 2.54                    | 36.33 $\pm$ 2.63                    | 35.06 $\pm$ 3.15                   | 37.89 $\pm$ 1.28                   | 35.83 $\pm$ 2.39  | 38.66 $\pm$ 2.99                   | 4.63 $\pm$ 0.75                    | <b>43.94 <math>\pm</math> 2.11</b>  |
| DiatomSizeReduction          | 78.43 $\pm$ 9.98                    | 84.44 $\pm$ 4.38                   | 89.74 $\pm$ 8.40                    | 83.73 $\pm$ 5.41                    | 63.01 $\pm$ 13.25                   | 84.42 $\pm$ 6.89                    | 59.91 $\pm$ 2.78                   | 84.97 $\pm$ 5.38                   | 59.28 $\pm$ 18.44 | 82.35 $\pm$ 7.47                   | 35.10 $\pm$ 12.79                  | <b>90.39 <math>\pm</math> 5.62</b>  |
| DistalPhalanxOutlineAgeGroup | 66.91 $\pm$ 6.08                    | <b>67.77 <math>\pm</math> 3.31</b> | 65.76 $\pm$ 6.53                    | 55.54 $\pm$ 12.17                   | 58.42 $\pm$ 11.15                   | <b>57.07 <math>\pm</math> 19.57</b> | 50.84 $\pm$ 14.56                  | 56.12 $\pm$ 13.32                  | 56.26 $\pm$ 12.92 | 57.07 $\pm$ 14.47                  | 38.99 $\pm$ 8.80                   | <b>67.77 <math>\pm</math> 3.39</b>  |
| DistalPhalanxOutlineCorrect  | 52.32 $\pm$ 12.15                   | 51.59 $\pm$ 11.49                  | 52.54 $\pm$ 12.44                   | 49.57 $\pm$ 11.83                   | 51.45 $\pm$ 15.36                   | <b>57.49 <math>\pm</math> 13.09</b> | 56.76 $\pm$ 10.57                  | 55.80 $\pm$ 11.93                  | 51.38 $\pm$ 12.34 | 56.64 $\pm$ 12.97                  | 50.65 $\pm$ 9.31                   | 55.80 $\pm$ 14.95                   |
| DistalPhalanxTW              | 49.50 $\pm$ 8.83                    | <b>55.54 <math>\pm</math> 4.21</b> | 54.10 $\pm$ 4.92                    | 39.28 $\pm$ 8.61                    | 53.96 $\pm$ 7.56                    | 41.97 $\pm$ 11.56                   | 42.45 $\pm$ 10.74                  | 39.81 $\pm$ 18.49                  | 47.05 $\pm$ 8.79  | 45.80 $\pm$ 16.17                  | 22.45 $\pm$ 8.85                   | 51.37 $\pm$ 8.97                    |
| DodgerLoopDay                | 28.75 $\pm$ 2.34                    | 23.75 $\pm$ 2.50                   | 28.00 $\pm$ 5.70                    | <b>44.00 <math>\pm</math> 4.09</b>  | 38.50 $\pm$ 7.42                    | 39.58 $\pm$ 4.02                    | 23.75 $\pm$ 2.50                   | 42.50 $\pm$ 7.60                   | 19.75 $\pm$ 4.09  | 42.50 $\pm$ 3.31                   | 38.50 $\pm$ 7.42                   | 35.50 $\pm$ 8.78                    |
| DodgerLoopGame               | 54.93 $\pm$ 6.45                    | 63.04 $\pm$ 8.02                   | 58.41 $\pm$ 11.09                   | 53.62 $\pm$ 8.51                    | 65.65 $\pm$ 18.08                   | 67.87 $\pm$ 7.26                    | 48.55 $\pm$ 4.53                   | <b>71.50 <math>\pm</math> 0.84</b> | 50.14 $\pm$ 3.81  | 68.36 $\pm$ 3.42                   | 65.65 $\pm$ 18.08                  | 61.01 $\pm$ 13.03                   |
| DodgerLoopWeekend            | 92.75 $\pm$ 2.40                    | 87.68 $\pm$ 5.59                   | 92.03 $\pm$ 3.69                    | 94.64 $\pm$ 2.74                    | 96.09 $\pm$ 1.41                    | 96.38 $\pm$ 1.26                    | 64.98 $\pm$ 4.71                   | <b>97.34 <math>\pm</math> 0.42</b> | 65.22 $\pm$ 8.45  | 96.14 $\pm$ 0.84                   | 96.09 $\pm$ 1.41                   | 84.64 $\pm$ 2.73                    |
| ECG200                       | 69.40 $\pm$ 7.77                    | 75.20 $\pm$ 7.05                   | 71.80 $\pm$ 5.54                    | 76.80 $\pm$ 3.56                    | 71.80 $\pm$ 4.38                    | <b>78.33 <math>\pm</math> 1.53</b>  | 74.33 $\pm$ 5.69                   | 75.67 $\pm$ 3.21                   | 73.60 $\pm$ 3.65  | 75.33 $\pm$ 0.58                   | 52.20 $\pm$ 18.20                  | 73.00 $\pm$ 3.46                    |
| ECG5000                      | 61.90 $\pm$ 13.09                   | 69.05 $\pm$ 11.38                  | 61.86 $\pm$ 10.65                   | <b>74.23 <math>\pm</math> 10.96</b> | 58.61 $\pm$ 14.82                   | 72.42 $\pm$ 4.03                    | 65.61 $\pm$ 11.76                  | 69.40 $\pm$ 4.84                   | 67.93 $\pm$ 11.62 | 66.36 $\pm$ 9.81                   | 30.14 $\pm$ 19.40                  | 67.68 $\pm$ 9.76                    |
| ECGFiveDays                  | 68.78 $\pm$ 9.46                    | 56.79 $\pm$ 4.91                   | 71.73 $\pm$ 8.76                    | 57.12 $\pm$ 2.23                    | 67.29 $\pm$ 3.58                    | 66.55 $\pm$ 6.20                    | 74.64 $\pm$ 13.99                  | <b>75.18 <math>\pm</math> 2.37</b> | 68.76 $\pm$ 10.24 | 68.52 $\pm$ 6.54                   | 57.79 $\pm$ 7.24                   | 69.12 $\pm$ 9.82                    |
| EOGHorizontalSignal          | 20.66 $\pm$ 1.70                    | 23.43 $\pm$ 5.48                   | 21.38 $\pm$ 4.60                    | 25.86 $\pm$ 4.54                    | 28.07 $\pm$ 2.90                    | 30.66 $\pm$ 2.91                    | 12.06 $\pm$ 1.96                   | 31.68 $\pm$ 2.61                   | 9.56 $\pm$ 1.50   | 29.01 $\pm$ 2.49                   | 29.83 $\pm$ 3.23                   | <b>37.35 <math>\pm</math> 4.07</b>  |
| EOGVerticalSignal            | 13.37 $\pm$ 1.32                    | 16.57 $\pm$ 3.25                   | 14.86 $\pm$ 2.77                    | 26.80 $\pm$ 1.98                    | 29.56 $\pm$ 4.11                    | 26.98 $\pm$ 4.28                    | 11.88 $\pm$ 1.54                   | 26.34 $\pm$ 4.16                   | 10.83 $\pm$ 1.45  | 26.61 $\pm$ 6.09                   | 20.88 $\pm$ 2.94                   | <b>31.82 <math>\pm</math> 3.47</b>  |
| Earthquakes                  | 52.95 $\pm$ 7.17                    | 57.84 $\pm$ 11.11                  | 49.35 $\pm$ 10.38                   | 53.24 $\pm$ 4.93                    | <b>63.88 <math>\pm</math> 6.42</b>  | 63.31 $\pm$ 10.45                   | 49.16 $\pm$ 1.10                   | 54.92 $\pm$ 17.17                  | 56.55 $\pm$ 8.34  | 60.43 $\pm$ 21.88                  | 57.12 $\pm$ 15.34                  | 55.25 $\pm$ 14.33                   |
| ElectricDevices              | 37.63 $\pm$ 9.44                    | 41.80 $\pm$ 2.08                   | 34.70 $\pm$ 5.95                    | 21.27 $\pm$ 5.73                    | 37.69 $\pm$ 10.00                   | 28.63 $\pm$ 3.21                    | 22.56 $\pm$ 2.12                   | 20.63 $\pm$ 2.87                   | 22.76 $\pm$ 4.51  | 29.88 $\pm$ 3.14                   | 14.36 $\pm$ 4.97                   | <b>47.14 <math>\pm</math> 3.99</b>  |
| EthanolLevel                 | 25.32 $\pm$ 0.64                    | <b>25.52 <math>\pm</math> 1.56</b> | 25.12 $\pm$ 2.09                    | 25.76 $\pm$ 1.71                    | 26.96 $\pm$ 0.82                    | 26.87 $\pm$ 2.61                    | <b>29.20 <math>\pm</math> 1.91</b> | 25.67 $\pm$ 3.18                   | 25.24 $\pm$ 0.98  | 26.47 $\pm$ 2.08                   | 25.12 $\pm$ 0.81                   | <b>28.00 <math>\pm</math> 3.11</b>  |
| FaceAll                      | <b>55.98 <math>\pm</math> 6.19</b>  | 45.22 $\pm$ 8.18                   | <b>55.98 <math>\pm</math> 6.19</b>  | 40.27 $\pm$ 5.76                    | 24.12 $\pm$ 4.85                    | 38.56 $\pm$ 3.11                    | 38.84 $\pm$ 3.81                   | 36.71 $\pm$ 2.55                   | 38.89 $\pm$ 3.64  | 41.40 $\pm$ 3.79                   | 11.35 $\pm$ 5.01                   | 39.09 $\pm$ 1.28                    |
| FaceFour                     | 77.73 $\pm$ 4.99                    | 52.05 $\pm$ 10.37                  | 81.82 $\pm$ 7.14                    | 74.09 $\pm$ 5.98                    | 59.32 $\pm$ 5.76                    | 79.55 $\pm$ 9.02                    | 57.58 $\pm$ 7.31                   | 80.68 $\pm$ 3.41                   | 62.50 $\pm$ 7.14  | <b>84.85 <math>\pm</math> 2.37</b> | 40.45 $\pm$ 14.16                  | 68.55 $\pm$ 4.91                    |
| FacesUCR                     | <b>67.20 <math>\pm</math> 6.42</b>  | 51.90 $\pm$ 6.85                   | <b>62.04 <math>\pm</math> 8.47</b>  | 41.67 $\pm$ 3.36                    | 25.42 $\pm$ 6.93                    | 44.55 $\pm$ 2.52                    | 45.95 $\pm$ 4.27                   | 49.07 $\pm$ 5.45                   | 48.56 $\pm$ 3.79  | 49.27 $\pm$ 6.47                   | 10.96 $\pm$ 5.26                   | 55.08 $\pm$ 2.32                    |
| FiftyWords                   | 19.03 $\pm$ 1.51                    | 13.49 $\pm$ 1.95                   | 19.03 $\pm$ 1.51                    | 40.09 $\pm$ 3.22                    | 28.35 $\pm$ 3.05                    | 44.69 $\pm$ 2.97                    | 5.05 $\pm$ 0.44                    | 46.74 $\pm$ 2.43                   | 16.53 $\pm$ 1.75  | 44.18 $\pm$ 3.70                   | 15.16 $\pm$ 3.75                   | <b>47.30 <math>\pm</math> 2.89</b>  |
| Fish                         | <b>50.97 <math>\pm</math> 10.77</b> | 26.97 $\pm$ 9.88                   | <b>50.51 <math>\pm</math> 16.65</b> | 50.17 $\pm$ 6.36                    | 31.54 $\pm$ 5.18                    | 45.90 $\pm$ 8.88                    | 20.00 $\pm$ 4.12                   | 39.05 $\pm$ 5.64                   | 22.97 $\pm$ 3.62  | 45.71 $\pm$ 10.47                  | 22.86 $\pm$ 2.83                   | 49.91 $\pm$ 10.25                   |
| FordA                        | <b>64.98 <math>\pm</math> 8.59</b>  | 62.27 $\pm$ 9.82                   | <b>62.82 <math>\pm</math> 9.33</b>  | 49.92 $\pm$ 1.84                    | 60.48 $\pm$ 7.29                    | 50.30 $\pm$ 1.31                    | 50.66 $\pm$ 0.90                   | 50.15 $\pm$ 0.79                   | 50.27 $\pm$ 1.01  | 50.58 $\pm$ 1.43                   | 50.76 $\pm$ 2.11                   | 53.35 $\pm$ 2.84                    |
| FordB                        | <b>61.19 <math>\pm</math> 3.67</b>  | 58.10 $\pm$ 1.40                   | 58.67 $\pm$ 2.63                    | 49.09 $\pm$ 1.35                    | 56.89 $\pm$ 2.99                    | 49.84 $\pm$ 0.6                     |                                    |                                    |                   |                                    |                                    |                                     |



Table 4: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

| Dataset                        | ResNet               | TapNet               | COSCO-ResNet         | DLinear             | PatchTST            | TimesNet             | Autoformer           | Crossformer          | FEDformer            | Informer            | GPT4TS               | DuoTSP               |
|--------------------------------|----------------------|----------------------|----------------------|---------------------|---------------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|----------------------|
| GesturePebbleZ1                | 62.44 ± 5.24         | 41.40 ± 5.40         | <b>64.88 ± 2.42</b>  | 53.26 ± 4.18        | 29.53 ± 10.46       | 58.91 ± 3.41         | 16.86 ± 3.54         | 57.17 ± 0.89         | 24.19 ± 4.00         | 61.82 ± 3.41        | 29.53 ± 10.46        | 37.91 ± 6.35         |
| GesturePebbleZ2                | <b>66.58 ± 5.17</b>  | 44.05 ± 14.70        | 61.70 ± 6.13         | 51.90 ± 9.19        | 35.32 ± 9.94        | 55.06 ± 7.46         | 15.82 ± 0.63         | 46.84 ± 8.79         | 20.76 ± 3.15         | 56.75 ± 4.12        | 35.32 ± 9.94         | 41.39 ± 4.24         |
| GunPoint                       | 67.07 ± 8.15         | 65.20 ± 5.95         | 66.00 ± 5.89         | 60.13 ± 7.74        | 65.47 ± 7.68        | 55.33 ± 6.57         | 64.44 ± 5.18         | 58.44 ± 9.46         | 68.13 ± 5.80         | 57.11 ± 7.95        | 54.00 ± 6.53         | <b>79.33 ± 16.25</b> |
| GunPointAgeSpan                | 65.38 ± 2.37         | 63.73 ± 4.38         | <b>68.61 ± 1.37</b>  | 67.41 ± 13.19       | 61.08 ± 13.41       | 61.08 ± 8.21         | 52.22 ± 8.74         | 55.49 ± 10.60        | 52.41 ± 8.13         | 58.76 ± 12.22       | 47.41 ± 14.13        | 64.49 ± 7.90         |
| GunPointMaleVersusFemale       | 79.30 ± 6.81         | 84.62 ± 8.74         | 81.65 ± 8.32         | 57.15 ± 7.06        | 65.95 ± 9.57        | 60.65 ± 4.86         | 50.11 ± 5.57         | 60.76 ± 10.27        | 52.59 ± 5.35         | 65.82 ± 3.12        | 65.00 ± 5.79         | <b>87.66 ± 5.66</b>  |
| GunPointOldVersusYoung         | <b>91.94 ± 11.58</b> | 89.97 ± 13.74        | 91.37 ± 12.08        | 72.89 ± 4.20        | 73.14 ± 5.24        | 73.54 ± 7.07         | 56.51 ± 2.29         | 72.49 ± 6.74         | 59.30 ± 3.02         | 75.87 ± 9.07        | 78.03 ± 17.26        | 70.92 ± 5.05         |
| Ham                            | 45.71 ± 9.69         | 50.86 ± 8.18         | 48.00 ± 10.27        | 54.86 ± 11.40       | 53.33 ± 7.28        | 54.92 ± 1.98         | <b>58.73 ± 5.25</b>  | 50.48 ± 13.83        | 55.24 ± 4.62         | 57.78 ± 9.77        | 52.57 ± 10.93        | 50.86 ± 8.07         |
| HandOutlines                   | 57.19 ± 12.29        | 63.30 ± 2.05         | 60.65 ± 3.52         | 69.19 ± 14.14       | 63.57 ± 17.16       | 67.21 ± 16.82        | 52.16 ± 12.39        | 68.11 ± 20.52        | 41.57 ± 12.57        | 67.48 ± 15.69       | <b>69.89 ± 13.52</b> | 64.59 ± 10.14        |
| Haptics                        | 29.16 ± 3.65         | 24.35 ± 4.42         | 25.32 ± 2.75         | 28.31 ± 7.72        | 28.25 ± 7.08        | <b>29.33 ± 7.26</b>  | 23.81 ± 0.68         | 29.22 ± 2.66         | 20.97 ± 3.07         | 29.00 ± 7.31        | 26.95 ± 8.77         | 26.62 ± 4.64         |
| Herring                        | 52.81 ± 11.18        | <b>56.88 ± 7.13</b>  | 52.81 ± 11.23        | 51.25 ± 5.89        | 52.19 ± 12.13       | 47.92 ± 6.51         | 46.35 ± 3.61         | 45.83 ± 3.25         | 50.00 ± 4.69         | 48.96 ± 4.51        | 45.00 ± 9.84         | <b>55.00 ± 4.34</b>  |
| HouseTwenty                    | <b>76.30 ± 9.04</b>  | 69.24 ± 11.83        | 67.39 ± 13.31        | 60.00 ± 7.10        | 63.03 ± 13.56       | 68.91 ± 7.47         | 47.34 ± 6.31         | 59.94 ± 1.75         | 52.10 ± 8.21         | 71.15 ± 7.81        | 60.34 ± 11.20        | <b>71.93 ± 10.30</b> |
| InlineSkate                    | 18.58 ± 2.99         | 19.56 ± 1.81         | <b>19.93 ± 2.98</b>  | 18.29 ± 2.68        | 17.05 ± 3.64        | 16.85 ± 1.00         | 15.76 ± 0.93         | 15.82 ± 1.62         | 16.29 ± 1.31         | 17.03 ± 0.46        | 16.36 ± 2.46         | 18.98 ± 2.54         |
| InsectEPGRegularTrain          | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | 51.97 ± 7.58        | 58.39 ± 6.76        | 58.37 ± 4.44         | 37.08 ± 10.70        | 54.89 ± 2.35         | 37.99 ± 13.89        | 59.71 ± 4.91        | 81.69 ± 17.66        | 68.76 ± 9.60         |
| InsectEPGSmallTrain            | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | <b>100.00 ± 0.00</b> | 51.33 ± 4.40        | 58.47 ± 6.25        | 57.56 ± 4.71         | 41.37 ± 7.24         | 58.23 ± 4.08         | 42.65 ± 10.05        | 58.10 ± 4.04        | <b>80.40 ± 17.17</b> | 67.79 ± 2.45         |
| InsectWingbeatSound            | 20.38 ± 3.08         | 16.62 ± 4.40         | 18.70 ± 3.47         | 47.87 ± 2.61        | 33.18 ± 5.98        | <b>50.44 ± 4.17</b>  | 20.69 ± 3.83         | 49.41 ± 1.90         | 27.56 ± 2.55         | <b>51.03 ± 4.83</b> | 16.68 ± 7.51         | 31.49 ± 3.72         |
| ItalyPowerDemand               | 68.88 ± 12.96        | 75.63 ± 13.47        | 73.61 ± 12.06        | 50.03 ± 0.16        | 79.18 ± 15.47       | <b>54.49 ± 7.77</b>  | 75.87 ± 16.00        | 76.35 ± 22.01        | <b>83.15 ± 15.39</b> | 73.57 ± 21.06       | 50.07 ± 0.34         | 65.03 ± 13.70        |
| LargeKitchenAppliances         | 45.65 ± 8.51         | <b>54.19 ± 10.41</b> | 50.93 ± 13.42        | 33.60 ± 7.89        | 44.85 ± 7.66        | 37.96 ± 12.60        | 32.00 ± 3.23         | 34.40 ± 7.77         | 32.96 ± 7.01         | 36.62 ± 11.05       | 32.00 ± 6.67         | 42.56 ± 5.00         |
| Lightning2                     | 57.05 ± 8.63         | 53.77 ± 7.90         | <b>59.34 ± 2.07</b>  | 54.10 ± 10.50       | 51.80 ± 7.11        | 57.38 ± 4.34         | 49.73 ± 9.88         | 51.91 ± 4.13         | 52.13 ± 11.90        | 55.19 ± 6.21        | 50.16 ± 8.80         | <b>63.28 ± 8.25</b>  |
| Lightning7                     | 45.48 ± 6.31         | 43.01 ± 5.94         | 42.74 ± 6.38         | 38.90 ± 8.91        | 47.95 ± 8.16        | <b>59.36 ± 11.07</b> | 19.63 ± 3.45         | 50.23 ± 12.43        | 27.40 ± 4.94         | 56.16 ± 10.34       | 20.82 ± 11.60        | 46.03 ± 6.17         |
| Mallat                         | 73.35 ± 15.31        | 47.53 ± 14.01        | 80.84 ± 4.13         | 86.79 ± 1.54        | 65.16 ± 19.15       | <b>91.78 ± 0.99</b>  | 63.31 ± 2.66         | 78.79 ± 4.89         | 70.90 ± 5.59         | <b>87.65 ± 2.43</b> | 49.21 ± 6.11         | 80.67 ± 4.06         |
| Meat                           | 54.33 ± 7.42         | 42.00 ± 7.76         | <b>73.00 ± 16.39</b> | <b>89.67 ± 7.85</b> | 52.00 ± 12.16       | 53.33 ± 6.01         | 43.33 ± 8.33         | 41.11 ± 7.52         | 51.33 ± 18.61        | 57.22 ± 7.88        | 36.33 ± 5.82         | 57.67 ± 17.30        |
| MedicalImages                  | 33.45 ± 8.28         | 25.74 ± 5.92         | <b>33.61 ± 5.62</b>  | 24.45 ± 6.71        | 27.45 ± 4.51        | 28.46 ± 3.05         | 27.85 ± 6.61         | 28.55 ± 2.80         | <b>34.53 ± 1.25</b>  | 30.00 ± 1.71        | 29.82 ± 16.29        | 31.63 ± 5.59         |
| MelbournePedestrian            | <b>63.62 ± 4.54</b>  | 57.50 ± 8.49         | 63.00 ± 2.07         | 14.05 ± 1.71        | 42.42 ± 4.22        | 40.29 ± 3.90         | 51.77 ± 5.29         | 50.24 ± 2.74         | 54.32 ± 5.40         | 46.37 ± 2.70        | 42.42 ± 4.22         | 49.78 ± 4.68         |
| MiddlePhalanxOutlineAgeGroup   | 47.01 ± 9.52         | 44.55 ± 12.64        | 45.45 ± 9.45         | <b>50.39 ± 7.58</b> | 44.29 ± 10.12       | 37.45 ± 10.11        | 46.75 ± 13.26        | 45.24 ± 16.76        | 48.57 ± 12.10        | 48.92 ± 14.26       | 22.86 ± 4.12         | 42.99 ± 12.95        |
| MiddlePhalanxOutlineCorrect    | 49.42 ± 7.12         | 51.13 ± 7.83         | 49.42 ± 8.07         | 53.75 ± 12.47       | 44.54 ± 7.47        | 52.92 ± 7.99         | 47.54 ± 10.52        | <b>54.07 ± 11.30</b> | 48.59 ± 8.66         | 53.26 ± 15.07       | 50.72 ± 11.13        | 48.87 ± 3.56         |
| MiddlePhalanxTW                | 37.92 ± 11.48        | 44.68 ± 3.99         | 42.73 ± 9.27         | 38.96 ± 14.59       | 39.22 ± 13.51       | 45.45 ± 5.95         | 43.07 ± 3.27         | 36.36 ± 13.61        | <b>46.36 ± 2.32</b>  | 45.24 ± 3.58        | 16.36 ± 7.24         | 44.81 ± 7.78         |
| MixedShapesRegularTrain        | 55.18 ± 4.21         | 33.76 ± 13.30        | 55.82 ± 8.68         | 58.38 ± 6.27        | 54.48 ± 12.34       | 62.60 ± 3.23         | 19.42 ± 0.39         | 63.00 ± 3.88         | 22.36 ± 6.85         | 63.64 ± 2.51        | 54.05 ± 4.44         | <b>67.98 ± 4.06</b>  |
| MixedShapesSmallTrain          | 45.73 ± 9.51         | 37.76 ± 12.10        | 51.77 ± 7.21         | 61.69 ± 3.51        | 54.28 ± 10.55       | 63.00 ± 2.46         | 19.05 ± 0.77         | 64.04 ± 1.32         | 24.01 ± 6.26         | 62.64 ± 3.39        | 57.53 ± 4.77         | <b>68.35 ± 3.26</b>  |
| MoteStrain                     | 81.95 ± 4.07         | 75.34 ± 9.74         | 78.95 ± 3.25         | 70.97 ± 10.57       | 75.59 ± 8.13        | 78.41 ± 9.87         | 68.53 ± 3.31         | <b>82.48 ± 1.96</b>  | 68.63 ± 10.07        | <b>83.65 ± 4.68</b> | 55.75 ± 11.79        | 58.56 ± 15.12        |
| NonInvasiveFetalECGThorax1     | 25.55 ± 2.02         | 14.44 ± 3.90         | 25.55 ± 2.02         | <b>55.96 ± 3.30</b> | 18.44 ± 9.38        | 50.77 ± 4.84         | 18.37 ± 2.16         | 51.65 ± 2.21         | 21.42 ± 2.01         | 53.89 ± 1.99        | 13.55 ± 0.51         | 50.58 ± 2.13         |
| NonInvasiveFetalECGThorax2     | 27.07 ± 5.56         | 22.30 ± 6.32         | 27.07 ± 5.56         | <b>62.09 ± 2.37</b> | 28.27 ± 8.71        | 55.69 ± 4.28         | 27.23 ± 2.03         | 56.05 ± 0.94         | 30.73 ± 3.23         | 58.63 ± 2.42        | 18.29 ± 1.75         | 59.72 ± 2.76         |
| OSULeaf                        | 55.70 ± 9.28         | 38.35 ± 6.31         | <b>56.03 ± 5.89</b>  | 25.79 ± 2.77        | 25.29 ± 5.19        | 29.89 ± 3.34         | 18.18 ± 1.43         | 28.10 ± 2.07         | 21.90 ± 3.33         | 28.51 ± 4.37        | 23.31 ± 3.85         | 36.28 ± 6.97         |
| OliveOil                       | 64.67 ± 10.70        | 32.67 ± 5.48         | 71.33 ± 10.70        | <b>80.67 ± 3.65</b> | 17.33 ± 10.90       | 32.22 ± 11.71        | 38.89 ± 7.70         | 25.56 ± 12.62        | 24.00 ± 10.90        | 58.89 ± 22.69       | 22.00 ± 13.04        | 23.33 ± 9.72         |
| PLAID                          | 19.29 ± 3.99         | 16.65 ± 2.54         | 20.89 ± 4.30         | 22.98 ± 3.80        | 35.53 ± 7.89        | 27.56 ± 4.82         | 17.44 ± 2.97         | 25.88 ± 4.40         | 19.81 ± 3.24         | 28.68 ± 3.91        | 35.53 ± 7.89         | <b>40.19 ± 5.06</b>  |
| PhalangesOutlinesCorrect       | 52.91 ± 9.24         | 53.03 ± 10.79        | 52.61 ± 9.16         | 56.64 ± 11.15       | 54.13 ± 11.04       | <b>60.64 ± 3.81</b>  | 58.70 ± 2.16         | 60.37 ± 4.20         | 52.26 ± 8.88         | 60.06 ± 3.62        | 49.04 ± 14.25        | 50.84 ± 2.93         |
| Phoneme                        | <b>12.22 ± 1.88</b>  | 10.75 ± 2.12         | 11.22 ± 1.88         | 5.41 ± 0.40         | 7.31 ± 0.69         | 7.77 ± 0.84          | 4.13 ± 1.74          | 7.00 ± 0.45          | 6.19 ± 2.34          | 5.29 ± 0.61         | 5.20 ± 0.75          | 10.02 ± 0.59         |
| PickupGestureWiimoteZ          | 43.20 ± 11.37        | 53.20 ± 9.55         | <b>62.40 ± 1.67</b>  | 43.20 ± 6.72        | 40.40 ± 8.29        | 52.67 ± 10.07        | 21.33 ± 1.15         | 53.33 ± 4.16         | 27.60 ± 3.58         | 51.33 ± 6.43        | 40.40 ± 8.29         | 58.80 ± 3.03         |
| PigAirwayPressure              | 28.37 ± 3.10         | 13.08 ± 1.90         | <b>49.52 ± 0.76</b>  | 5.00 ± 0.26         | 3.65 ± 0.87         | 4.49 ± 0.28          | 3.21 ± 1.21          | 7.05 ± 0.73          | 4.62 ± 1.05          | 6.09 ± 0.73         | 5.10 ± 0.80          | 9.81 ± 0.43          |
| PigArtPressure                 | 50.58 ± 2.62         | 23.65 ± 3.56         | <b>69.42 ± 3.86</b>  | 7.88 ± 0.26         | 5.29 ± 1.63         | 10.10 ± 0.48         | 8.01 ± 1.39          | 9.78 ± 1.11          | 7.12 ± 1.24          | 8.33 ± 0.56         | 19.13 ± 1.04         | 30.67 ± 1.38         |
| PigCVP                         | 31.54 ± 0.94         | 18.08 ± 2.27         | <b>61.92 ± 1.78</b>  | 7.02 ± 0.43         | 3.94 ± 1.10         | 6.73 ± 0.00          | 4.01 ± 0.56          | 7.85 ± 1.00          | 4.13 ± 0.80          | 5.29 ± 0.83         | 10.10 ± 1.40         | 19.33 ± 0.99         |
| Plane                          | <b>100.00 ± 0.00</b> | 94.29 ± 3.75         | 99.05 ± 1.17         | 86.10 ± 6.69        | 91.62 ± 3.39        | 86.98 ± 7.02         | 84.44 ± 7.02         | 81.90 ± 1.90         | 87.24 ± 4.98         | 86.98 ± 6.34        | 21.71 ± 8.89         | <b>99.24 ± 0.80</b>  |
| PowerCons                      | 82.11 ± 6.61         | <b>83.78 ± 3.63</b>  | 82.33 ± 7.91         | 72.11 ± 2.98        | 70.78 ± 14.65       | 69.81 ± 4.10         | 52.22 ± 1.47         | 70.93 ± 7.46         | 50.67 ± 4.38         | 70.00 ± 8.73        | 65.89 ± 16.73        | 66.78 ± 15.52        |
| ProximalPhalanxOutlineAgeGroup | <b>68.10 ± 13.07</b> | 64.29 ± 28.01        | 64.78 ± 22.07        | 58.44 ± 28.63       | 68.00 ± 16.32       | 53.17 ± 31.21        | 56.26 ± 31.37        | 56.91 ± 37.90        | 59.22 ± 23.54        | 51.38 ± 35.67       | 33.95 ± 22.63        | 66.24 ± 17.89        |
| ProximalPhalanxOutlineCorrect  | <b>65.43 ± 5.55</b>  | 60.07 ± 16.64        | 63.85 ± 6.55         | 63.37 ± 8.02        | 62.96 ± 6.50        | 54.75 ± 14.29        | 54.30 ± 16.96        | 60.94 ± 6.65         | 59.86 ± 17.09        | 55.21 ± 16.86       | 44.88 ± 18.17        | 59.93 ± 12.82        |
| ProximalPhalanxTW              | 50.83 ± 4.70         | 56.00 ± 9.92         | <b>67.12 ± 8.12</b>  | 49.76 ± 11.43       | 53.37 ± 7.43        | 58.70 ± 5.35         | 53.98 ± 16.76        | 42.28 ± 24.98        | 56.00 ± 8.06         | 48.29 ± 19.01       | 16.98 ± 12.37        | 57.76 ± 7.59         |
| RefrigerationDevices           | <b>41.39 ± 9.49</b>  | 40.37 ± 10.57        | 41.23 ± 9.97         | 33.92 ± 0.58        | 35.31 ± 3.72        | 36.71 ± 1.61         | 33.87 ± 4.03         | 35.64 ± 1.78         | 33.81 ± 1.50         | 33.60 ± 3.03        | 33.07 ± 2.39         | 33.39 ± 7.13         |
| Rock                           | 32.00 ± 5.48         | 38.40 ± 8.65         | 37.80 ± 13.31        | 67.60 ± 10.53       | 44.80 ± 18.25       | 58.67 ± 13.61        | <b>70.00 ± 11.14</b> | 66.67 ± 8.08         | 47.60 ± 12.28        | 60.00 ± 14.00       | 44.40 ± 14.24        | 64.00 ± 7.07         |
| ScreenType                     | 36.91 ± 4.95         | <b>40.21 ± 6.72</b>  | 38.40 ± 6.31         | 35.25 ± 3.07        | 36.75 ± 2.52        | 39.56 ± 4.47         | 33.78 ± 3.74         | 38.67 ± 2.63         | 33.71 ± 2.21         | 39.11 ± 4.89        | 33.81 ± 3.79         | 35.63 ± 4.40         |
| SemgHandGenderCh2              | 55.13 ± 5.29         | 58.23 ± 11.97        | 54.27 ± 6.20         | 53.27 ± 13.91       | <b>62.87 ± 8.65</b> | 49.22 ± 10.20        | 55.56 ± 1.55         | 62.44 ± 3.53         | 43.87 ± 10.45        | 56.22 ± 13.04       | 53.70 ± 4.90         | 59.73 ± 10.87        |
| SemgHandMovementCh2            | 33.51 ± 7.12         | 31.29 ± 5.18         | <b>36.00 ± 6.56</b>  | 29.78 ± 3.16        | 23.56 ± 5.26        | 24.44 ± 2.52         | 18.89 ± 0.59         | 24.44 ± 1.39         | 18.71 ± 1.13         | 24.07 ± 5.35        | 27.91 ± 3.59         | 31.16 ± 4.70         |

Continued on next page

Table 4: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

| Dataset                | ResNet                              | TapNet                             | COSCO-ResNet                       | DLinear                            | PatchTST          | TimesNet                           | Autoformer                         | Crossformer                        | FEDformer         | Informer                           | GPT4TS            | DuoTSP                              |
|------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|-------------------|-------------------------------------|
| SemgHandSubjectCh2     | 32.00 $\pm$ 3.88                    | 30.71 $\pm$ 4.13                   | 33.56 $\pm$ 5.04                   | <b>45.91 <math>\pm</math> 5.66</b> | 40.67 $\pm$ 4.39  | 29.56 $\pm$ 4.02                   | 23.56 $\pm$ 2.47                   | 32.52 $\pm$ 7.62                   | 24.09 $\pm$ 3.07  | 30.81 $\pm$ 0.84                   | 29.47 $\pm$ 4.28  | 43.20 $\pm$ 6.72                    |
| ShakeGestureWiimoteZ   | 76.00 $\pm$ 6.63                    | 76.80 $\pm$ 7.43                   | <b>81.60 <math>\pm</math> 0.89</b> | 47.20 $\pm$ 7.95                   | 58.40 $\pm$ 5.18  | 54.00 $\pm$ 6.00                   | 26.00 $\pm$ 4.00                   | 54.67 $\pm$ 4.62                   | 32.40 $\pm$ 2.61  | 56.00 $\pm$ 5.29                   | 58.40 $\pm$ 5.18  | <b>81.60 <math>\pm</math> 2.61</b>  |
| ShapeletSim            | 78.89 $\pm$ 14.16                   | <b>98.33 <math>\pm</math> 2.08</b> | 92.11 $\pm$ 5.26                   | 50.11 $\pm$ 2.87                   | 80.22 $\pm$ 16.23 | 47.59 $\pm$ 3.16                   | 50.74 $\pm$ 3.06                   | 48.15 $\pm$ 1.79                   | 49.22 $\pm$ 2.93  | 46.48 $\pm$ 3.06                   | 49.67 $\pm$ 1.50  | 67.44 $\pm$ 10.92                   |
| ShapesAll              | 36.80 $\pm$ 3.60                    | 28.17 $\pm$ 3.75                   | 36.80 $\pm$ 3.60                   | 50.73 $\pm$ 1.71                   | 32.23 $\pm$ 5.06  | 55.78 $\pm$ 1.50                   | 6.11 $\pm$ 0.63                    | 53.17 $\pm$ 1.00                   | 18.33 $\pm$ 5.95  | 53.33 $\pm$ 2.62                   | 27.63 $\pm$ 3.10  | <b>63.03 <math>\pm</math> 2.22</b>  |
| SmallKitchenAppliances | 48.85 $\pm$ 7.91                    | 50.40 $\pm$ 8.87                   | 50.19 $\pm$ 7.71                   | 41.49 $\pm$ 3.33                   | 47.15 $\pm$ 6.04  | 49.60 $\pm$ 6.54                   | 35.20 $\pm$ 2.93                   | 40.09 $\pm$ 2.14                   | 42.03 $\pm$ 4.49  | 44.09 $\pm$ 8.36                   | 34.61 $\pm$ 4.49  | <b>55.68 <math>\pm</math> 8.74</b>  |
| SmoothSubspace         | 78.93 $\pm$ 10.17                   | 77.87 $\pm$ 5.55                   | 80.13 $\pm$ 5.80                   | 33.33 $\pm$ 0.00                   | 51.20 $\pm$ 5.65  | 58.22 $\pm$ 4.73                   | 68.89 $\pm$ 7.31                   | 59.11 $\pm$ 4.23                   | 70.67 $\pm$ 4.88  | 81.33 $\pm$ 6.36                   | 48.27 $\pm$ 9.05  | <b>85.07 <math>\pm</math> 5.20</b>  |
| SonyAIBORobotSurface1  | 90.42 $\pm$ 4.34                    | <b>94.08 <math>\pm</math> 1.77</b> | 89.38 $\pm$ 5.35                   | 64.83 $\pm$ 12.29                  | 70.05 $\pm$ 3.77  | 62.90 $\pm$ 13.91                  | 61.90 $\pm$ 7.89                   | 66.61 $\pm$ 10.89                  | 59.10 $\pm$ 5.18  | 62.73 $\pm$ 9.45                   | 52.65 $\pm$ 8.42  | 71.85 $\pm$ 8.28                    |
| SonyAIBORobotSurface2  | <b>88.00 <math>\pm</math> 2.82</b>  | 84.87 $\pm$ 6.58                   | 83.67 $\pm$ 9.41                   | 74.59 $\pm$ 3.28                   | 78.70 $\pm$ 2.68  | 69.60 $\pm$ 7.04                   | 77.26 $\pm$ 1.79                   | 78.49 $\pm$ 0.86                   | 78.30 $\pm$ 1.49  | 79.29 $\pm$ 1.17                   | 56.81 $\pm$ 17.39 | 68.21 $\pm$ 5.28                    |
| StarLightCurves        | <b>76.49 <math>\pm</math> 11.97</b> | 66.10 $\pm$ 9.91                   | 66.22 $\pm$ 10.70                  | 67.92 $\pm$ 13.52                  | 71.57 $\pm$ 12.97 | 63.90 $\pm$ 14.70                  | 35.37 $\pm$ 11.93                  | 65.31 $\pm$ 14.87                  | 34.69 $\pm$ 14.98 | 63.78 $\pm$ 14.15                  | 68.45 $\pm$ 12.08 | 74.47 $\pm$ 15.08                   |
| Strawberry             | 62.81 $\pm$ 6.83                    | 57.14 $\pm$ 6.89                   | 60.22 $\pm$ 9.39                   | 51.84 $\pm$ 8.28                   | 56.59 $\pm$ 2.41  | 53.33 $\pm$ 7.10                   | <b>66.58 <math>\pm</math> 2.89</b> | 52.97 $\pm$ 8.18                   | 59.62 $\pm$ 11.05 | 54.59 $\pm$ 4.47                   | 51.03 $\pm$ 11.13 | 59.41 $\pm$ 6.38                    |
| SwedishLeaf            | <b>72.45 <math>\pm</math> 3.37</b>  | 54.30 $\pm$ 6.89                   | <b>72.45 <math>\pm</math> 3.37</b> | 34.02 $\pm$ 3.00                   | 43.26 $\pm$ 7.26  | 48.91 $\pm$ 2.97                   | 33.28 $\pm$ 3.94                   | 46.24 $\pm$ 1.25                   | 38.85 $\pm$ 4.16  | 47.52 $\pm$ 2.78                   | 7.94 $\pm$ 0.74   | 64.99 $\pm$ 3.83                    |
| Symbols                | <b>90.49 <math>\pm</math> 1.42</b>  | 75.88 $\pm$ 7.72                   | 88.48 $\pm$ 4.41                   | 81.23 $\pm$ 0.99                   | 76.18 $\pm$ 6.87  | 81.37 $\pm$ 1.84                   | 37.42 $\pm$ 3.12                   | 82.04 $\pm$ 0.58                   | 58.95 $\pm$ 6.86  | 80.00 $\pm$ 1.74                   | 51.50 $\pm$ 7.93  | 82.69 $\pm$ 3.43                    |
| SyntheticControl       | 79.67 $\pm$ 4.05                    | 85.07 $\pm$ 4.63                   | 82.60 $\pm$ 6.01                   | 41.40 $\pm$ 6.37                   | 83.80 $\pm$ 5.24  | 55.00 $\pm$ 3.38                   | 44.11 $\pm$ 1.39                   | 59.89 $\pm$ 3.15                   | 47.93 $\pm$ 4.58  | 62.89 $\pm$ 3.15                   | 20.80 $\pm$ 4.95  | <b>87.93 <math>\pm</math> 3.95</b>  |
| ToeSegmentation1       | <b>68.42 <math>\pm</math> 7.02</b>  | 60.79 $\pm$ 8.14                   | 61.58 $\pm$ 7.54                   | 53.95 $\pm$ 2.19                   | 58.60 $\pm$ 10.97 | 51.90 $\pm$ 4.30                   | 53.80 $\pm$ 2.21                   | 53.95 $\pm$ 0.44                   | 52.37 $\pm$ 4.44  | 51.61 $\pm$ 3.08                   | 50.70 $\pm$ 2.53  | 60.26 $\pm$ 8.54                    |
| ToeSegmentation2       | 76.15 $\pm$ 9.44                    | <b>79.69 <math>\pm</math> 9.24</b> | 72.15 $\pm$ 7.68                   | 48.92 $\pm$ 5.64                   | 68.62 $\pm$ 12.68 | 66.41 $\pm$ 15.71                  | 51.54 $\pm$ 5.38                   | 63.08 $\pm$ 15.78                  | 47.08 $\pm$ 8.33  | 63.59 $\pm$ 15.79                  | 54.31 $\pm$ 14.96 | 76.15 $\pm$ 12.44                   |
| Trace                  | 76.00 $\pm$ 3.00                    | 76.40 $\pm$ 3.05                   | 73.80 $\pm$ 2.95                   | 52.40 $\pm$ 4.04                   | 76.60 $\pm$ 4.39  | 47.33 $\pm$ 3.21                   | 51.00 $\pm$ 4.00                   | 49.00 $\pm$ 3.00                   | 55.60 $\pm$ 4.62  | 49.00 $\pm$ 5.00                   | 46.00 $\pm$ 6.78  | <b>98.40 <math>\pm</math> 3.05</b>  |
| TwoLeadECG             | <b>82.53 <math>\pm</math> 6.47</b>  | 79.19 $\pm$ 7.45                   | 70.63 $\pm$ 4.92                   | 55.68 $\pm$ 3.84                   | 59.95 $\pm$ 3.74  | 62.36 $\pm$ 5.86                   | 56.60 $\pm$ 1.27                   | 55.90 $\pm$ 1.72                   | 55.49 $\pm$ 2.80  | 60.14 $\pm$ 7.03                   | 48.48 $\pm$ 5.21  | 74.63 $\pm$ 8.09                    |
| TwoPatterns            | 52.71 $\pm$ 4.44                    | 71.75 $\pm$ 5.17                   | 65.30 $\pm$ 11.45                  | 32.62 $\pm$ 1.84                   | 44.61 $\pm$ 9.45  | 31.66 $\pm$ 1.59                   | 24.88 $\pm$ 1.30                   | 28.57 $\pm$ 1.14                   | 25.14 $\pm$ 0.61  | 29.88 $\pm$ 2.34                   | 27.30 $\pm$ 2.15  | <b>78.23 <math>\pm</math> 0.41</b>  |
| UMD                    | 54.86 $\pm$ 7.95                    | 71.67 $\pm$ 9.36                   | 68.75 $\pm$ 16.74                  | 54.17 $\pm$ 4.55                   | 62.50 $\pm$ 12.00 | 52.31 $\pm$ 6.22                   | 47.69 $\pm$ 9.93                   | 60.19 $\pm$ 1.45                   | 53.06 $\pm$ 4.83  | 56.25 $\pm$ 6.05                   | 47.50 $\pm$ 4.16  | <b>78.86 <math>\pm</math> 14.11</b> |
| UWaveGestureLibraryAll | 32.50 $\pm$ 3.38                    | 28.30 $\pm$ 5.01                   | 31.51 $\pm$ 4.14                   | <b>69.44 <math>\pm</math> 6.55</b> | 43.65 $\pm$ 6.39  | 66.54 $\pm$ 6.96                   | 26.72 $\pm$ 0.50                   | 66.58 $\pm$ 7.48                   | 30.82 $\pm$ 2.10  | <b>66.61 <math>\pm</math> 7.59</b> | 54.22 $\pm$ 7.86  | 42.54 $\pm$ 0.28                    |
| UWaveGestureLibraryX   | 31.94 $\pm$ 2.72                    | 28.97 $\pm$ 3.44                   | 30.99 $\pm$ 3.77                   | 50.42 $\pm$ 3.39                   | 42.01 $\pm$ 2.19  | <b>51.69 <math>\pm</math> 1.72</b> | 21.35 $\pm$ 1.20                   | 51.05 $\pm$ 1.76                   | 28.72 $\pm$ 2.91  | 50.93 $\pm$ 1.76                   | 22.53 $\pm$ 5.80  | 42.54 $\pm$ 0.28                    |
| UWaveGestureLibraryY   | 28.00 $\pm$ 4.17                    | 26.42 $\pm$ 2.97                   | 27.96 $\pm$ 5.26                   | 43.36 $\pm$ 3.42                   | 39.43 $\pm$ 5.20  | 42.54 $\pm$ 7.86                   | 19.88 $\pm$ 2.66                   | <b>43.61 <math>\pm</math> 6.37</b> | 28.02 $\pm$ 2.78  | 41.75 $\pm$ 7.40                   | 28.22 $\pm$ 7.42  | 42.54 $\pm$ 0.28                    |
| UWaveGestureLibraryZ   | 31.19 $\pm$ 2.60                    | 30.19 $\pm$ 2.10                   | 28.42 $\pm$ 4.12                   | 44.60 $\pm$ 3.60                   | 41.59 $\pm$ 3.78  | 42.17 $\pm$ 5.87                   | 23.93 $\pm$ 2.40                   | 43.23 $\pm$ 7.30                   | 29.59 $\pm$ 1.71  | 40.53 $\pm$ 6.44                   | 23.38 $\pm$ 7.02  | <b>59.88 <math>\pm</math> 2.03</b>  |
| Wafer                  | 47.08 $\pm$ 18.17                   | 57.95 $\pm$ 22.06                  | 52.16 $\pm$ 16.40                  | 56.77 $\pm$ 13.38                  | 49.59 $\pm$ 17.01 | 67.22 $\pm$ 1.00                   | 55.02 $\pm$ 6.93                   | 66.32 $\pm$ 1.28                   | 56.21 $\pm$ 9.73  | <b>67.64 <math>\pm</math> 1.29</b> | 54.99 $\pm$ 20.43 | 55.01 $\pm$ 15.62                   |
| Wine                   | <b>60.74 <math>\pm</math> 8.43</b>  | 50.74 $\pm$ 7.92                   | 60.48 $\pm$ 12.52                  | 53.70 $\pm$ 8.88                   | 47.41 $\pm$ 11.54 | 53.70 $\pm$ 3.21                   | 53.09 $\pm$ 5.35                   | 50.00 $\pm$ 0.00                   | 50.74 $\pm$ 1.66  | 54.94 $\pm$ 26.28                  | 50.74 $\pm$ 1.66  | 48.89 $\pm$ 5.34                    |
| WordSynonyms           | 20.63 $\pm$ 4.03                    | 16.05 $\pm$ 1.67                   | 20.63 $\pm$ 4.03                   | 21.35 $\pm$ 4.96                   | 20.06 $\pm$ 3.25  | 29.05 $\pm$ 0.77                   | 6.11 $\pm$ 1.90                    | 29.73 $\pm$ 1.73                   | 13.92 $\pm$ 2.48  | 29.57 $\pm$ 1.41                   | 6.43 $\pm$ 4.20   | <b>34.92 <math>\pm</math> 2.64</b>  |
| Worms                  | <b>54.03 <math>\pm</math> 11.78</b> | 44.94 $\pm$ 11.42                  | 52.73 $\pm$ 7.32                   | 22.60 $\pm$ 2.99                   | 28.57 $\pm$ 6.43  | 19.48 $\pm$ 1.30                   | 17.32 $\pm$ 5.41                   | 17.75 $\pm$ 5.25                   | 15.58 $\pm$ 4.00  | 20.35 $\pm$ 3.75                   | 18.70 $\pm$ 5.00  | 38.70 $\pm$ 4.63                    |
| WormsTwoClass          | 55.32 $\pm$ 9.03                    | 52.47 $\pm$ 8.50                   | <b>55.84 <math>\pm</math> 3.31</b> | 48.57 $\pm$ 2.69                   | 49.09 $\pm$ 4.63  | 49.78 $\pm$ 2.70                   | 51.52 $\pm$ 8.84                   | 47.19 $\pm$ 3.97                   | 47.27 $\pm$ 4.37  | 46.32 $\pm$ 1.50                   | 48.83 $\pm$ 7.82  | 54.03 $\pm$ 7.72                    |
| Yoga                   | <b>52.99 <math>\pm</math> 4.56</b>  | 50.68 $\pm$ 2.33                   | 50.48 $\pm$ 3.16                   | 50.86 $\pm$ 0.91                   | 51.49 $\pm$ 3.75  | 50.12 $\pm$ 2.91                   | 51.48 $\pm$ 2.93                   | 50.70 $\pm$ 2.90                   | 51.46 $\pm$ 3.58  | 50.58 $\pm$ 2.12                   | 51.69 $\pm$ 4.15  | 50.64 $\pm$ 4.90                    |
| Avg. Acc.              | 53.98                               | 50.79                              | <u>54.95</u>                       | 48.01                              | 48.43             | 49.95                              | 39.45                              | 49.45                              | 41.71             | 51.14                              | 38.30             | <b>55.55</b>                        |
| Avg. Rank              | 4.84                                | 5.63                               | <u>4.83</u>                        | 6.78                               | 6.42              | 6.13                               | 9.36                               | 6.29                               | 8.63              | 5.64                               | 9.17              | <b>4.28</b>                         |
| #Best                  | <b>33</b>                           | 13                                 | 19                                 | 10                                 | 2                 | 9                                  | 4                                  | 6                                  | 3                 | 7                                  | 2                 | <u>32</u>                           |

Table 5: Per-dataset 5-shot classification results on 128 shared UCR datasets. Each value is reported as mean  $\pm$  standard deviation over 5 runs.

| Dataset                      | ResNet                             | TapNet                             | COSCO-ResNet                        | DLinear                            | PatchTST                           | TimesNet                           | Autoformer                         | Crossformer                        | FEDformer                           | Informer                           | GPT4TS                             | DuoTSP                             |
|------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| ACSF1                        | <b>76.00 <math>\pm</math> 2.92</b> | 64.60 $\pm$ 5.18                   | 72.80 $\pm$ 1.64                    | 36.20 $\pm$ 3.49                   | 50.40 $\pm$ 2.61                   | 60.00 $\pm$ 2.00                   | 40.00 $\pm$ 1.00                   | 62.67 $\pm$ 2.52                   | 46.80 $\pm$ 3.70                    | 58.67 $\pm$ 1.53                   | 30.60 $\pm$ 4.83                   | 68.33 $\pm$ 2.89                   |
| Adiac                        | 56.21 $\pm$ 4.18                   | 42.81 $\pm$ 4.46                   | 56.21 $\pm$ 4.18                    | 35.29 $\pm$ 1.32                   | 34.58 $\pm$ 2.33                   | 43.82 $\pm$ 1.54                   | 46.97 $\pm$ 3.07                   | 38.62 $\pm$ 1.68                   | 64.50 $\pm$ 5.47                    | 43.99 $\pm$ 3.86                   | 7.11 $\pm$ 1.15                    | <b>69.51 <math>\pm</math> 2.42</b> |
| AllGestureWiimoteX           | <b>40.46 <math>\pm</math> 4.50</b> | 34.14 $\pm$ 3.58                   | 38.71 $\pm$ 5.08                    | 23.63 $\pm$ 2.19                   | 35.40 $\pm$ 4.31                   | 32.24 $\pm$ 3.43                   | 13.57 $\pm$ 1.57                   | 29.95 $\pm$ 4.86                   | 18.17 $\pm$ 3.28                    | 29.38 $\pm$ 3.89                   | 35.40 $\pm$ 4.31                   | 39.40 $\pm$ 2.86                   |
| AllGestureWiimoteY           | 43.20 $\pm$ 1.70                   | 40.60 $\pm$ 1.51                   | 36.20 $\pm$ 3.42                    | 27.89 $\pm$ 2.07                   | 40.97 $\pm$ 3.06                   | 36.76 $\pm$ 2.07                   | 15.38 $\pm$ 0.81                   | 34.43 $\pm$ 2.02                   | 19.49 $\pm$ 2.37                    | 35.43 $\pm$ 1.55                   | 40.97 $\pm$ 3.06                   | <b>48.11 <math>\pm</math> 2.89</b> |
| AllGestureWiimoteZ           | <b>41.46 <math>\pm</math> 4.33</b> | 35.71 $\pm$ 5.12                   | 36.46 $\pm$ 5.30                    | 20.20 $\pm$ 1.22                   | 27.11 $\pm$ 2.51                   | 26.57 $\pm$ 0.25                   | 13.86 $\pm$ 1.71                   | 26.67 $\pm$ 2.65                   | 17.37 $\pm$ 2.23                    | 26.86 $\pm$ 0.99                   | 27.11 $\pm$ 2.51                   | 36.14 $\pm$ 0.97                   |
| ArrowHead                    | 66.40 $\pm$ 5.44                   | 61.03 $\pm$ 5.46                   | 64.23 $\pm$ 7.62                    | 59.43 $\pm$ 5.27                   | 58.86 $\pm$ 12.66                  | 60.19 $\pm$ 7.46                   | 42.67 $\pm$ 0.87                   | 60.76 $\pm$ 1.19                   | 48.11 $\pm$ 2.87                    | 63.43 $\pm$ 8.42                   | 54.06 $\pm$ 9.79                   | <b>73.60 <math>\pm</math> 3.71</b> |
| BME                          | 55.60 $\pm$ 7.80                   | 56.93 $\pm$ 3.90                   | 68.00 $\pm$ 9.90                    | 58.53 $\pm$ 7.20                   | 89.60 $\pm$ 10.65                  | 58.89 $\pm$ 5.55                   | 73.33 $\pm$ 8.11                   | 76.89 $\pm$ 7.34                   | 80.00 $\pm$ 6.70                    | 73.33 $\pm$ 5.93                   | 53.07 $\pm$ 5.63                   | <b>97.73 <math>\pm</math> 1.53</b> |
| Beef                         | 54.67 $\pm$ 4.47                   | 47.33 $\pm$ 2.79                   | 55.33 $\pm$ 2.98                    | 52.67 $\pm$ 7.60                   | 43.33 $\pm$ 4.71                   | 64.44 $\pm$ 3.85                   | 50.00 $\pm$ 3.33                   | 52.22 $\pm$ 8.39                   | 62.00 $\pm$ 8.03                    | <b>84.44 <math>\pm</math> 1.92</b> | 35.33 $\pm$ 4.47                   | 51.33 $\pm$ 6.91                   |
| BeetleFly                    | 72.00 $\pm$ 7.58                   | 60.00 $\pm$ 11.73                  | 79.00 $\pm$ 10.84                   | 71.00 $\pm$ 14.32                  | 57.00 $\pm$ 8.37                   | 70.00 $\pm$ 8.66                   | 51.67 $\pm$ 2.89                   | 66.67 $\pm$ 16.07                  | 49.00 $\pm$ 6.52                    | 70.00 $\pm$ 10.00                  | 59.00 $\pm$ 18.51                  | <b>91.67 <math>\pm</math> 2.89</b> |
| BirdChicken                  | 69.00 $\pm$ 12.94                  | 77.00 $\pm$ 9.75                   | <b>90.00 <math>\pm</math> 3.54</b>  | 57.00 $\pm$ 5.70                   | 61.00 $\pm$ 16.73                  | 63.33 $\pm$ 7.64                   | 36.67 $\pm$ 2.89                   | 65.00 $\pm$ 10.00                  | 54.00 $\pm$ 8.94                    | 68.33 $\pm$ 15.28                  | 54.00 $\pm$ 4.18                   | 85.00 $\pm$ 11.73                  |
| CBF                          | 95.87 $\pm$ 2.59                   | 95.58 $\pm$ 3.67                   | <b>95.89 <math>\pm</math> 3.27</b>  | 61.47 $\pm$ 7.78                   | 95.67 $\pm$ 4.48                   | 74.52 $\pm$ 5.66                   | 39.04 $\pm$ 0.80                   | 80.11 $\pm$ 4.97                   | 42.29 $\pm$ 1.09                    | 80.59 $\pm$ 4.09                   | 41.20 $\pm$ 4.62                   | 93.76 $\pm$ 3.90                   |
| Car                          | 59.33 $\pm$ 9.02                   | 51.00 $\pm$ 7.96                   | 65.67 $\pm$ 5.73                    | 52.33 $\pm$ 6.41                   | 45.67 $\pm$ 2.79                   | 68.89 $\pm$ 6.31                   | 31.67 $\pm$ 5.00                   | 48.89 $\pm$ 4.19                   | 41.00 $\pm$ 12.56                   | 57.78 $\pm$ 5.85                   | 36.00 $\pm$ 4.50                   | <b>69.67 <math>\pm</math> 3.98</b> |
| Chinatown                    | 96.15 $\pm$ 2.16                   | 96.97 $\pm$ 0.49                   | <b>97.43 <math>\pm</math> 0.38</b>  | 50.96 $\pm$ 33.10                  | 96.03 $\pm$ 2.24                   | 84.94 $\pm$ 15.12                  | 91.06 $\pm$ 1.94                   | 93.00 $\pm$ 5.85                   | 89.56 $\pm$ 2.00                    | 87.95 $\pm$ 8.75                   | 73.00 $\pm$ 0.76                   | 93.46 $\pm$ 10.56                  |
| ChlorineConcentration        | 35.97 $\pm$ 2.09                   | 35.63 $\pm$ 3.63                   | 36.15 $\pm$ 2.53                    | 35.57 $\pm$ 3.02                   | 35.80 $\pm$ 3.16                   | 35.02 $\pm$ 2.94                   | 36.15 $\pm$ 1.24                   | 35.49 $\pm$ 4.56                   | 37.02 $\pm$ 1.05                    | 35.96 $\pm$ 1.50                   | 38.60 $\pm$ 5.65                   | <b>39.58 <math>\pm</math> 3.20</b> |
| CinCECGTorso                 | 49.41 $\pm$ 5.43                   | 57.77 $\pm$ 5.88                   | 56.17 $\pm$ 3.91                    | 43.81 $\pm$ 1.84                   | <b>69.26 <math>\pm</math> 4.67</b> | 63.41 $\pm$ 2.73                   | 37.80 $\pm$ 1.42                   | 64.95 $\pm$ 2.40                   | 64.83 $\pm$ 11.83                   | 68.41 $\pm$ 5.18                   | 54.58 $\pm$ 3.46                   | <b>82.56 <math>\pm</math> 7.22</b> |
| Coffee                       | 86.43 $\pm$ 14.81                  | 97.86 $\pm$ 1.96                   | <b>100.00 <math>\pm</math> 0.00</b> | 92.86 $\pm$ 3.57                   | <b>97.86 <math>\pm</math> 1.96</b> | 86.90 $\pm$ 2.06                   | 76.19 $\pm$ 2.06                   | 82.14 $\pm$ 9.45                   | 92.86 $\pm$ 10.41                   | 85.71 $\pm$ 0.00                   | 50.00 $\pm$ 6.19                   | 94.29 $\pm$ 1.96                   |
| Computers                    | 58.80 $\pm$ 7.81                   | 61.60 $\pm$ 5.76                   | 60.96 $\pm$ 6.58                    | 46.64 $\pm$ 4.37                   | 54.48 $\pm$ 7.54                   | 48.97 $\pm$ 5.33                   | 50.53 $\pm$ 1.29                   | 48.00 $\pm$ 8.43                   | 49.04 $\pm$ 4.09                    | 48.67 $\pm$ 4.31                   | 46.56 $\pm$ 5.23                   | <b>62.67 <math>\pm</math> 4.74</b> |
| CricketX                     | 45.90 $\pm$ 4.41                   | 43.08 $\pm$ 3.97                   | 43.08 $\pm$ 2.15                    | 25.23 $\pm$ 1.80                   | 36.10 $\pm$ 5.19                   | 34.87 $\pm$ 4.89                   | 11.79 $\pm$ 1.43                   | 30.77 $\pm$ 0.51                   | 16.92 $\pm$ 1.32                    | 33.68 $\pm$ 3.44                   | 11.33 $\pm$ 1.41                   | <b>49.06 <math>\pm</math> 3.21</b> |
| CricketY                     | 42.67 $\pm$ 4.49                   | 43.08 $\pm$ 3.40                   | 43.28 $\pm$ 6.01                    | 30.41 $\pm$ 3.78                   | 36.92 $\pm$ 6.92                   | 37.86 $\pm$ 2.42                   | 9.06 $\pm$ 1.21                    | 36.58 $\pm$ 2.06                   | 15.23 $\pm$ 1.85                    | 37.26 $\pm$ 2.44                   | 12.05 $\pm$ 2.18                   | <b>49.66 <math>\pm</math> 1.41</b> |
| CricketZ                     | <b>51.59 <math>\pm</math> 3.02</b> | 49.64 $\pm$ 3.11                   | 47.79 $\pm$ 6.26                    | 26.31 $\pm$ 2.67                   | 36.31 $\pm$ 4.35                   | 37.18 $\pm$ 1.18                   | 10.77 $\pm$ 0.51                   | 34.87 $\pm$ 1.43                   | 15.44 $\pm$ 1.54                    | 37.18 $\pm$ 0.77                   | 10.92 $\pm$ 1.59                   | <b>53.68 <math>\pm</math> 2.06</b> |
| Crop                         | 49.93 $\pm$ 1.45                   | 46.93 $\pm$ 2.27                   | 49.93 $\pm$ 1.45                    | 31.91 $\pm$ 2.21                   | 38.13 $\pm$ 2.76                   | 46.92 $\pm$ 2.69                   | 44.47 $\pm$ 0.97                   | 47.82 $\pm$ 2.55                   | 47.76 $\pm$ 1.51                    | 48.69 $\pm$ 2.69                   | 8.15 $\pm$ 3.22                    | <b>52.18 <math>\pm</math> 2.24</b> |
| DiatomSizeReduction          | 85.75 $\pm$ 6.98                   | 70.26 $\pm$ 16.84                  | <b>95.95 <math>\pm</math> 1.26</b>  | 61.57 $\pm$ 2.74                   | 65.49 $\pm$ 10.17                  | 87.69 $\pm$ 0.50                   | 68.63 $\pm$ 2.55                   | 66.34 $\pm$ 8.52                   | 73.53 $\pm$ 9.37                    | 87.25 $\pm$ 0.00                   | 41.83 $\pm$ 13.83                  | 93.66 $\pm$ 1.97                   |
| DistalPhalanxOutlineAgeGroup | 65.76 $\pm$ 4.54                   | 68.63 $\pm$ 3.96                   | 67.19 $\pm$ 5.75                    | 58.99 $\pm$ 14.81                  | 63.31 $\pm$ 4.46                   | 64.75 $\pm$ 3.74                   | 67.63 $\pm$ 2.16                   | 69.30 $\pm$ 3.24                   | 68.20 $\pm$ 2.35                    | 68.59 $\pm$ 2.31                   | 53.67 $\pm$ 12.57                  | <b>69.35 <math>\pm</math> 1.20</b> |
| DistalPhalanxOutlineCorrect  | 66.81 $\pm$ 3.92                   | 61.96 $\pm$ 7.85                   | <b>68.62 <math>\pm</math> 6.57</b>  | 56.23 $\pm$ 10.81                  | 60.07 $\pm$ 6.07                   | 62.44 $\pm$ 3.65                   | 62.68 $\pm$ 3.32                   | 58.94 $\pm$ 3.37                   | 63.41 $\pm$ 7.40                    | 59.30 $\pm$ 3.08                   | 55.87 $\pm$ 8.58                   | 63.77 $\pm$ 11.63                  |
| DistalPhalanxTW              | 58.99 $\pm$ 2.69                   | 57.12 $\pm$ 1.40                   | 58.56 $\pm$ 3.40                    | 49.64 $\pm$ 7.01                   | <b>60.58 <math>\pm</math> 2.61</b> | 57.31 $\pm$ 0.42                   | 51.32 $\pm$ 3.62                   | 59.47 $\pm$ 4.40                   | 54.10 $\pm$ 4.24                    | 56.59 $\pm$ 5.05                   | 31.94 $\pm$ 15.91                  | 56.40 $\pm$ 4.87                   |
| DodgerLoopDay                | 37.75 $\pm$ 3.35                   | 40.50 $\pm$ 4.11                   | 37.00 $\pm$ 3.26                    | 46.75 $\pm$ 4.01                   | 44.25 $\pm$ 2.09                   | <b>52.50 <math>\pm</math> 5.45</b> | 22.08 $\pm$ 0.72                   | 45.42 $\pm$ 2.60                   | 24.25 $\pm$ 3.49                    | 50.83 $\pm$ 4.73                   | 44.25 $\pm$ 2.09                   | 45.75 $\pm$ 3.26                   |
| DodgerLoopGame               | 66.52 $\pm$ 4.02                   | 82.32 $\pm$ 3.38                   | 72.17 $\pm$ 3.64                    | 76.09 $\pm$ 5.15                   | <b>84.49 <math>\pm</math> 1.89</b> | 75.36 $\pm$ 5.02                   | 50.00 $\pm$ 2.61                   | 78.99 $\pm$ 7.63                   | 53.33 $\pm$ 2.09                    | 77.29 $\pm$ 4.82                   | <b>84.49 <math>\pm</math> 1.89</b> | 72.61 $\pm$ 10.25                  |
| DodgerLoopWeekend            | 95.51 $\pm$ 1.65                   | 92.46 $\pm$ 3.46                   | 94.78 $\pm$ 1.88                    | 97.39 $\pm$ 0.40                   | 97.25 $\pm$ 0.61                   | 96.62 $\pm$ 0.84                   | 66.43 $\pm$ 4.25                   | <b>97.83 <math>\pm</math> 0.00</b> | 72.90 $\pm$ 1.82                    | 96.86 $\pm$ 0.42                   | 97.25 $\pm$ 0.61                   | 93.62 $\pm$ 2.37                   |
| ECG200                       | 76.40 $\pm$ 3.58                   | 77.80 $\pm$ 6.26                   | 76.20 $\pm$ 4.60                    | 75.00 $\pm$ 1.58                   | 62.80 $\pm$ 16.36                  | 77.67 $\pm$ 3.06                   | 76.67 $\pm$ 3.79                   | 78.00 $\pm$ 2.00                   | 77.60 $\pm$ 3.21                    | <b>79.33 <math>\pm</math> 4.93</b> | 69.60 $\pm$ 4.51                   | 67.40 $\pm$ 8.91                   |
| ECG5000                      | 78.85 $\pm$ 9.03                   | 75.51 $\pm$ 9.29                   | 75.56 $\pm$ 6.88                    | <b>84.88 <math>\pm</math> 3.51</b> | 71.69 $\pm$ 7.99                   | 82.01 $\pm$ 5.12                   | 76.33 $\pm$ 7.90                   | <b>83.48 <math>\pm</math> 5.73</b> | 74.58 $\pm$ 8.80                    | 82.23 $\pm$ 5.84                   | 18.64 $\pm$ 16.84                  | 73.89 $\pm$ 8.21                   |
| ECGFiveDays                  | 71.54 $\pm$ 13.30                  | 70.27 $\pm$ 8.41                   | 70.22 $\pm$ 9.52                    | 69.38 $\pm$ 6.83                   | 77.54 $\pm$ 3.65                   | 72.94 $\pm$ 9.66                   | <b>87.65 <math>\pm</math> 8.42</b> | <b>75.22 <math>\pm</math> 2.60</b> | <b>83.30 <math>\pm</math> 11.51</b> | 75.61 $\pm$ 4.26                   | 60.49 $\pm$ 6.59                   | 70.96 $\pm$ 3.91                   |
| EOGHorizontalSignal          | 24.97 $\pm$ 1.88                   | 32.71 $\pm$ 1.62                   | 27.35 $\pm$ 3.43                    | 29.56 $\pm$ 3.95                   | 37.73 $\pm$ 6.23                   | 36.92 $\pm$ 1.62                   | 10.96 $\pm$ 1.15                   | 38.40 $\pm$ 2.19                   | 9.94 $\pm$ 1.90                     | 35.17 $\pm$ 4.76                   | 31.99 $\pm$ 1.89                   | <b>46.35 <math>\pm</math> 2.27</b> |
| EOGVerticalSignal            | 18.18 $\pm$ 3.62                   | 22.10 $\pm$ 3.22                   | 18.78 $\pm$ 4.22                    | 29.01 $\pm$ 2.45                   | <b>38.23 <math>\pm</math> 4.43</b> | 31.77 $\pm$ 5.00                   | 8.47 $\pm$ 1.80                    | 31.95 $\pm$ 3.43                   | 10.39 $\pm$ 1.59                    | 31.22 $\pm$ 4.29                   | 22.82 $\pm$ 2.72                   | 36.74 $\pm$ 2.24                   |
| Earthquakes                  | 50.94 $\pm$ 12.41                  | 50.22 $\pm$ 8.85                   | 52.23 $\pm$ 11.61                   | <b>56.83 <math>\pm</math> 2.92</b> | 52.09 $\pm$ 5.05                   | 56.35 $\pm$ 11.87                  | 51.32 $\pm$ 9.11                   | 56.12 $\pm$ 3.14                   | 53.38 $\pm$ 4.56                    | 54.68 $\pm$ 7.61                   | 52.81 $\pm$ 9.99                   | 54.68 $\pm$ 9.31                   |
| ElectricDevices              | 36.69 $\pm$ 5.52                   | 36.63 $\pm$ 6.97                   | 36.08 $\pm$ 5.74                    | 23.65 $\pm$ 1.86                   | 38.77 $\pm$ 8.33                   | 38.73 $\pm$ 7.16                   | 22.89 $\pm$ 3.05                   | 25.09 $\pm$ 5.42                   | 31.29 $\pm$ 6.08                    | 38.61 $\pm$ 5.70                   | 11.47 $\pm$ 4.02                   | <b>47.06 <math>\pm</math> 4.18</b> |
| EthanolLevel                 | 27.52 $\pm$ 1.92                   | 26.36 $\pm$ 0.86                   | 26.60 $\pm$ 1.66                    | 27.88 $\pm$ 1.25                   | 24.88 $\pm$ 0.18                   | 28.40 $\pm$ 1.73                   | 28.60 $\pm$ 2.78                   | 26.53 $\pm$ 0.12                   | 25.40 $\pm$ 0.75                    | <b>29.53 <math>\pm</math> 0.64</b> | 26.32 $\pm$ 1.07                   | 25.92 $\pm$ 1.40                   |
| FaceAll                      | 67.63 $\pm$ 4.66                   | <b>67.75 <math>\pm</math> 7.28</b> | 67.63 $\pm$ 4.66                    | 53.96 $\pm$ 2.14                   | 41.22 $\pm$ 3.29                   | 55.76 $\pm$ 2.36                   | 53.79 $\pm$ 5.19                   | 55.44 $\pm$ 2.51                   | 51.51 $\pm$ 5.08                    | 54.30 $\pm$ 2.41                   | 17.64 $\pm$ 1.73                   | 54.54 $\pm$ 1.26                   |
| FaceFour                     | 88.18 $\pm$ 2.85                   | 61.82 $\pm$ 13.74                  | <b>89.77 <math>\pm</math> 4.62</b>  | 78.86 $\pm$ 3.17                   | 57.50 $\pm$ 8.82                   | 86.36 $\pm$ 1.14                   | 64.77 $\pm$ 2.27                   | 87.88 $\pm$ 1.74                   | 71.82 $\pm$ 1.24                    | 85.61 $\pm$ 1.74                   | 41.14 $\pm$ 6.88                   | 75.68 $\pm$ 4.37                   |
| FacesUCR                     | <b>78.67 <math>\pm</math> 2.69</b> | 73.00 $\pm$ 4.70                   | 76.25 $\pm$ 3.16                    | 50.94 $\pm$ 2.11                   | 43.00 $\pm$ 7.53                   | 68.33 $\pm$ 3.31                   | 58.46 $\pm$ 2.45                   | 68.99 $\pm$ 1.59                   | 60.62 $\pm$ 1.49                    | 67.61 $\pm$ 2.78                   | 27.20 $\pm$ 3.34                   | 68.58 $\pm$ 2.57                   |
| FiftyWords                   | 30.07 $\pm$ 3.14                   | 27.91 $\pm$ 3.50                   | 30.07 $\pm$ 3.14                    | 47.03 $\pm$ 2.26                   | 37.71 $\pm$ 3.50                   | 52.23 $\pm$ 1.46                   | 9.45 $\pm$ 3.91                    | 56.19 $\pm$ 3.63                   | 25.10 $\pm$ 2.19                    | 49.38 $\pm$ 1.41                   | 40.84 $\pm$ 2.35                   | <b>61.58 <math>\pm</math> 2.55</b> |
| Fish                         | 61.26 $\pm$ 15.63                  | 48.69 $\pm$ 6.91                   | 67.66 $\pm$ 6.55                    | 56.80 $\pm$ 6.22                   | 42.97 $\pm$ 5.32                   | <b>68.76 <math>\pm</math> 3.49</b> | 25.52 $\pm$ 2.93                   | 56.57 $\pm$ 5.51                   | 40.80 $\pm$ 3.78                    | 67.05 $\pm$ 1.44                   | 26.40 $\pm$ 8.24                   | 64.57 $\pm$ 5.92                   |
| FordA                        | 71.41 $\pm$ 10.22                  | <b>71.94 <math>\pm</math> 9.09</b> | 70.55 $\pm$ 10.29                   | 50.33 $\pm$ 1.09                   | 63.62 $\pm$ 6.82                   | 50.28 $\pm$ 2.11                   | 51.67 $\pm$ 1.61                   | 49.67 $\pm$ 1.72                   | 51.11 $\pm$ 2.04                    | 49.82 $\pm$ 1.42                   | 50.91 $\pm$ 1.25                   | 49.80 $\pm$ 1.22                   |
| FordB                        | <b>61.26 <math>\pm</math> 2.75</b> | 58.96 $\pm$ 1.47                   | 59.04 $\pm$ 3.16                    | 50.00 $\pm$ 1.41                   | 55.88 $\pm$ 4.39                   | 50.86 $\pm$ 0.77                   | 49.51 $\pm$ 0.81                   | 48.81 $\pm$ 1.26                   | 49.56 $\pm$ 0.80                    | 50.21                              |                                    |                                    |

Table 5: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

| Dataset                        | ResNet                              | TapNet                              | COSCO-ResNet                        | DLinear                             | PatchTST          | TimesNet                           | Autoformer                         | Crossformer                        | FEDformer                          | Informer                            | GPT4TS                             | DuoTSP                              |
|--------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|
| GesturePebbleZ1                | 71.74 $\pm$ 3.41                    | 64.53 $\pm$ 3.05                    | 73.60 $\pm$ 2.12                    | 65.35 $\pm$ 3.15                    | 63.95 $\pm$ 8.94  | 65.89 $\pm$ 3.50                   | 23.45 $\pm$ 1.21                   | 73.64 $\pm$ 2.20                   | 28.95 $\pm$ 4.06                   | <b>76.16 <math>\pm</math> 6.07</b>  | 63.95 $\pm$ 8.94                   | 63.84 $\pm$ 8.99                    |
| GesturePebbleZ2                | 67.34 $\pm$ 3.97                    | 52.53 $\pm$ 17.40                   | <b>73.04 <math>\pm</math> 4.07</b>  | 58.35 $\pm$ 6.18                    | 48.10 $\pm$ 7.12  | 62.03 $\pm$ 7.30                   | 19.41 $\pm$ 4.45                   | 64.98 $\pm$ 9.05                   | 24.81 $\pm$ 3.02                   | 60.34 $\pm$ 8.06                    | 48.10 $\pm$ 7.12                   | 60.89 $\pm$ 8.06                    |
| GunPoint                       | <b>82.53 <math>\pm</math> 12.46</b> | 77.33 $\pm$ 12.19                   | 79.87 $\pm$ 13.86                   | 59.60 $\pm$ 4.87                    | 73.07 $\pm$ 2.03  | 66.00 $\pm$ 6.36                   | 55.11 $\pm$ 6.19                   | 70.22 $\pm$ 2.52                   | 65.87 $\pm$ 7.58                   | 66.00 $\pm$ 2.91                    | 62.40 $\pm$ 5.79                   | <b>90.13 <math>\pm</math> 10.49</b> |
| GunPointAgeSpan                | 75.70 $\pm$ 7.16                    | 69.81 $\pm$ 6.17                    | 76.46 $\pm$ 8.05                    | 75.51 $\pm$ 2.81                    | 73.67 $\pm$ 12.74 | 76.79 $\pm$ 5.43                   | 49.26 $\pm$ 6.34                   | 75.74 $\pm$ 4.37                   | 60.38 $\pm$ 3.16                   | 69.20 $\pm$ 7.84                    | 65.70 $\pm$ 10.14                  | <b>81.52 <math>\pm</math> 9.45</b>  |
| GunPointMaleVersusFemale       | 91.20 $\pm$ 8.64                    | <b>96.65 <math>\pm</math> 4.17</b>  | 90.32 $\pm$ 9.07                    | 61.14 $\pm$ 7.95                    | 87.53 $\pm$ 12.91 | 74.37 $\pm$ 13.69                  | 55.17 $\pm$ 0.48                   | 75.00 $\pm$ 13.22                  | 61.20 $\pm$ 4.73                   | 77.95 $\pm$ 14.78                   | 63.92 $\pm$ 13.57                  | 94.62 $\pm$ 6.24                    |
| GunPointOldVersusYoung         | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 75.87 $\pm$ 2.49                    | 70.29 $\pm$ 8.33  | 70.79 $\pm$ 4.48                   | 56.93 $\pm$ 3.97                   | 73.23 $\pm$ 4.31                   | 58.98 $\pm$ 5.43                   | 70.48 $\pm$ 8.59                    | 92.38 $\pm$ 8.67                   | 80.44 $\pm$ 8.63                    |
| Ham                            | 60.95 $\pm$ 6.21                    | 60.00 $\pm$ 6.77                    | 62.10 $\pm$ 4.59                    | 62.86 $\pm$ 11.45                   | 58.86 $\pm$ 7.48  | 60.00 $\pm$ 11.70                  | 63.49 $\pm$ 0.55                   | 56.51 $\pm$ 8.59                   | 63.62 $\pm$ 3.26                   | <b>66.35 <math>\pm</math> 16.28</b> | 56.76 $\pm$ 9.56                   | 60.57 $\pm$ 9.91                    |
| HandOutlines                   | 46.86 $\pm$ 11.04                   | 52.86 $\pm$ 13.07                   | 67.24 $\pm$ 3.85                    | 80.76 $\pm$ 4.40                    | 74.49 $\pm$ 7.28  | <b>83.06 <math>\pm</math> 3.92</b> | 47.84 $\pm$ 8.25                   | 81.62 $\pm$ 4.95                   | 35.95 $\pm$ 0.00                   | 82.34 $\pm$ 4.91                    | 75.57 $\pm$ 5.75                   | 68.54 $\pm$ 6.21                    |
| Haptics                        | 32.14 $\pm$ 2.79                    | 28.25 $\pm$ 2.44                    | 31.49 $\pm$ 1.92                    | 35.39 $\pm$ 2.49                    | 28.18 $\pm$ 3.44  | 36.58 $\pm$ 0.19                   | 21.32 $\pm$ 3.65                   | <b>37.12 <math>\pm</math> 1.14</b> | 24.68 $\pm$ 3.46                   | <b>37.12 <math>\pm</math> 0.94</b>  | 31.82 $\pm$ 7.39                   | 26.95 $\pm$ 4.88                    |
| Herring                        | 50.94 $\pm$ 5.70                    | <b>55.00 <math>\pm</math> 8.93</b>  | 52.50 $\pm$ 7.46                    | 51.88 $\pm$ 6.76                    | 50.94 $\pm$ 6.68  | 48.96 $\pm$ 5.02                   | <b>57.29 <math>\pm</math> 4.51</b> | 51.56 $\pm$ 7.16                   | 46.88 $\pm$ 13.58                  | 49.48 $\pm$ 2.39                    | 49.06 $\pm$ 7.13                   | 45.00 $\pm$ 4.87                    |
| HouseTwenty                    | 83.70 $\pm$ 6.31                    | 81.18 $\pm$ 9.08                    | 81.85 $\pm$ 9.95                    | 58.99 $\pm$ 5.13                    | 76.81 $\pm$ 7.55  | 73.67 $\pm$ 7.34                   | 54.90 $\pm$ 2.57                   | 66.95 $\pm$ 4.78                   | 59.66 $\pm$ 6.85                   | 72.83 $\pm$ 4.23                    | 63.03 $\pm$ 4.28                   | <b>84.37 <math>\pm</math> 7.51</b>  |
| InlineSkate                    | 20.15 $\pm$ 4.70                    | 22.95 $\pm$ 2.93                    | 19.75 $\pm$ 2.66                    | 21.05 $\pm$ 2.12                    | 18.98 $\pm$ 2.35  | 24.85 $\pm$ 1.07                   | 16.00 $\pm$ 1.31                   | 24.06 $\pm$ 0.93                   | 16.51 $\pm$ 0.15                   | 24.06 $\pm$ 0.90                    | 19.20 $\pm$ 1.41                   | <b>28.07 <math>\pm</math> 1.19</b>  |
| InsectEPGRegularTrain          | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 56.47 $\pm$ 1.83                    | 78.96 $\pm$ 9.81  | 59.57 $\pm$ 2.67                   | 38.96 $\pm$ 2.13                   | 57.30 $\pm$ 1.41                   | 53.17 $\pm$ 6.32                   | 61.31 $\pm$ 3.42                    | 89.88 $\pm$ 9.24                   | 80.24 $\pm$ 3.80                    |
| InsectEPGSmallTrain            | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 53.98 $\pm$ 1.90                    | 77.03 $\pm$ 8.38  | 59.44 $\pm$ 2.44                   | 52.61 $\pm$ 2.78                   | 60.51 $\pm$ 1.98                   | 55.66 $\pm$ 1.65                   | 62.12 $\pm$ 1.62                    | <b>83.13 <math>\pm</math> 0.00</b> | 83.05 $\pm$ 2.80                    |
| InsectWingbeatSound            | 25.39 $\pm$ 1.91                    | 24.63 $\pm$ 1.27                    | 26.10 $\pm$ 5.93                    | 53.74 $\pm$ 0.54                    | 42.18 $\pm$ 1.58  | 56.25 $\pm$ 3.65                   | 25.84 $\pm$ 1.75                   | <b>58.96 <math>\pm</math> 0.98</b> | 38.75 $\pm$ 2.42                   | 57.51 $\pm$ 1.69                    | 20.02 $\pm$ 5.68                   | 41.44 $\pm$ 2.25                    |
| ItalyPowerDemand               | 87.66 $\pm$ 4.88                    | 89.29 $\pm$ 4.25                    | 88.36 $\pm$ 4.24                    | 62.66 $\pm$ 11.58                   | 93.39 $\pm$ 3.05  | 58.50 $\pm$ 7.42                   | 89.57 $\pm$ 5.31                   | <b>93.78 <math>\pm</math> 2.95</b> | <b>95.72 <math>\pm</math> 1.14</b> | <b>95.72 <math>\pm</math> 1.14</b>  | 56.64 $\pm$ 14.68                  | 84.26 $\pm$ 5.94                    |
| LargeKitchenAppliances         | 60.00 $\pm$ 4.27                    | 58.51 $\pm$ 9.39                    | <b>62.67 <math>\pm</math> 10.64</b> | 37.55 $\pm$ 2.23                    | 60.48 $\pm$ 4.88  | 44.09 $\pm$ 3.07                   | 36.98 $\pm$ 2.46                   | 40.44 $\pm$ 3.60                   | 37.71 $\pm$ 3.74                   | 44.09 $\pm$ 6.61                    | 37.17 $\pm$ 1.99                   | 50.13 $\pm$ 3.07                    |
| Lightning2                     | 64.92 $\pm$ 6.82                    | 62.30 $\pm$ 7.42                    | 67.87 $\pm$ 5.51                    | 60.33 $\pm$ 3.91                    | 64.92 $\pm$ 4.86  | <b>69.95 <math>\pm</math> 2.50</b> | 50.82 $\pm$ 5.68                   | 68.85 $\pm$ 5.68                   | 61.64 $\pm$ 7.91                   | 68.31 $\pm$ 3.41                    | 59.02 $\pm$ 6.86                   | 69.51 $\pm$ 10.47                   |
| Lightning7                     | 58.90 $\pm$ 7.12                    | 54.25 $\pm$ 7.09                    | 60.55 $\pm$ 3.80                    | 51.23 $\pm$ 2.84                    | 50.96 $\pm$ 8.07  | 67.58 $\pm$ 3.45                   | 18.26 $\pm$ 3.95                   | 61.64 $\pm$ 3.62                   | 30.41 $\pm$ 0.61                   | <b>70.32 <math>\pm</math> 2.85</b>  | 22.19 $\pm$ 2.63                   | 66.30 $\pm$ 3.83                    |
| Mallat                         | 71.03 $\pm$ 13.07                   | 70.33 $\pm$ 5.20                    | <b>93.72 <math>\pm</math> 2.08</b>  | 88.67 $\pm$ 2.40                    | 77.56 $\pm$ 9.50  | <b>94.66 <math>\pm</math> 0.33</b> | 65.67 $\pm$ 5.01                   | 87.51 $\pm$ 2.78                   | 79.72 $\pm$ 5.50                   | 90.06 $\pm$ 1.99                    | 56.21 $\pm$ 5.78                   | 90.88 $\pm$ 1.15                    |
| Meat                           | 64.33 $\pm$ 14.22                   | 50.00 $\pm$ 9.93                    | 81.00 $\pm$ 8.47                    | <b>85.33 <math>\pm</math> 10.70</b> | 49.33 $\pm$ 15.48 | 76.67 $\pm$ 10.41                  | 47.78 $\pm$ 15.03                  | 37.22 $\pm$ 5.36                   | 50.00 $\pm$ 11.61                  | <b>88.89 <math>\pm</math> 2.55</b>  | 34.00 $\pm$ 1.49                   | 64.00 $\pm$ 11.82                   |
| MedicalImages                  | <b>45.61 <math>\pm</math> 3.94</b>  | 43.95 $\pm$ 4.74                    | 42.63 $\pm$ 4.22                    | 27.26 $\pm$ 7.75                    | 32.26 $\pm$ 2.93  | 37.41 $\pm$ 0.59                   | 36.84 $\pm$ 3.23                   | 38.03 $\pm$ 3.53                   | 40.16 $\pm$ 6.60                   | 38.20 $\pm$ 1.38                    | 5.24 $\pm$ 3.29                    | 42.03 $\pm$ 2.75                    |
| MelbournePedestrian            | 76.79 $\pm$ 2.31                    | 74.96 $\pm$ 2.28                    | <b>77.84 <math>\pm</math> 3.11</b>  | 11.65 $\pm$ 1.29                    | 58.25 $\pm$ 3.82  | 62.66 $\pm$ 1.83                   | 70.73 $\pm$ 4.35                   | 71.76 $\pm$ 3.72                   | 70.23 $\pm$ 2.95                   | 67.90 $\pm$ 5.27                    | 58.25 $\pm$ 3.82                   | 65.81 $\pm$ 3.77                    |
| MiddlePhalanxOutlineAgeGroup   | 47.01 $\pm$ 8.99                    | 54.03 $\pm$ 8.97                    | 50.78 $\pm$ 7.01                    | 52.08 $\pm$ 8.63                    | 49.09 $\pm$ 6.85  | 50.22 $\pm$ 4.61                   | 51.73 $\pm$ 7.75                   | 53.90 $\pm$ 7.87                   | <b>55.06 <math>\pm</math> 4.76</b> | 54.11 $\pm$ 9.46                    | 31.82 $\pm$ 16.75                  | 49.74 $\pm$ 4.72                    |
| MiddlePhalanxOutlineCorrect    | 53.20 $\pm$ 7.83                    | 52.71 $\pm$ 15.62                   | 58.14 $\pm$ 12.45                   | 46.94 $\pm$ 12.76                   | 48.93 $\pm$ 10.44 | 49.60 $\pm$ 3.67                   | <b>58.65 <math>\pm</math> 6.95</b> | 55.44 $\pm$ 15.07                  | 51.68 $\pm$ 10.94                  | 49.60 $\pm$ 9.23                    | 50.93 $\pm$ 8.48                   | 48.45 $\pm$ 7.93                    |
| MiddlePhalanxTW                | <b>45.84 <math>\pm</math> 9.64</b>  | <b>45.84 <math>\pm</math> 3.45</b>  | <b>45.71 <math>\pm</math> 4.74</b>  | <b>45.84 <math>\pm</math> 4.10</b>  | 43.12 $\pm$ 4.67  | 44.37 $\pm$ 0.37                   | 40.04 $\pm$ 2.62                   | 43.72 $\pm$ 4.56                   | 44.16 $\pm$ 4.95                   | 44.59 $\pm$ 2.28                    | 22.86 $\pm$ 6.13                   | 43.64 $\pm$ 5.71                    |
| MixedShapesRegularTrain        | 63.93 $\pm$ 9.77                    | 61.94 $\pm$ 4.99                    | 68.21 $\pm$ 8.35                    | 68.04 $\pm$ 1.34                    | 68.63 $\pm$ 3.58  | 67.15 $\pm$ 2.55                   | 20.78 $\pm$ 0.21                   | 70.05 $\pm$ 0.70                   | 30.67 $\pm$ 6.49                   | 67.15 $\pm$ 1.60                    | 64.11 $\pm$ 1.37                   | <b>76.68 <math>\pm</math> 5.11</b>  |
| MixedShapesSmallTrain          | 66.24 $\pm$ 12.23                   | 57.21 $\pm$ 4.26                    | 72.37 $\pm$ 7.91                    | 66.37 $\pm$ 2.62                    | 66.13 $\pm$ 3.66  | 68.30 $\pm$ 1.42                   | 22.61 $\pm$ 0.88                   | 67.60 $\pm$ 3.67                   | 25.23 $\pm$ 4.04                   | 66.78 $\pm$ 1.26                    | 60.00 $\pm$ 5.54                   | <b>78.71 <math>\pm</math> 1.70</b>  |
| MoteStrain                     | 84.74 $\pm$ 4.04                    | 83.85 $\pm$ 4.63                    | 85.10 $\pm$ 2.03                    | 80.19 $\pm$ 8.46                    | 77.96 $\pm$ 3.58  | <b>85.73 <math>\pm</math> 4.20</b> | 72.68 $\pm$ 2.25                   | 80.09 $\pm$ 3.68                   | 77.11 $\pm$ 6.04                   | 85.36 $\pm$ 1.87                    | 60.30 $\pm$ 15.64                  | 80.37 $\pm$ 4.50                    |
| NonInvasiveFetalECGThorax1     | 40.80 $\pm$ 1.72                    | 31.63 $\pm$ 3.55                    | 40.80 $\pm$ 1.72                    | <b>72.10 <math>\pm</math> 1.96</b>  | 51.79 $\pm$ 9.05  | 70.09 $\pm$ 3.02                   | 22.04 $\pm$ 0.40                   | 70.72 $\pm$ 1.13                   | 32.89 $\pm$ 2.08                   | 69.19 $\pm$ 0.16                    | 28.56 $\pm$ 4.30                   | 67.42 $\pm$ 0.56                    |
| NonInvasiveFetalECGThorax2     | 41.15 $\pm$ 5.46                    | 43.13 $\pm$ 8.61                    | 41.15 $\pm$ 5.46                    | <b>78.55 <math>\pm</math> 0.96</b>  | 60.71 $\pm$ 3.55  | 75.84 $\pm$ 1.88                   | 37.61 $\pm$ 1.37                   | 75.40 $\pm$ 1.50                   | 50.66 $\pm$ 2.46                   | 74.93 $\pm$ 2.49                    | 37.30 $\pm$ 2.25                   | 75.31 $\pm$ 0.82                    |
| OSULeaf                        | 57.85 $\pm$ 5.97                    | 50.74 $\pm$ 10.99                   | <b>72.23 <math>\pm</math> 4.84</b>  | 32.40 $\pm$ 1.86                    | 35.37 $\pm$ 3.13  | 38.84 $\pm$ 1.09                   | 16.80 $\pm$ 1.04                   | 32.64 $\pm$ 2.71                   | 24.05 $\pm$ 2.60                   | 36.91 $\pm$ 1.04                    | 23.06 $\pm$ 4.07                   | 49.92 $\pm$ 6.46                    |
| OliveOil                       | 59.33 $\pm$ 13.21                   | 28.67 $\pm$ 13.66                   | <b>74.00 <math>\pm</math> 6.83</b>  | 31.33 $\pm$ 9.60                    | 26.67 $\pm$ 10.00 | 42.22 $\pm$ 25.02                  | 40.00 $\pm$ 10.00                  | 32.22 $\pm$ 13.47                  | 40.00 $\pm$ 21.73                  | 32.22 $\pm$ 22.19                   | 26.67 $\pm$ 10.00                  | 28.67 $\pm$ 11.69                   |
| PLAID                          | 24.02 $\pm$ 4.96                    | 24.51 $\pm$ 3.44                    | 22.87 $\pm$ 2.95                    | 30.06 $\pm$ 4.34                    | 47.93 $\pm$ 3.93  | 36.56 $\pm$ 2.69                   | 26.88 $\pm$ 2.01                   | 40.16 $\pm$ 2.24                   | 27.26 $\pm$ 3.48                   | 37.93 $\pm$ 4.09                    | 47.93 $\pm$ 3.93                   | <b>52.96 <math>\pm</math> 5.39</b>  |
| PhalangesOutlinesCorrect       | 56.22 $\pm$ 7.89                    | 55.22 $\pm$ 9.82                    | <b>56.85 <math>\pm</math> 6.96</b>  | 52.75 $\pm$ 11.57                   | 53.15 $\pm$ 7.53  | 51.83 $\pm$ 11.55                  | 50.35 $\pm$ 9.47                   | 51.20 $\pm$ 13.21                  | 53.05 $\pm$ 9.33                   | 50.85 $\pm$ 11.66                   | 53.61 $\pm$ 9.97                   | 53.61 $\pm$ 8.69                    |
| Phoneme                        | <b>19.05 <math>\pm</math> 2.97</b>  | 18.15 $\pm$ 2.47                    | <b>19.05 <math>\pm</math> 2.97</b>  | 6.61 $\pm$ 0.25                     | 9.00 $\pm$ 1.08   | 10.58 $\pm$ 0.95                   | 6.87 $\pm$ 0.83                    | 9.00 $\pm$ 0.29                    | 9.61 $\pm$ 1.88                    | 9.30 $\pm$ 1.34                     | 8.21 $\pm$ 0.75                    | 15.63 $\pm$ 1.56                    |
| PickupGestureWiimoteZ          | 56.00 $\pm$ 10.20                   | 66.80 $\pm$ 5.93                    | <b>73.60 <math>\pm</math> 4.56</b>  | 52.40 $\pm$ 1.67                    | 56.80 $\pm$ 9.55  | 57.33 $\pm$ 3.06                   | 27.33 $\pm$ 1.15                   | 58.00 $\pm$ 3.46                   | 42.00 $\pm$ 2.00                   | 58.67 $\pm$ 1.15                    | 56.80 $\pm$ 9.55                   | 68.00 $\pm$ 2.45                    |
| PigAirwayPressure              | 25.00 $\pm$ 2.61                    | 13.37 $\pm$ 2.05                    | <b>48.17 <math>\pm</math> 0.40</b>  | 5.10 $\pm$ 0.26                     | 4.62 $\pm$ 0.55   | 4.81 $\pm$ 0.00                    | 4.17 $\pm$ 2.22                    | 6.73 $\pm$ 0.48                    | 4.04 $\pm$ 0.87                    | 5.93 $\pm$ 0.56                     | 4.42 $\pm$ 0.86                    | 9.81 $\pm$ 0.73                     |
| PigArtPressure                 | 50.96 $\pm$ 5.81                    | 20.77 $\pm$ 3.29                    | <b>68.17 <math>\pm</math> 4.63</b>  | 8.46 $\pm$ 0.65                     | 5.10 $\pm$ 1.82   | 8.97 $\pm$ 1.21                    | 6.57 $\pm$ 2.00                    | 8.97 $\pm$ 0.73                    | 6.92 $\pm$ 1.65                    | 9.29 $\pm$ 1.11                     | 18.17 $\pm$ 2.24                   | 33.08 $\pm$ 1.33                    |
| PigCVP                         | 29.33 $\pm$ 2.90                    | 18.75 $\pm$ 2.40                    | <b>62.98 <math>\pm</math> 1.73</b>  | 6.83 $\pm$ 0.22                     | 3.37 $\pm$ 1.32   | 7.53 $\pm$ 0.56                    | 4.65 $\pm$ 1.11                    | 7.53 $\pm$ 0.28                    | 3.75 $\pm$ 1.24                    | 4.81 $\pm$ 0.48                     | 10.38 $\pm$ 1.21                   | 19.52 $\pm$ 0.55                    |
| Plane                          | <b>100.00 <math>\pm</math> 0.00</b> | 98.10 $\pm$ 2.43                    | <b>99.62 <math>\pm</math> 0.52</b>  | 88.95 $\pm$ 7.93                    | 95.24 $\pm$ 3.16  | 97.78 $\pm$ 0.55                   | 96.51 $\pm$ 1.98                   | 95.87 $\pm$ 1.45                   | 97.14 $\pm$ 1.51                   | 95.87 $\pm$ 0.55                    | 23.05 $\pm$ 11.32                  | 99.24 $\pm$ 1.24                    |
| PowerCons                      | 80.56 $\pm$ 3.36                    | <b>84.44 <math>\pm</math> 5.02</b>  | 84.00 $\pm$ 6.30                    | 70.11 $\pm$ 5.26                    | 81.11 $\pm$ 10.74 | 73.37 $\pm$ 5.45                   | 54.26 $\pm$ 3.90                   | 80.93 $\pm$ 10.72                  | 52.22 $\pm$ 4.39                   | 78.52 $\pm$ 10.32                   | 68.89 $\pm$ 15.97                  | 76.11 $\pm$ 6.55                    |
| ProximalPhalanxOutlineAgeGroup | 73.37 $\pm$ 8.19                    | 81.76 $\pm$ 3.12                    | 79.90 $\pm$ 3.95                    | 65.56 $\pm$ 9.78                    | 75.41 $\pm$ 5.61  | 79.02 $\pm$ 7.19                   | 79.02 $\pm$ 8.4                    |                                    |                                    |                                     |                                    |                                     |

Table 5: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

| Dataset                | ResNet                             | TapNet                             | COSCO-ResNet                       | DLinear                            | PatchTST          | TimesNet                           | Autoformer                         | Crossformer                         | FEDformer         | Informer          | GPT4TS            | DuoTSP                              |
|------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|------------------------------------|-------------------------------------|-------------------|-------------------|-------------------|-------------------------------------|
| SemgHandSubjectCh2     | 37.11 $\pm$ 4.58                   | 44.09 $\pm$ 1.20                   | 35.60 $\pm$ 3.13                   | 50.76 $\pm$ 1.90                   | 49.02 $\pm$ 0.83  | 46.67 $\pm$ 2.12                   | 27.78 $\pm$ 0.59                   | 51.19 $\pm$ 2.74                    | 30.62 $\pm$ 1.93  | 41.48 $\pm$ 6.03  | 36.62 $\pm$ 4.30  | <b>54.31 <math>\pm</math> 3.37</b>  |
| ShakeGestureWiimoteZ   | 80.40 $\pm$ 4.34                   | 91.60 $\pm$ 2.61                   | 89.20 $\pm$ 1.10                   | 56.40 $\pm$ 1.67                   | 72.80 $\pm$ 5.22  | 62.00 $\pm$ 0.00                   | 35.33 $\pm$ 1.15                   | 60.67 $\pm$ 3.06                    | 44.00 $\pm$ 2.45  | 55.33 $\pm$ 2.31  | 72.80 $\pm$ 5.22  | <b>92.00 <math>\pm</math> 0.00</b>  |
| ShapeletSim            | 82.78 $\pm$ 10.96                  | <b>99.67 <math>\pm</math> 0.75</b> | 92.56 $\pm$ 6.55                   | 49.44 $\pm$ 2.69                   | 88.22 $\pm$ 10.06 | 55.93 $\pm$ 2.31                   | 49.81 $\pm$ 2.80                   | 49.63 $\pm$ 4.21                    | 50.67 $\pm$ 1.82  | 49.44 $\pm$ 3.33  | 49.67 $\pm$ 3.98  | 92.00 $\pm$ 3.88                    |
| ShapesAll              | 57.57 $\pm$ 3.68                   | 48.33 $\pm$ 5.58                   | 57.57 $\pm$ 3.68                   | 58.13 $\pm$ 1.68                   | 51.50 $\pm$ 3.69  | 64.67 $\pm$ 1.64                   | 9.22 $\pm$ 0.79                    | 65.83 $\pm$ 3.88                    | 13.83 $\pm$ 3.47  | 59.22 $\pm$ 2.28  | 56.47 $\pm$ 1.46  | <b>76.20 <math>\pm</math> 1.83</b>  |
| SmallKitchenAppliances | 59.47 $\pm$ 8.51                   | 57.39 $\pm$ 6.38                   | <b>62.35 <math>\pm</math> 6.85</b> | 39.04 $\pm$ 4.11                   | 57.55 $\pm$ 2.52  | 57.51 $\pm$ 6.62                   | 38.67 $\pm$ 2.54                   | 39.56 $\pm$ 5.26                    | 46.67 $\pm$ 3.53  | 56.62 $\pm$ 3.96  | 44.00 $\pm$ 7.15  | 51.63 $\pm$ 12.25                   |
| SmoothSubspace         | 86.80 $\pm$ 2.13                   | 82.00 $\pm$ 2.83                   | <b>88.67 <math>\pm</math> 2.36</b> | 33.87 $\pm$ 1.19                   | 57.07 $\pm$ 4.58  | 65.78 $\pm$ 2.04                   | 69.33 $\pm$ 3.71                   | 61.11 $\pm$ 0.77                    | 77.07 $\pm$ 5.51  | 84.00 $\pm$ 4.81  | 37.07 $\pm$ 6.74  | 88.00 $\pm$ 2.87                    |
| SonyAIBORobotSurface1  | 93.94 $\pm$ 2.22                   | <b>95.71 <math>\pm</math> 1.41</b> | 92.85 $\pm$ 2.46                   | 61.83 $\pm$ 10.72                  | 84.73 $\pm$ 2.02  | 71.21 $\pm$ 3.16                   | 67.39 $\pm$ 1.20                   | 71.60 $\pm$ 6.11                    | 64.96 $\pm$ 2.40  | 64.89 $\pm$ 3.03  | 54.54 $\pm$ 12.34 | 84.86 $\pm$ 2.67                    |
| SonyAIBORobotSurface2  | <b>92.51 <math>\pm</math> 3.32</b> | 91.14 $\pm$ 0.89                   | 91.54 $\pm$ 4.19                   | 78.15 $\pm$ 2.59                   | 85.83 $\pm$ 4.10  | 79.96 $\pm$ 1.73                   | 83.11 $\pm$ 2.73                   | 81.78 $\pm$ 2.73                    | 80.76 $\pm$ 4.57  | 83.21 $\pm$ 2.88  | 61.15 $\pm$ 14.84 | 77.21 $\pm$ 3.67                    |
| StarLightCurves        | 81.52 $\pm$ 8.85                   | 60.67 $\pm$ 16.70                  | 76.44 $\pm$ 1.22                   | 74.70 $\pm$ 4.25                   | 76.72 $\pm$ 3.78  | 75.63 $\pm$ 4.10                   | 28.48 $\pm$ 9.45                   | 75.33 $\pm$ 3.39                    | 27.99 $\pm$ 0.00  | 75.69 $\pm$ 5.23  | 75.95 $\pm$ 2.54  | <b>81.86 <math>\pm</math> 7.93</b>  |
| Strawberry             | 63.51 $\pm$ 2.99                   | 64.22 $\pm$ 4.90                   | 60.22 $\pm$ 9.83                   | 58.49 $\pm$ 2.90                   | 63.03 $\pm$ 7.09  | 61.89 $\pm$ 4.92                   | <b>67.21 <math>\pm</math> 4.90</b> | 59.82 $\pm$ 1.80                    | 61.03 $\pm$ 3.64  | 65.77 $\pm$ 7.40  | 52.92 $\pm$ 6.34  | 58.05 $\pm$ 7.04                    |
| SwedishLeaf            | <b>79.81 <math>\pm</math> 1.70</b> | 76.00 $\pm$ 4.97                   | <b>79.81 <math>\pm</math> 1.70</b> | 47.97 $\pm$ 1.68                   | 65.66 $\pm$ 4.54  | 62.61 $\pm$ 4.53                   | 49.23 $\pm$ 4.26                   | 67.15 $\pm$ 4.00                    | 56.93 $\pm$ 3.43  | 62.29 $\pm$ 1.68  | 15.90 $\pm$ 6.80  | 79.78 $\pm$ 2.56                    |
| Symbols                | 88.78 $\pm$ 7.99                   | 88.74 $\pm$ 7.74                   | <b>91.24 <math>\pm</math> 0.96</b> | 83.44 $\pm$ 1.29                   | 84.92 $\pm$ 3.31  | 84.29 $\pm$ 1.01                   | 36.11 $\pm$ 0.23                   | 86.40 $\pm$ 1.63                    | 58.19 $\pm$ 7.40  | 83.75 $\pm$ 0.90  | 27.56 $\pm$ 4.43  | 88.04 $\pm$ 1.30                    |
| SyntheticControl       | 87.80 $\pm$ 2.38                   | <b>96.33 <math>\pm</math> 0.33</b> | 92.60 $\pm$ 1.01                   | 48.53 $\pm$ 3.68                   | 92.73 $\pm$ 2.94  | 73.22 $\pm$ 3.10                   | 52.11 $\pm$ 3.20                   | 66.22 $\pm$ 3.66                    | 56.27 $\pm$ 3.45  | 72.78 $\pm$ 3.66  | 20.93 $\pm$ 4.87  | 95.07 $\pm$ 1.62                    |
| ToeSegmentation1       | <b>85.96 <math>\pm</math> 7.05</b> | 77.37 $\pm$ 5.14                   | 81.58 $\pm$ 5.14                   | 54.74 $\pm$ 3.58                   | 68.16 $\pm$ 3.79  | 55.56 $\pm$ 2.16                   | 53.95 $\pm$ 2.74                   | 51.61 $\pm$ 0.91                    | 53.25 $\pm$ 3.38  | 54.82 $\pm$ 1.32  | 48.77 $\pm$ 4.31  | 70.53 $\pm$ 10.39                   |
| ToeSegmentation2       | 74.77 $\pm$ 12.53                  | <u>77.85 <math>\pm</math> 7.33</u> | <u>74.77 <math>\pm</math> 8.79</u> | 50.77 $\pm$ 2.88                   | 68.46 $\pm$ 6.25  | 58.97 $\pm$ 1.60                   | 50.26 $\pm$ 0.44                   | 59.49 $\pm$ 6.72                    | 37.54 $\pm$ 3.14  | 64.10 $\pm$ 2.70  | 52.31 $\pm$ 9.12  | <b>78.00 <math>\pm</math> 6.36</b>  |
| Trace                  | 88.60 $\pm$ 7.37                   | <u>99.00 <math>\pm</math> 1.00</u> | 93.00 $\pm$ 8.00                   | 53.00 $\pm$ 4.42                   | 76.40 $\pm$ 5.77  | 51.67 $\pm$ 4.93                   | 53.33 $\pm$ 2.08                   | 49.67 $\pm$ 3.79                    | 59.80 $\pm$ 2.59  | 55.67 $\pm$ 8.50  | 48.80 $\pm$ 10.71 | <b>100.00 <math>\pm</math> 0.00</b> |
| TwoLeadECG             | <b>97.98 <math>\pm</math> 4.17</b> | 95.47 $\pm$ 5.89                   | 75.84 $\pm$ 5.58                   | 58.75 $\pm$ 2.38                   | 82.55 $\pm$ 14.28 | 62.16 $\pm$ 1.52                   | 60.81 $\pm$ 0.10                   | 56.34 $\pm$ 2.59                    | 61.88 $\pm$ 1.65  | 68.51 $\pm$ 9.61  | 52.10 $\pm$ 2.73  | 96.68 $\pm$ 5.58                    |
| TwoPatterns            | 64.62 $\pm$ 3.64                   | <b>74.98 <math>\pm</math> 5.04</b> | 74.94 $\pm$ 1.43                   | 38.09 $\pm$ 3.07                   | 51.91 $\pm$ 7.48  | 40.83 $\pm$ 3.89                   | 26.00 $\pm$ 0.92                   | 39.83 $\pm$ 3.80                    | 26.50 $\pm$ 1.83  | 37.54 $\pm$ 1.63  | 27.50 $\pm$ 4.13  | 74.22 $\pm$ 2.81                    |
| UMD                    | 87.08 $\pm$ 10.10                  | 91.53 $\pm$ 5.28                   | <b>96.67 <math>\pm</math> 2.05</b> | 56.53 $\pm$ 3.69                   | 74.44 $\pm$ 17.00 | 61.57 $\pm$ 1.45                   | 62.96 $\pm$ 5.82                   | 68.52 $\pm$ 10.79                   | 71.81 $\pm$ 2.17  | 70.14 $\pm$ 10.69 | 48.47 $\pm$ 7.32  | 95.78 $\pm$ 2.33                    |
| UWaveGestureLibraryAll | 42.53 $\pm$ 1.15                   | 39.33 $\pm$ 5.11                   | 44.53 $\pm$ 1.84                   | 77.02 $\pm$ 5.21                   | 57.17 $\pm$ 2.24  | <b>78.18 <math>\pm</math> 3.39</b> | 26.42 $\pm$ 1.62                   | <u>77.47 <math>\pm</math> 3.61</u>  | 35.67 $\pm$ 1.53  | 76.96 $\pm$ 4.32  | 65.94 $\pm$ 6.48  | 48.36 $\pm$ 0.30                    |
| UWaveGestureLibraryX   | 39.64 $\pm$ 3.12                   | 46.31 $\pm$ 4.46                   | 46.23 $\pm$ 4.40                   | 56.88 $\pm$ 2.64                   | 50.95 $\pm$ 2.22  | 58.42 $\pm$ 2.37                   | 19.27 $\pm$ 3.18                   | <b>58.58 <math>\pm</math> 2.34</b>  | 31.22 $\pm$ 3.55  | 57.86 $\pm$ 1.47  | 26.48 $\pm$ 5.59  | 47.36 $\pm$ 0.30                    |
| UWaveGestureLibraryY   | 37.00 $\pm$ 1.20                   | 38.44 $\pm$ 2.57                   | 40.23 $\pm$ 3.63                   | <b>52.69 <math>\pm</math> 2.24</b> | 48.41 $\pm$ 3.12  | 51.42 $\pm$ 3.10                   | 21.22 $\pm$ 1.11                   | 51.72 $\pm$ 2.98                    | 30.92 $\pm$ 5.43  | 50.81 $\pm$ 4.53  | 19.79 $\pm$ 3.07  | 51.36 $\pm$ 0.31                    |
| UWaveGestureLibraryZ   | 38.74 $\pm$ 2.41                   | 44.95 $\pm$ 4.51                   | 40.85 $\pm$ 4.13                   | 48.35 $\pm$ 3.41                   | 47.76 $\pm$ 4.10  | 50.34 $\pm$ 2.37                   | 20.23 $\pm$ 1.68                   | 49.48 $\pm$ 1.66                    | 31.23 $\pm$ 4.16  | 48.05 $\pm$ 3.36  | 25.20 $\pm$ 6.78  | <b>55.54 <math>\pm</math> 2.61</b>  |
| Wafer                  | 76.83 $\pm$ 9.15                   | 69.99 $\pm$ 14.88                  | 71.79 $\pm$ 15.46                  | 55.60 $\pm$ 16.77                  | 73.34 $\pm$ 19.66 | 80.42 $\pm$ 16.85                  | 76.25 $\pm$ 9.34                   | <b>83.86 <math>\pm</math> 16.37</b> | 75.80 $\pm$ 12.54 | 83.26 $\pm$ 17.85 | 64.01 $\pm$ 22.29 | 71.65 $\pm$ 9.58                    |
| Wine                   | 57.41 $\pm$ 13.03                  | 48.15 $\pm$ 3.93                   | <b>61.11 <math>\pm</math> 5.40</b> | 55.93 $\pm$ 3.56                   | 53.33 $\pm$ 5.62  | 47.53 $\pm$ 8.35                   | 54.94 $\pm$ 8.55                   | 50.00 $\pm$ 0.00                    | 46.67 $\pm$ 6.47  | 37.65 $\pm$ 19.80 | 50.00 $\pm$ 0.00  | 51.48 $\pm$ 2.03                    |
| WordSynonyms           | 25.08 $\pm$ 1.54                   | 21.22 $\pm$ 1.56                   | 25.08 $\pm$ 1.54                   | 29.56 $\pm$ 2.06                   | 30.38 $\pm$ 2.74  | 45.87 $\pm$ 4.63                   | 7.99 $\pm$ 1.24                    | 44.67 $\pm$ 4.83                    | 18.81 $\pm$ 2.07  | 41.33 $\pm$ 4.67  | 22.26 $\pm$ 2.64  | <b>47.05 <math>\pm</math> 3.87</b>  |
| Worms                  | 57.14 $\pm$ 9.76                   | 55.58 $\pm$ 5.91                   | <b>62.86 <math>\pm</math> 4.55</b> | 24.68 $\pm$ 4.68                   | 40.26 $\pm$ 5.11  | 22.94 $\pm$ 0.75                   | 19.91 $\pm$ 3.27                   | 25.97 $\pm$ 0.00                    | 16.62 $\pm$ 2.32  | 22.08 $\pm$ 1.30  | 22.08 $\pm$ 5.11  | 44.42 $\pm$ 6.26                    |
| WormsTwoClass          | <b>68.05 <math>\pm</math> 7.21</b> | 59.74 $\pm$ 6.69                   | 62.08 $\pm$ 7.97                   | 48.83 $\pm$ 3.13                   | 51.17 $\pm$ 8.09  | 52.38 $\pm$ 2.70                   | 42.86 $\pm$ 3.44                   | 51.08 $\pm$ 5.25                    | 48.83 $\pm$ 3.26  | 52.81 $\pm$ 3.27  | 48.83 $\pm$ 5.92  | 54.81 $\pm$ 7.59                    |
| Yoga                   | <b>55.71 <math>\pm</math> 3.37</b> | 53.57 $\pm$ 4.72                   | <u>55.19 <math>\pm</math> 2.57</u> | 50.13 $\pm$ 2.22                   | 53.89 $\pm$ 3.78  | 49.79 $\pm$ 3.21                   | 52.94 $\pm$ 1.57                   | 47.50 $\pm$ 3.55                    | 51.67 $\pm$ 1.99  | 49.68 $\pm$ 4.28  | 48.87 $\pm$ 2.05  | 51.31 $\pm$ 4.34                    |
| Avg. Acc.              | 61.27                              | 59.47                              | <u>63.55</u>                       | 52.30                              | 56.94             | 57.86                              | 43.20                              | 57.20                               | 47.30             | 58.35             | 42.94             | <b>64.14</b>                        |
| Avg. Rank              | 4.77                               | 5.34                               | <u>4.23</u>                        | 7.73                               | 6.60              | 5.69                               | 9.55                               | 5.94                                | 8.70              | 5.79              | 9.48              | <b>4.19</b>                         |
| #Best                  | 21                                 | 13                                 | <u>30</u>                          | 6                                  | 4                 | 8                                  | 4                                  | 7                                   | 2                 | 8                 | 1                 | <b>36</b>                           |

Table 6: Per-dataset 10-shot classification results on 128 shared UCR datasets. Each value is reported as mean  $\pm$  standard deviation over 5 runs.

| Dataset                      | ResNet                             | TapNet                             | COSCO-ResNet                        | DLinear                            | PatchTST                           | TimesNet                           | Autoformer                         | Crossformer                         | FEDformer         | Informer                           | GPT4TS                             | DuoTSP                             |
|------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------|-------------------|------------------------------------|------------------------------------|------------------------------------|
| ACSF1                        | <b>85.40 <math>\pm</math> 2.19</b> | 76.80 $\pm$ 3.35                   | 80.80 $\pm$ 0.84                    | 38.60 $\pm$ 0.89                   | 58.80 $\pm$ 2.59                   | 64.67 $\pm$ 0.58                   | 42.67 $\pm$ 1.53                   | 67.00 $\pm$ 2.00                    | 53.60 $\pm$ 2.51  | 67.00 $\pm$ 2.00                   | 36.80 $\pm$ 3.27                   | 81.33 $\pm$ 1.53                   |
| Adiac                        | 62.46 $\pm$ 4.53                   | 56.98 $\pm$ 3.84                   | 62.46 $\pm$ 4.53                    | 45.32 $\pm$ 1.06                   | 47.72 $\pm$ 5.48                   | 56.01 $\pm$ 2.77                   | 48.68 $\pm$ 0.78                   | 47.91 $\pm$ 6.68                    | 71.71 $\pm$ 1.94  | 48.00 $\pm$ 3.58                   | 12.84 $\pm$ 1.42                   | <b>76.42 <math>\pm</math> 1.68</b> |
| AllGestureWiimoteX           | 49.37 $\pm$ 3.15                   | 43.11 $\pm$ 2.57                   | 39.26 $\pm$ 2.50                    | 23.91 $\pm$ 1.60                   | 43.54 $\pm$ 2.85                   | 37.14 $\pm$ 0.52                   | 15.00 $\pm$ 0.87                   | 35.19 $\pm$ 1.33                    | 25.80 $\pm$ 2.24  | 31.00 $\pm$ 1.57                   | 43.54 $\pm$ 2.85                   | <b>52.31 <math>\pm</math> 2.26</b> |
| AllGestureWiimoteY           | <b>54.89 <math>\pm</math> 2.09</b> | 50.74 $\pm$ 4.62                   | 40.17 $\pm$ 2.24                    | 27.26 $\pm$ 1.11                   | 48.03 $\pm$ 1.58                   | 42.57 $\pm$ 3.19                   | 16.95 $\pm$ 2.22                   | 42.57 $\pm$ 3.81                    | 30.37 $\pm$ 4.65  | 41.24 $\pm$ 2.84                   | 48.03 $\pm$ 1.58                   | <b>56.34 <math>\pm</math> 3.54</b> |
| AllGestureWiimoteZ           | <b>49.34 <math>\pm</math> 1.38</b> | 46.03 $\pm$ 2.76                   | 39.80 $\pm$ 2.97                    | 22.43 $\pm$ 1.80                   | 34.54 $\pm$ 4.20                   | 31.29 $\pm$ 1.13                   | 16.67 $\pm$ 1.15                   | 29.67 $\pm$ 3.32                    | 21.71 $\pm$ 1.69  | 32.19 $\pm$ 0.59                   | 34.54 $\pm$ 4.20                   | 44.43 $\pm$ 2.91                   |
| ArrowHead                    | 72.23 $\pm$ 5.22                   | 63.89 $\pm$ 7.06                   | 65.49 $\pm$ 9.09                    | 64.34 $\pm$ 0.65                   | 62.86 $\pm$ 7.57                   | 60.38 $\pm$ 1.19                   | 48.19 $\pm$ 2.58                   | 59.62 $\pm$ 2.88                    | 57.83 $\pm$ 5.80  | 63.43 $\pm$ 3.48                   | 45.83 $\pm$ 13.23                  | <b>81.49 <math>\pm</math> 3.22</b> |
| BME                          | 64.27 $\pm$ 9.29                   | 69.33 $\pm$ 1.56                   | 78.67 $\pm$ 4.64                    | 64.53 $\pm$ 0.30                   | <b>97.60 <math>\pm</math> 3.29</b> | 71.11 $\pm$ 2.34                   | 78.89 $\pm$ 1.02                   | 88.44 $\pm$ 1.02                    | 91.33 $\pm$ 2.31  | 86.67 $\pm$ 0.00                   | 46.40 $\pm$ 7.46                   | <b>99.20 <math>\pm</math> 0.30</b> |
| Beef                         | 51.33 $\pm$ 6.91                   | 44.00 $\pm$ 6.83                   | 52.00 $\pm$ 2.98                    | 61.33 $\pm$ 4.47                   | 46.00 $\pm$ 6.41                   | 70.00 $\pm$ 3.33                   | 51.11 $\pm$ 5.09                   | 54.44 $\pm$ 3.85                    | 64.00 $\pm$ 7.23  | <b>80.00 <math>\pm</math> 0.00</b> | 37.33 $\pm$ 4.35                   | 68.00 $\pm$ 1.83                   |
| BeetleFly                    | 71.00 $\pm$ 8.94                   | 74.00 $\pm$ 6.52                   | 75.00 $\pm$ 6.12                    | 86.00 $\pm$ 4.18                   | 86.00 $\pm$ 2.24                   | 81.67 $\pm$ 2.89                   | 48.33 $\pm$ 2.89                   | 83.33 $\pm$ 2.89                    | 49.00 $\pm$ 11.94 | 75.00 $\pm$ 5.00                   | 70.00 $\pm$ 12.25                  | <b>96.67 <math>\pm</math> 2.89</b> |
| BirdChicken                  | 84.00 $\pm$ 12.45                  | <b>92.00 <math>\pm</math> 5.70</b> | <b>94.00 <math>\pm</math> 4.18</b>  | 68.00 $\pm$ 2.74                   | 81.00 $\pm$ 9.62                   | 71.67 $\pm$ 7.64                   | 41.67 $\pm$ 7.64                   | 68.33 $\pm$ 16.07                   | 56.00 $\pm$ 5.48  | 56.67 $\pm$ 7.64                   | 56.00 $\pm$ 4.18                   | 89.00 $\pm$ 5.48                   |
| CBF                          | 98.96 $\pm$ 0.23                   | 98.27 $\pm$ 0.96                   | <b>99.24 <math>\pm</math> 0.32</b>  | 67.49 $\pm$ 2.09                   | 95.98 $\pm$ 4.64                   | 90.07 $\pm$ 1.09                   | 39.37 $\pm$ 1.51                   | 91.44 $\pm$ 1.93                    | 51.24 $\pm$ 7.49  | 90.11 $\pm$ 0.88                   | 49.31 $\pm$ 9.05                   | 97.71 $\pm$ 0.69                   |
| Car                          | 57.67 $\pm$ 10.84                  | 58.67 $\pm$ 3.21                   | 65.33 $\pm$ 2.17                    | 73.00 $\pm$ 2.98                   | 62.67 $\pm$ 5.22                   | 70.00 $\pm$ 2.89                   | 27.22 $\pm$ 5.85                   | 65.56 $\pm$ 0.96                    | 37.33 $\pm$ 3.84  | 72.22 $\pm$ 1.92                   | 50.33 $\pm$ 8.28                   | <b>76.33 <math>\pm</math> 5.45</b> |
| Chinatown                    | 98.19 $\pm$ 0.13                   | 98.13 $\pm$ 0.16                   | <b>98.54 <math>\pm</math> 0.21</b>  | 54.58 $\pm$ 24.80                  | 97.73 $\pm$ 0.75                   | 94.75 $\pm$ 2.10                   | 95.63 $\pm$ 1.01                   | 97.28 $\pm$ 0.61                    | 96.62 $\pm$ 0.60  | 96.60 $\pm$ 1.10                   | 63.56 $\pm$ 20.21                  | 94.56 $\pm$ 3.09                   |
| ChlorineConcentration        | <b>41.18 <math>\pm</math> 3.66</b> | 34.41 $\pm$ 8.63                   | 37.86 $\pm$ 5.20                    | 34.42 $\pm$ 4.40                   | 35.51 $\pm$ 6.32                   | 35.06 $\pm$ 1.39                   | 38.85 $\pm$ 1.62                   | 32.99 $\pm$ 4.01                    | 39.05 $\pm$ 4.08  | 39.50 $\pm$ 4.07                   | 33.01 $\pm$ 11.68                  | 34.77 $\pm$ 6.93                   |
| CinCECGTorso                 | 54.03 $\pm$ 3.03                   | 65.94 $\pm$ 5.62                   | 64.64 $\pm$ 5.11                    | 45.36 $\pm$ 1.09                   | 80.65 $\pm$ 7.24                   | 81.59 $\pm$ 1.76                   | 35.82 $\pm$ 2.41                   | 84.42 $\pm$ 2.94                    | 63.93 $\pm$ 1.27  | <b>85.19 <math>\pm</math> 1.24</b> | 52.77 $\pm$ 4.47                   | <b>94.71 <math>\pm</math> 4.65</b> |
| Coffee                       | 97.14 $\pm$ 6.39                   | 97.14 $\pm$ 1.60                   | <b>100.00 <math>\pm</math> 0.00</b> | 96.43 $\pm$ 4.37                   | 97.86 $\pm$ 1.96                   | 91.67 $\pm$ 2.06                   | 86.90 $\pm$ 16.50                  | 92.86 $\pm$ 0.00                    | 98.57 $\pm$ 3.19  | 90.48 $\pm$ 2.06                   | 56.43 $\pm$ 7.74                   | 95.71 $\pm$ 2.99                   |
| Computers                    | 66.48 $\pm$ 3.03                   | 63.20 $\pm$ 6.81                   | <b>68.08 <math>\pm</math> 4.62</b>  | 48.88 $\pm$ 2.52                   | 56.40 $\pm$ 10.98                  | 52.13 $\pm$ 9.60                   | 52.13 $\pm$ 6.02                   | 50.93 $\pm$ 12.73                   | 49.68 $\pm$ 5.04  | 53.87 $\pm$ 12.94                  | 51.52 $\pm$ 3.69                   | 65.60 $\pm$ 6.66                   |
| CricketX                     | 60.72 $\pm$ 2.35                   | 55.18 $\pm$ 4.36                   | 51.28 $\pm$ 3.40                    | 30.15 $\pm$ 2.84                   | 42.92 $\pm$ 1.99                   | 46.50 $\pm$ 1.21                   | 14.27 $\pm$ 1.71                   | 45.38 $\pm$ 2.45                    | 20.87 $\pm$ 2.94  | 45.38 $\pm$ 1.94                   | 19.33 $\pm$ 2.95                   | <b>66.92 <math>\pm</math> 2.71</b> |
| CricketY                     | 56.92 $\pm$ 2.81                   | 49.44 $\pm$ 1.94                   | 47.69 $\pm$ 2.43                    | 32.82 $\pm$ 4.21                   | 42.15 $\pm$ 5.17                   | 44.62 $\pm$ 0.44                   | 9.32 $\pm$ 1.04                    | 44.02 $\pm$ 2.09                    | 17.90 $\pm$ 2.44  | 41.11 $\pm$ 0.53                   | 23.95 $\pm$ 3.97                   | <b>57.95 <math>\pm</math> 1.33</b> |
| CricketZ                     | 63.18 $\pm$ 1.66                   | 57.38 $\pm$ 2.58                   | 48.82 $\pm$ 4.98                    | 30.56 $\pm$ 3.52                   | 43.18 $\pm$ 5.27                   | 46.32 $\pm$ 1.71                   | 12.91 $\pm$ 2.15                   | 47.78 $\pm$ 1.41                    | 18.26 $\pm$ 2.08  | 45.30 $\pm$ 1.50                   | 21.95 $\pm$ 3.02                   | <b>65.30 <math>\pm</math> 1.16</b> |
| Crop                         | 57.92 $\pm$ 1.22                   | 54.52 $\pm$ 1.67                   | 57.92 $\pm$ 1.22                    | 42.46 $\pm$ 2.10                   | 44.96 $\pm$ 1.35                   | 54.14 $\pm$ 2.07                   | 52.84 $\pm$ 0.15                   | 54.82 $\pm$ 1.79                    | 54.74 $\pm$ 0.79  | 54.60 $\pm$ 1.45                   | 25.02 $\pm$ 2.10                   | <b>57.97 <math>\pm</math> 1.25</b> |
| DiatomSizeReduction          | 87.32 $\pm$ 3.50                   | 71.05 $\pm$ 13.58                  | <b>96.80 <math>\pm</math> 0.91</b>  | 44.77 $\pm$ 1.16                   | 62.68 $\pm$ 4.64                   | 87.04 $\pm$ 1.80                   | 66.88 $\pm$ 2.86                   | 51.96 $\pm$ 5.03                    | 48.89 $\pm$ 18.63 | 87.47 $\pm$ 0.19                   | 30.13 $\pm$ 0.15                   | <b>92.61 <math>\pm</math> 2.00</b> |
| DistalPhalanxOutlineAgeGroup | 68.20 $\pm$ 2.98                   | 68.78 $\pm$ 4.24                   | 70.50 $\pm$ 1.61                    | 69.50 $\pm$ 1.66                   | 69.50 $\pm$ 2.81                   | 66.67 $\pm$ 1.10                   | 65.71 $\pm$ 3.62                   | 70.50 $\pm$ 1.25                    | 68.78 $\pm$ 2.72  | 66.91 $\pm$ 1.25                   | 48.06 $\pm$ 11.76                  | <b>70.65 <math>\pm</math> 2.18</b> |
| DistalPhalanxOutlineCorrect  | 64.78 $\pm$ 12.49                  | 65.07 $\pm$ 11.29                  | 63.62 $\pm$ 12.87                   | 50.00 $\pm$ 13.55                  | 60.14 $\pm$ 9.92                   | 58.94 $\pm$ 4.04                   | <b>67.15 <math>\pm</math> 2.41</b> | <b>55.43 <math>\pm</math> 14.84</b> | 62.39 $\pm$ 9.46  | 61.71 $\pm$ 6.72                   | 47.61 $\pm$ 8.80                   | 60.51 $\pm$ 11.85                  |
| DistalPhalanxTW              | 55.83 $\pm$ 3.65                   | 59.14 $\pm$ 2.12                   | <b>60.58 <math>\pm</math> 3.58</b>  | 47.34 $\pm$ 6.42                   | 57.55 $\pm$ 6.08                   | 53.96 $\pm$ 3.14                   | 53.00 $\pm$ 2.91                   | 54.92 $\pm$ 0.83                    | 52.52 $\pm$ 4.35  | 54.92 $\pm$ 2.91                   | 32.37 $\pm$ 14.50                  | 54.10 $\pm$ 3.11                   |
| DodgerLoopDay                | 49.25 $\pm$ 3.71                   | 47.25 $\pm$ 4.28                   | 39.75 $\pm$ 3.11                    | 41.50 $\pm$ 2.24                   | 47.00 $\pm$ 4.11                   | <b>49.58 <math>\pm</math> 5.64</b> | 28.75 $\pm$ 2.50                   | 45.42 $\pm$ 1.91                    | 29.00 $\pm$ 2.40  | 48.75 $\pm$ 1.25                   | 47.00 $\pm$ 4.11                   | 48.75 $\pm$ 1.77                   |
| DodgerLoopGame               | 68.70 $\pm$ 6.09                   | 83.33 $\pm$ 5.28                   | 75.07 $\pm$ 2.64                    | 79.28 $\pm$ 0.97                   | <b>88.84 <math>\pm</math> 1.67</b> | 76.81 $\pm$ 1.26                   | 48.79 $\pm$ 3.42                   | 81.40 $\pm$ 1.51                    | 53.48 $\pm$ 2.83  | 78.99 $\pm$ 1.45                   | <b>88.84 <math>\pm</math> 1.67</b> | 83.62 $\pm$ 0.65                   |
| DodgerLoopWeekend            | 94.06 $\pm$ 1.39                   | 94.20 $\pm$ 2.11                   | 96.81 $\pm$ 0.83                    | <b>97.83 <math>\pm</math> 0.00</b> | 96.81 $\pm$ 0.65                   | 96.86 $\pm$ 0.42                   | 72.71 $\pm$ 1.82                   | <b>97.58 <math>\pm</math> 0.42</b>  | 76.38 $\pm$ 1.82  | 97.10 $\pm$ 0.00                   | 96.81 $\pm$ 0.65                   | <b>97.83 <math>\pm</math> 0.00</b> |
| ECG200                       | 80.00 $\pm$ 4.00                   | <b>83.00 <math>\pm</math> 2.00</b> | 80.40 $\pm$ 4.28                    | 77.60 $\pm$ 4.34                   | 79.80 $\pm$ 6.53                   | 79.33 $\pm$ 3.21                   | 78.00 $\pm$ 3.61                   | <b>81.00 <math>\pm</math> 5.00</b>  | 82.00 $\pm$ 2.92  | 82.00 $\pm$ 3.61                   | 63.00 $\pm$ 15.26                  | 77.40 $\pm$ 4.88                   |
| ECG5000                      | 83.56 $\pm$ 5.09                   | 78.23 $\pm$ 2.20                   | 80.72 $\pm$ 4.37                    | <b>88.37 <math>\pm</math> 1.22</b> | 76.31 $\pm$ 5.83                   | 79.59 $\pm$ 2.75                   | 73.64 $\pm$ 2.21                   | <b>84.75 <math>\pm</math> 1.16</b>  | 79.28 $\pm$ 5.11  | 82.19 $\pm$ 1.79                   | 24.56 $\pm$ 15.15                  | 73.78 $\pm$ 6.18                   |
| ECGFiveDays                  | 95.40 $\pm$ 1.88                   | 88.99 $\pm$ 4.44                   | <b>97.56 <math>\pm</math> 1.20</b>  | 74.87 $\pm$ 4.61                   | 94.61 $\pm$ 2.00                   | 83.82 $\pm$ 7.05                   | 96.44 $\pm$ 1.42                   | <b>91.64 <math>\pm</math> 1.90</b>  | 97.51 $\pm$ 1.12  | 86.18 $\pm$ 8.29                   | 57.47 $\pm$ 4.38                   | 75.28 $\pm$ 1.71                   |
| EOGHorizontalSignal          | 30.61 $\pm$ 3.99                   | 43.15 $\pm$ 5.84                   | 30.94 $\pm$ 3.48                    | 29.89 $\pm$ 1.60                   | 38.78 $\pm$ 6.14                   | 39.59 $\pm$ 1.67                   | 8.01 $\pm$ 1.27                    | 41.62 $\pm$ 1.42                    | 10.61 $\pm$ 3.44  | 40.06 $\pm$ 3.72                   | 39.56 $\pm$ 4.17                   | <b>49.17 <math>\pm</math> 1.62</b> |
| EOGVerticalSignal            | 21.71 $\pm$ 2.53                   | 28.23 $\pm$ 2.47                   | 19.83 $\pm$ 3.08                    | 31.55 $\pm$ 2.19                   | 40.06 $\pm$ 1.98                   | 36.00 $\pm$ 1.12                   | 8.47 $\pm$ 0.84                    | 37.75 $\pm$ 1.67                    | 11.99 $\pm$ 1.33  | 37.02 $\pm$ 1.99                   | 30.55 $\pm$ 3.17                   | <b>42.21 <math>\pm</math> 2.26</b> |
| Earthquakes                  | 53.81 $\pm$ 8.62                   | 53.96 $\pm$ 9.50                   | 51.37 $\pm$ 3.47                    | 52.52 $\pm$ 6.49                   | <b>54.82 <math>\pm</math> 2.85</b> | 60.91 $\pm$ 4.63                   | 49.16 $\pm$ 8.34                   | 59.71 $\pm$ 6.86                    | 54.68 $\pm$ 5.26  | <b>61.87 <math>\pm</math> 3.60</b> | 47.91 $\pm$ 6.66                   | 54.68 $\pm$ 7.63                   |
| ElectricDevices              | 40.12 $\pm$ 4.42                   | 38.65 $\pm$ 3.96                   | 36.00 $\pm$ 4.64                    | 28.04 $\pm$ 1.63                   | 39.08 $\pm$ 8.26                   | 48.59 $\pm$ 6.21                   | 30.07 $\pm$ 3.26                   | 33.74 $\pm$ 3.43                    | 37.44 $\pm$ 1.49  | 48.14 $\pm$ 2.51                   | 16.08 $\pm$ 5.57                   | <b>49.42 <math>\pm</math> 5.02</b> |
| EthanolLevel                 | 26.56 $\pm$ 1.96                   | 25.76 $\pm$ 1.05                   | 26.16 $\pm$ 0.71                    | 29.28 $\pm$ 1.78                   | 25.92 $\pm$ 1.70                   | 28.07 $\pm$ 2.55                   | 25.20 $\pm$ 3.34                   | 26.80 $\pm$ 0.69                    | 25.60 $\pm$ 1.13  | <b>29.87 <math>\pm</math> 2.47</b> | 25.56 $\pm$ 1.05                   | 28.24 $\pm$ 0.50                   |
| FaceAll                      | 77.73 $\pm$ 3.48                   | <b>80.30 <math>\pm</math> 1.67</b> | 77.73 $\pm$ 3.48                    | 67.09 $\pm$ 2.40                   | 54.49 $\pm$ 5.41                   | 67.46 $\pm$ 1.82                   | 59.68 $\pm$ 1.83                   | 67.32 $\pm$ 2.82                    | 60.69 $\pm$ 2.63  | 64.06 $\pm$ 2.32                   | 32.89 $\pm$ 2.71                   | 63.01 $\pm$ 1.34                   |
| FaceFour                     | 87.95 $\pm$ 3.07                   | 59.32 $\pm$ 7.51                   | <b>92.95 <math>\pm</math> 2.03</b>  | 74.77 $\pm$ 0.95                   | 66.36 $\pm$ 10.77                  | 85.98 $\pm$ 1.31                   | 65.91 $\pm$ 2.27                   | <b>88.26 <math>\pm</math> 0.66</b>  | 69.55 $\pm$ 1.69  | 83.71 $\pm$ 1.31                   | 34.55 $\pm$ 3.82                   | 79.09 $\pm$ 1.90                   |
| FacesUCR                     | <b>87.80 <math>\pm</math> 1.60</b> | 84.26 $\pm$ 2.06                   | 82.41 $\pm$ 2.48                    | 63.80 $\pm$ 1.79                   | 54.40 $\pm$ 3.05                   | 80.54 $\pm$ 1.15                   | 70.11 $\pm$ 1.05                   | <b>83.69 <math>\pm</math> 1.13</b>  | 72.08 $\pm$ 1.16  | 79.51 $\pm$ 1.53                   | 34.67 $\pm$ 3.28                   | 79.08 $\pm$ 1.41                   |
| FiftyWords                   | 45.36 $\pm$ 2.39                   | 36.70 $\pm$ 1.89                   | 45.36 $\pm$ 2.39                    | 53.67 $\pm$ 1.47                   | 50.46 $\pm$ 2.83                   | 64.32 $\pm$ 1.87                   | 16.85 $\pm$ 0.67                   | 67.99 $\pm$ 1.84                    | 37.05 $\pm$ 3.62  | 57.36 $\pm$ 1.54                   | 56.40 $\pm$ 1.98                   | <b>71.34 <math>\pm</math> 2.10</b> |
| Fish                         | <b>83.31 <math>\pm</math> 3.78</b> | 56.00 $\pm$ 4.28                   | 71.54 $\pm$ 6.28                    | 65.60 $\pm$ 3.49                   | 62.74 $\pm$ 2.15                   | 68.95 $\pm$ 4.29                   | 24.38 $\pm$ 3.80                   | 71.05 $\pm$ 1.44                    | 57.14 $\pm$ 3.45  | 75.05 $\pm$ 2.31                   | 38.17 $\pm$ 5.05                   | <b>78.29 <math>\pm</math> 3.92</b> |
| FordA                        | 78.26 $\pm$ 5.14                   | 75.61 $\pm$ 0.71                   | <b>78.70 <math>\pm</math> 3.26</b>  | 49.09 $\pm$ 0.93                   | 67.88 $\pm$ 3.90                   | 51.99 $\pm$ 2.52                   | 52.22 $\pm$ 0.95                   | 48.81 $\pm$ 1.31                    | 53.05 $\pm$ 1.09  | 48.21 $\pm$ 1.67                   | 50.06 $\pm$ 2.65                   | 57.15 $\pm$ 1.85                   |
| FordB                        | <b>63.83 <math>\pm</math> 5.74</b> | 62.17 $\pm$ 3.33                   | 61.70 $\pm$ 3.19                    | 48.54 $\pm$ 0.72                   | 59.06 $\pm$ 1.85                   | 46.75 $\pm$ 0.61                   | 49.79 $\pm$ 1.45                   | 47.24 $\pm$ 1.24                    | 50.               |                                    |                                    |                                    |

Table 6: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

| Dataset                        | ResNet                              | TapNet                              | COSCO-ResNet                        | DLinear                            | PatchTST                           | TimesNet                           | Autoformer        | Crossformer                        | FEDformer                          | Informer                           | GPT4TS            | DuoTSP                             |
|--------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|
| GesturePebbleZ1                | 76.86 $\pm$ 5.40                    | 75.47 $\pm$ 8.13                    | 77.67 $\pm$ 2.20                    | 66.63 $\pm$ 2.78                   | 68.84 $\pm$ 4.61                   | 75.00 $\pm$ 1.16                   | 22.29 $\pm$ 3.87  | 77.33 $\pm$ 2.10                   | 38.95 $\pm$ 10.62                  | <b>81.40 <math>\pm</math> 2.53</b> | 68.84 $\pm$ 4.61  | 75.12 $\pm$ 5.19                   |
| GesturePebbleZ2                | 73.16 $\pm$ 4.51                    | 63.42 $\pm$ 5.15                    | 73.16 $\pm$ 2.82                    | 60.38 $\pm$ 3.46                   | 67.59 $\pm$ 4.18                   | 69.62 $\pm$ 2.90                   | 19.20 $\pm$ 0.73  | 74.26 $\pm$ 2.56                   | 28.86 $\pm$ 9.85                   | 73.00 $\pm$ 1.59                   | 67.59 $\pm$ 4.18  | <b>75.57 <math>\pm</math> 4.64</b> |
| GunPoint                       | 85.73 $\pm$ 9.32                    | 82.13 $\pm$ 8.02                    | 91.73 $\pm$ 11.16                   | 68.27 $\pm$ 5.09                   | 77.60 $\pm$ 4.39                   | 71.78 $\pm$ 3.79                   | 64.00 $\pm$ 6.57  | 72.00 $\pm$ 1.15                   | 77.73 $\pm$ 11.90                  | 76.67 $\pm$ 6.67                   | 62.00 $\pm$ 16.53 | <b>97.47 <math>\pm</math> 2.23</b> |
| GunPointAgeSpan                | 89.56 $\pm$ 7.42                    | 72.47 $\pm$ 9.28                    | 74.87 $\pm$ 11.59                   | 83.54 $\pm$ 2.13                   | 82.22 $\pm$ 4.04                   | 85.44 $\pm$ 1.67                   | 52.22 $\pm$ 3.12  | 85.13 $\pm$ 2.39                   | 71.08 $\pm$ 9.43                   | 83.54 $\pm$ 3.02                   | 53.61 $\pm$ 10.43 | <b>92.72 <math>\pm</math> 5.80</b> |
| GunPointMaleVersusFemale       | 92.78 $\pm$ 8.49                    | 93.48 $\pm$ 5.65                    | 92.22 $\pm$ 9.65                    | 67.09 $\pm$ 4.97                   | 95.57 $\pm$ 2.50                   | 91.14 $\pm$ 4.91                   | 60.97 $\pm$ 6.06  | 93.57 $\pm$ 3.96                   | 74.05 $\pm$ 7.11                   | 94.41 $\pm$ 2.22                   | 67.22 $\pm$ 11.33 | <b>98.54 <math>\pm</math> 0.94</b> |
| GunPointOldVersusYoung         | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 76.19 $\pm$ 1.96                   | 79.17 $\pm$ 3.15                   | 78.20 $\pm$ 2.30                   | 57.46 $\pm$ 1.38  | 77.25 $\pm$ 5.77                   | 61.90 $\pm$ 5.09                   | 79.26 $\pm$ 2.47                   | 85.78 $\pm$ 19.21 | 86.79 $\pm$ 4.71                   |
| Ham                            | 60.76 $\pm$ 5.19                    | 66.48 $\pm$ 2.64                    | 65.52 $\pm$ 4.83                    | <b>71.24 <math>\pm</math> 5.65</b> | 62.86 $\pm$ 4.86                   | 66.03 $\pm$ 7.40                   | 63.49 $\pm$ 10.09 | 68.89 $\pm$ 6.69                   | 61.71 $\pm$ 8.23                   | 68.89 $\pm$ 7.02                   | 61.52 $\pm$ 3.79  | 62.86 $\pm$ 3.37                   |
| HandOutlines                   | 50.76 $\pm$ 12.61                   | 50.92 $\pm$ 12.38                   | 57.46 $\pm$ 2.17                    | 81.57 $\pm$ 4.57                   | <b>82.65 <math>\pm</math> 3.66</b> | 81.62 $\pm$ 2.31                   | 38.65 $\pm$ 4.09  | 78.47 $\pm$ 5.74                   | 35.95 $\pm$ 0.00                   | 81.62 $\pm$ 1.95                   | 76.97 $\pm$ 8.74  | 71.51 $\pm$ 5.35                   |
| Haptics                        | 29.68 $\pm$ 4.36                    | 29.16 $\pm$ 4.62                    | 33.31 $\pm$ 5.65                    | 37.73 $\pm$ 3.74                   | 31.95 $\pm$ 2.95                   | 37.66 $\pm$ 2.89                   | 23.81 $\pm$ 5.22  | 37.66 $\pm$ 2.77                   | 25.45 $\pm$ 4.31                   | 38.20 $\pm$ 2.76                   | 35.32 $\pm$ 3.26  | <b>39.22 <math>\pm</math> 4.29</b> |
| Herring                        | 55.00 $\pm$ 9.20                    | 55.62 $\pm$ 5.91                    | 55.00 $\pm$ 7.28                    | 50.31 $\pm$ 6.57                   | <b>59.69 <math>\pm</math> 3.39</b> | 52.60 $\pm$ 3.93                   | 47.92 $\pm$ 2.39  | 53.65 $\pm$ 8.02                   | 49.06 $\pm$ 4.50                   | 50.52 $\pm$ 1.80                   | 57.19 $\pm$ 6.68  | 55.31 $\pm$ 5.01                   |
| HouseTwenty                    | <b>90.92 <math>\pm</math> 1.10</b>  | 88.91 $\pm$ 1.82                    | 89.58 $\pm$ 3.18                    | 65.55 $\pm$ 5.48                   | 81.51 $\pm$ 3.71                   | 81.23 $\pm$ 2.43                   | 60.50 $\pm$ 3.36  | 74.79 $\pm$ 1.46                   | 64.20 $\pm$ 2.95                   | 81.79 $\pm$ 4.63                   | 70.76 $\pm$ 6.04  | 89.24 $\pm$ 2.81                   |
| InlineSkate                    | 30.04 $\pm$ 2.77                    | 24.04 $\pm$ 1.78                    | 21.27 $\pm$ 3.03                    | 25.16 $\pm$ 1.55                   | 21.67 $\pm$ 3.27                   | 28.42 $\pm$ 1.51                   | 16.91 $\pm$ 0.91  | 28.73 $\pm$ 0.83                   | 18.22 $\pm$ 1.24                   | 29.09 $\pm$ 2.20                   | 22.62 $\pm$ 1.07  | <b>33.49 <math>\pm</math> 2.77</b> |
| InsectEPGRegularTrain          | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 54.62 $\pm$ 4.82                   | 79.12 $\pm$ 3.30                   | 61.31 $\pm$ 3.90                   | 42.70 $\pm$ 1.23  | 60.37 $\pm$ 1.29                   | 55.18 $\pm$ 4.83                   | 64.39 $\pm$ 5.02                   | 96.63 $\pm$ 7.54  | 86.59 $\pm$ 4.06                   |
| InsectEPGSmallTrain            | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 60.00 $\pm$ 1.05                   | 81.69 $\pm$ 6.46                   | 69.08 $\pm$ 1.20                   | 55.82 $\pm$ 1.06  | 69.88 $\pm$ 0.80                   | 58.71 $\pm$ 4.87                   | 66.40 $\pm$ 1.81                   | 83.13 $\pm$ 0.00  | 85.06 $\pm$ 2.08                   |
| InsectWingbeatSound            | 34.13 $\pm$ 0.95                    | 34.52 $\pm$ 3.87                    | 30.58 $\pm$ 3.83                    | 60.48 $\pm$ 0.59                   | 47.36 $\pm$ 1.34                   | 60.99 $\pm$ 1.05                   | 39.46 $\pm$ 3.03  | <b>61.67 <math>\pm</math> 1.18</b> | 49.66 $\pm$ 3.26                   | 61.20 $\pm$ 0.34                   | 46.64 $\pm$ 2.91  | 51.36 $\pm$ 2.05                   |
| ItalyPowerDemand               | 92.21 $\pm$ 1.02                    | 92.56 $\pm$ 1.30                    | 92.75 $\pm$ 1.07                    | 53.43 $\pm$ 4.80                   | 93.97 $\pm$ 0.64                   | 87.66 $\pm$ 8.42                   | 94.69 $\pm$ 1.03  | <b>95.59 <math>\pm</math> 0.11</b> | <b>95.82 <math>\pm</math> 0.32</b> | 94.33 $\pm$ 0.49                   | 49.23 $\pm$ 14.84 | 92.19 $\pm$ 2.62                   |
| LargeKitchenAppliances         | 64.05 $\pm$ 4.75                    | 66.13 $\pm$ 5.93                    | 66.08 $\pm$ 5.44                    | 40.85 $\pm$ 3.41                   | <b>67.95 <math>\pm</math> 4.05</b> | 46.22 $\pm$ 6.49                   | 39.82 $\pm$ 1.74  | 43.11 $\pm$ 5.30                   | 40.53 $\pm$ 2.82                   | 45.33 $\pm$ 5.07                   | 41.71 $\pm$ 0.52  | 55.09 $\pm$ 5.30                   |
| Lightning2                     | 70.49 $\pm$ 3.84                    | 64.92 $\pm$ 7.48                    | 65.57 $\pm$ 4.34                    | 58.03 $\pm$ 2.74                   | 69.18 $\pm$ 5.84                   | 71.04 $\pm$ 1.89                   | 51.37 $\pm$ 4.13  | 63.93 $\pm$ 3.28                   | 58.36 $\pm$ 5.26                   | 68.85 $\pm$ 4.34                   | 60.00 $\pm$ 3.77  | <b>77.05 <math>\pm</math> 4.18</b> |
| Lightning7                     | 64.38 $\pm$ 5.97                    | 66.03 $\pm$ 3.41                    | 60.00 $\pm$ 4.37                    | 60.27 $\pm$ 3.36                   | 61.10 $\pm$ 9.76                   | 74.43 $\pm$ 1.58                   | 20.09 $\pm$ 4.81  | 65.75 $\pm$ 1.37                   | 36.71 $\pm$ 3.41                   | <b>75.34 <math>\pm</math> 2.37</b> | 26.30 $\pm$ 5.76  | 66.85 $\pm$ 7.34                   |
| Mallat                         | 68.22 $\pm$ 8.96                    | 69.46 $\pm$ 5.32                    | <b>95.78 <math>\pm</math> 0.57</b>  | 90.38 $\pm$ 0.87                   | 87.49 $\pm$ 9.31                   | 92.79 $\pm$ 0.83                   | 66.92 $\pm$ 5.03  | 85.20 $\pm$ 1.21                   | 79.12 $\pm$ 6.08                   | 89.69 $\pm$ 0.98                   | 25.09 $\pm$ 0.73  | 89.72 $\pm$ 1.50                   |
| Meat                           | 53.00 $\pm$ 18.31                   | 59.33 $\pm$ 6.08                    | 85.00 $\pm$ 4.86                    | <b>87.00 <math>\pm</math> 7.94</b> | 60.00 $\pm$ 5.27                   | 61.11 $\pm$ 4.19                   | 68.89 $\pm$ 7.52  | 55.56 $\pm$ 19.25                  | 55.33 $\pm$ 20.66                  | 86.11 $\pm$ 7.88                   | 38.67 $\pm$ 15.92 | 64.00 $\pm$ 6.08                   |
| MedicalImages                  | <b>56.61 <math>\pm</math> 3.74</b>  | 54.08 $\pm$ 3.55                    | 52.39 $\pm$ 5.17                    | 34.84 $\pm$ 5.66                   | 48.24 $\pm$ 3.52                   | 46.27 $\pm$ 5.02                   | 40.00 $\pm$ 3.41  | 45.09 $\pm$ 2.04                   | 47.45 $\pm$ 1.06                   | 47.72 $\pm$ 6.48                   | 18.58 $\pm$ 8.91  | 54.68 $\pm$ 2.10                   |
| MedbournePedestrian            | <b>84.60 <math>\pm</math> 1.98</b>  | 81.55 $\pm$ 1.62                    | 80.47 $\pm$ 2.59                    | 30.74 $\pm$ 5.50                   | 68.14 $\pm$ 1.82                   | 73.08 $\pm$ 1.13                   | 75.36 $\pm$ 1.75  | 78.01 $\pm$ 1.38                   | 77.48 $\pm$ 0.93                   | 76.77 $\pm$ 0.49                   | 68.14 $\pm$ 1.82  | 70.88 $\pm$ 1.49                   |
| MiddlePhalanxOutlineAgeGroup   | 44.29 $\pm$ 12.74                   | 54.68 $\pm$ 6.95                    | 55.19 $\pm$ 4.97                    | 54.16 $\pm$ 10.18                  | 53.51 $\pm$ 5.82                   | <b>58.66 <math>\pm</math> 1.35</b> | 50.43 $\pm$ 7.47  | 54.11 $\pm$ 4.79                   | 50.13 $\pm$ 8.45                   | 56.49 $\pm$ 1.12                   | 41.82 $\pm$ 18.66 | 51.04 $\pm$ 7.11                   |
| MiddlePhalanxOutlineCorrect    | 60.00 $\pm$ 5.84                    | <b>65.02 <math>\pm</math> 2.98</b>  | 60.62 $\pm$ 1.86                    | 54.23 $\pm$ 4.23                   | 56.36 $\pm$ 2.69                   | 50.86 $\pm$ 7.18                   | 64.72 $\pm$ 6.49  | 54.98 $\pm$ 7.77                   | 58.97 $\pm$ 7.04                   | 52.23 $\pm$ 9.34                   | 52.44 $\pm$ 10.59 | 59.18 $\pm$ 5.65                   |
| MiddlePhalanxTW                | <b>49.74 <math>\pm</math> 5.84</b>  | 48.05 $\pm$ 4.11                    | 49.48 $\pm$ 2.12                    | 45.71 $\pm$ 2.70                   | 46.23 $\pm$ 6.05                   | 46.10 $\pm$ 6.40                   | 37.23 $\pm$ 5.86  | 42.64 $\pm$ 7.61                   | 35.97 $\pm$ 2.62                   | 41.34 $\pm$ 8.07                   | 23.64 $\pm$ 8.52  | 44.42 $\pm$ 6.27                   |
| MixedShapesRegularTrain        | 67.65 $\pm$ 7.57                    | 70.16 $\pm$ 5.08                    | 72.53 $\pm$ 6.92                    | 70.49 $\pm$ 2.59                   | 73.35 $\pm$ 4.70                   | 73.87 $\pm$ 1.26                   | 24.40 $\pm$ 1.33  | 74.08 $\pm$ 2.13                   | 28.76 $\pm$ 6.79                   | 73.65 $\pm$ 2.23                   | 63.69 $\pm$ 2.98  | <b>79.60 <math>\pm</math> 1.74</b> |
| MixedShapesSmallTrain          | 70.61 $\pm$ 10.30                   | 69.46 $\pm$ 11.49                   | 75.93 $\pm$ 3.55                    | 73.56 $\pm$ 2.63                   | 73.16 $\pm$ 8.70                   | 78.08 $\pm$ 2.18                   | 23.04 $\pm$ 1.80  | 77.03 $\pm$ 1.02                   | 27.06 $\pm$ 6.78                   | 77.29 $\pm$ 1.92                   | 65.95 $\pm$ 4.04  | <b>81.61 <math>\pm</math> 2.47</b> |
| MoteStrain                     | 89.38 $\pm$ 0.88                    | 89.14 $\pm$ 3.04                    | 90.18 $\pm$ 0.68                    | 86.26 $\pm$ 0.20                   | 86.58 $\pm$ 1.30                   | 87.19 $\pm$ 1.69                   | 77.40 $\pm$ 1.04  | 87.43 $\pm$ 0.48                   | 85.32 $\pm$ 0.75                   | <b>90.65 <math>\pm</math> 0.48</b> | 67.41 $\pm$ 11.72 | 89.23 $\pm$ 1.53                   |
| NonInvasiveFetalECGThorax1     | 55.91 $\pm$ 3.81                    | 50.28 $\pm$ 5.45                    | 55.91 $\pm$ 3.81                    | <b>81.90 <math>\pm</math> 0.59</b> | 63.56 $\pm$ 4.40                   | 77.13 $\pm$ 1.08                   | 36.83 $\pm$ 0.46  | 78.18 $\pm$ 0.43                   | 46.96 $\pm$ 3.44                   | 75.71 $\pm$ 1.08                   | 58.08 $\pm$ 1.54  | 76.84 $\pm$ 1.40                   |
| NonInvasiveFetalECGThorax2     | 62.50 $\pm$ 1.19                    | 67.13 $\pm$ 2.71                    | 62.50 $\pm$ 1.19                    | <b>86.79 <math>\pm</math> 1.25</b> | 75.21 $\pm$ 2.50                   | 83.55 $\pm$ 0.48                   | 57.79 $\pm$ 2.45  | 83.27 $\pm$ 1.00                   | 69.23 $\pm$ 2.65                   | 80.68 $\pm$ 0.66                   | 70.22 $\pm$ 2.79  | 82.53 $\pm$ 1.33                   |
| OSULeaf                        | <b>72.64 <math>\pm</math> 5.62</b>  | 69.26 $\pm$ 4.17                    | 69.01 $\pm$ 3.75                    | 34.88 $\pm$ 3.54                   | 39.34 $\pm$ 3.85                   | 46.42 $\pm$ 2.52                   | 19.83 $\pm$ 4.60  | 40.50 $\pm$ 2.19                   | 29.83 $\pm$ 2.82                   | 37.19 $\pm$ 2.07                   | 28.10 $\pm$ 5.12  | 59.42 $\pm$ 3.58                   |
| OliveOil                       | 56.00 $\pm$ 14.79                   | 40.00 $\pm$ 0.00                    | <b>66.67 <math>\pm</math> 9.72</b>  | 40.00 $\pm$ 0.00                   | 40.00 $\pm$ 0.00                   | 40.00 $\pm$ 0.00                   | 45.56 $\pm$ 9.62  | 40.00 $\pm$ 0.00                   | 40.00 $\pm$ 0.00                   | 40.00 $\pm$ 0.00                   | 40.00 $\pm$ 0.00  | 40.00 $\pm$ 0.00                   |
| PLAID                          | 28.04 $\pm$ 4.01                    | 25.14 $\pm$ 3.42                    | 22.46 $\pm$ 6.07                    | 34.45 $\pm$ 1.13                   | 54.53 $\pm$ 5.93                   | 50.16 $\pm$ 2.34                   | 32.59 $\pm$ 0.99  | 44.82 $\pm$ 3.36                   | 37.77 $\pm$ 2.12                   | 46.80 $\pm$ 0.84                   | 54.53 $\pm$ 5.93  | <b>59.89 <math>\pm</math> 3.28</b> |
| PhalangesOutlinesCorrect       | 56.88 $\pm$ 6.98                    | 56.74 $\pm$ 8.68                    | 57.48 $\pm$ 7.09                    | 54.27 $\pm$ 11.23                  | 49.91 $\pm$ 6.62                   | <b>58.78 <math>\pm</math> 4.01</b> | 55.79 $\pm$ 3.74  | 55.36 $\pm$ 17.14                  | 52.42 $\pm$ 4.71                   | 54.23 $\pm$ 12.22                  | 53.99 $\pm$ 12.03 | 55.73 $\pm$ 8.07                   |
| Phoneme                        | <b>28.10 <math>\pm</math> 1.26</b>  | 25.96 $\pm$ 1.60                    | <b>28.10 <math>\pm</math> 1.26</b>  | 8.58 $\pm$ 0.50                    | 11.88 $\pm$ 0.54                   | 13.82 $\pm$ 0.86                   | 9.55 $\pm$ 1.16   | 11.71 $\pm$ 0.45                   | 10.91 $\pm$ 1.85                   | 10.88 $\pm$ 0.47                   | 10.94 $\pm$ 0.56  | 21.95 $\pm$ 1.27                   |
| PickupGestureWiimoteZ          | 50.80 $\pm$ 15.01                   | 68.40 $\pm$ 2.61                    | <b>73.60 <math>\pm</math> 2.19</b>  | 51.60 $\pm$ 1.67                   | 54.40 $\pm$ 6.84                   | 56.67 $\pm$ 3.06                   | 26.67 $\pm$ 1.15  | 54.67 $\pm$ 1.15                   | 42.00 $\pm$ 5.10                   | 60.67 $\pm$ 1.15                   | 54.40 $\pm$ 6.84  | 70.40 $\pm$ 2.61                   |
| PigAirwayPressure              | 26.35 $\pm$ 2.05                    | 12.88 $\pm$ 0.53                    | <b>50.58 <math>\pm</math> 0.92</b>  | 4.81 $\pm$ 0.00                    | 4.13 $\pm$ 0.43                    | 4.81 $\pm$ 0.48                    | 2.08 $\pm$ 1.00   | 6.25 $\pm$ 1.27                    | 4.62 $\pm$ 0.87                    | 6.73 $\pm$ 0.48                    | 5.00 $\pm$ 0.65   | 9.62 $\pm$ 0.48                    |
| PigArtPressure                 | 54.71 $\pm$ 1.57                    | 25.19 $\pm$ 5.85                    | <b>69.81 <math>\pm</math> 3.29</b>  | 8.27 $\pm$ 0.40                    | 6.06 $\pm$ 2.62                    | 8.97 $\pm$ 0.28                    | 6.41 $\pm$ 1.00   | 9.78 $\pm$ 0.73                    | 6.35 $\pm$ 0.92                    | 8.65 $\pm$ 1.27                    | 18.08 $\pm$ 2.06  | 31.73 $\pm$ 1.92                   |
| PigCVP                         | 28.85 $\pm$ 2.68                    | 19.62 $\pm$ 1.84                    | <b>61.35 <math>\pm</math> 2.42</b>  | 7.12 $\pm$ 0.40                    | 3.27 $\pm$ 1.61                    | 7.37 $\pm$ 0.73                    | 7.05 $\pm$ 1.21   | 8.33 $\pm$ 1.69                    | 4.13 $\pm$ 0.80                    | 5.93 $\pm$ 1.11                    | 11.63 $\pm$ 1.24  | 19.52 $\pm$ 1.34                   |
| Plane                          | <b>100.00 <math>\pm</math> 0.00</b> | 99.43 $\pm$ 1.28                    | <b>100.00 <math>\pm</math> 0.00</b> | 96.57 $\pm$ 1.28                   | 96.95 $\pm$ 1.24                   | 98.10 $\pm$ 0.00                   | 98.73 $\pm$ 0.55  | 97.14 $\pm$ 0.95                   | 98.48 $\pm$ 0.52                   | 97.14 $\pm$ 0.95                   | 49.71 $\pm$ 5.36  | 99.24 $\pm$ 1.24                   |
| PowerCons                      | 84.00 $\pm$ 3.74                    | 85.67 $\pm$ 2.68                    | <b>88.11 <math>\pm</math> 1.91</b>  | 73.22 $\pm$ 7.34                   | 83.67 $\pm$ 3.78                   | 85.93 $\pm$ 2.85                   | 53.33 $\pm$ 0.56  | 86.30 $\pm$ 2.25                   | 59.67 $\pm$ 5.39                   | 84.63 $\pm$ 0.85                   | 74.33 $\pm$ 14.06 | 82.33 $\pm$ 5.46                   |
| ProximalPhalanxOutlineAgeGroup | 69.07 $\pm$ 5.10                    | 80.49 $\pm$ 4.77                    | 80.98 $\pm$ 2.76                    | 78.98 $\pm$ 2.95                   | 75.90 $\pm$ 8.78                   | 81.30 $\pm$ 5.55                   | 81.14 $\pm$ 3.66  | <b>81.63 <math>\pm</math> 4.89</b> | 76.88 $\pm$ 9.31                   | 77.56 $\$                          |                   |                                    |

Table 6: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

| Dataset                | ResNet                             | TapNet                              | COSCO-ResNet                        | DLinear          | PatchTST                            | TimesNet                           | Autoformer                         | Crossformer                        | FEDformer         | Informer                           | GPT4TS            | DuoTSP                              |
|------------------------|------------------------------------|-------------------------------------|-------------------------------------|------------------|-------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------|------------------------------------|-------------------|-------------------------------------|
| SemgHandSubjectCh2     | 43.56 $\pm$ 2.78                   | 45.20 $\pm$ 5.06                    | 40.22 $\pm$ 5.31                    | 62.80 $\pm$ 4.54 | 54.13 $\pm$ 4.47                    | 60.37 $\pm$ 0.71                   | 34.22 $\pm$ 3.27                   | <b>68.07 <math>\pm</math> 2.65</b> | 39.96 $\pm$ 2.48  | 60.96 $\pm$ 1.10                   | 38.18 $\pm$ 2.42  | 66.71 $\pm$ 4.45                    |
| ShakeGestureWiimoteZ   | 82.00 $\pm$ 6.16                   | 91.20 $\pm$ 2.28                    | 88.80 $\pm$ 1.10                    | 55.60 $\pm$ 2.97 | 68.40 $\pm$ 7.67                    | 60.00 $\pm$ 4.00                   | 33.33 $\pm$ 3.06                   | 62.67 $\pm$ 3.06                   | 45.20 $\pm$ 7.29  | 56.67 $\pm$ 1.15                   | 68.40 $\pm$ 7.67  | <b>91.60 <math>\pm</math> 2.97</b>  |
| ShapeletSim            | 86.33 $\pm$ 6.80                   | <b>100.00 <math>\pm</math> 0.00</b> | <b>100.00 <math>\pm</math> 0.00</b> | 48.78 $\pm$ 0.82 | 95.22 $\pm$ 4.26                    | 59.07 $\pm$ 5.56                   | 48.70 $\pm$ 1.40                   | 50.19 $\pm$ 2.80                   | 51.00 $\pm$ 1.98  | 49.81 $\pm$ 1.95                   | 52.33 $\pm$ 4.64  | 96.67 $\pm$ 1.96                    |
| ShapesAll              | 68.60 $\pm$ 2.31                   | 66.63 $\pm$ 1.82                    | 68.60 $\pm$ 2.31                    | 62.47 $\pm$ 0.34 | 65.47 $\pm$ 2.49                    | 72.11 $\pm$ 0.69                   | 12.89 $\pm$ 4.16                   | 73.28 $\pm$ 0.38                   | 31.00 $\pm$ 5.80  | 65.56 $\pm$ 1.08                   | 70.00 $\pm$ 1.20  | <b>83.13 <math>\pm</math> 0.32</b>  |
| SmallKitchenAppliances | <b>67.20 <math>\pm</math> 8.07</b> | 61.87 $\pm$ 4.21                    | 64.69 $\pm$ 7.08                    | 36.59 $\pm$ 1.57 | 60.00 $\pm$ 3.85                    | 55.11 $\pm$ 4.59                   | 41.51 $\pm$ 2.40                   | 38.76 $\pm$ 2.56                   | 47.25 $\pm$ 3.08  | 51.02 $\pm$ 1.54                   | 41.33 $\pm$ 5.03  | 57.60 $\pm$ 5.56                    |
| SmoothSubspace         | <b>92.00 <math>\pm</math> 1.56</b> | 87.07 $\pm$ 3.67                    | <b>92.00 <math>\pm</math> 3.02</b>  | 38.13 $\pm$ 6.59 | 67.73 $\pm$ 10.31                   | 79.56 $\pm$ 2.78                   | 80.67 $\pm$ 0.67                   | 60.00 $\pm$ 11.02                  | 86.13 $\pm$ 3.38  | <b>90.67 <math>\pm</math> 4.00</b> | 43.60 $\pm$ 9.96  | 89.47 $\pm$ 2.60                    |
| SonyAIBORobotSurface1  | 95.11 $\pm$ 2.42                   | <b>96.01 <math>\pm</math> 0.87</b>  | 92.45 $\pm$ 1.40                    | 42.96 $\pm$ 0.07 | 87.95 $\pm$ 1.92                    | 64.28 $\pm$ 1.83                   | 67.05 $\pm$ 1.01                   | 76.43 $\pm$ 0.69                   | 64.86 $\pm$ 0.43  | 67.83 $\pm$ 1.06                   | 45.09 $\pm$ 4.84  | 86.09 $\pm$ 3.97                    |
| SonyAIBORobotSurface2  | <b>96.73 <math>\pm</math> 3.11</b> | 92.82 $\pm$ 3.91                    | 94.67 $\pm$ 2.40                    | 77.50 $\pm$ 0.71 | 90.95 $\pm$ 2.30                    | 81.25 $\pm$ 2.70                   | 83.07 $\pm$ 2.21                   | 86.08 $\pm$ 2.22                   | 82.71 $\pm$ 1.50  | 85.73 $\pm$ 2.18                   | 61.15 $\pm$ 12.88 | 84.11 $\pm$ 2.29                    |
| StarLightCurves        | 71.47 $\pm$ 14.56                  | 82.56 $\pm$ 3.87                    | 75.10 $\pm$ 5.06                    | 74.16 $\pm$ 3.22 | 77.51 $\pm$ 4.09                    | 75.29 $\pm$ 4.53                   | 29.60 $\pm$ 9.11                   | 76.31 $\pm$ 3.24                   | 30.20 $\pm$ 4.95  | 74.55 $\pm$ 5.89                   | 70.71 $\pm$ 14.11 | <b>85.26 <math>\pm</math> 3.54</b>  |
| Strawberry             | 79.73 $\pm$ 8.80                   | 76.70 $\pm$ 3.26                    | 76.70 $\pm$ 4.55                    | 60.81 $\pm$ 1.94 | <b>79.89 <math>\pm</math> 6.30</b>  | 61.98 $\pm$ 1.58                   | <b>80.63 <math>\pm</math> 1.13</b> | 59.10 $\pm$ 1.09                   | 77.51 $\pm$ 4.36  | 68.11 $\pm$ 2.15                   | 50.54 $\pm$ 10.66 | 75.84 $\pm$ 0.99                    |
| SwedishLeaf            | <b>88.03 <math>\pm</math> 1.39</b> | 83.68 $\pm$ 1.26                    | <b>88.03 <math>\pm</math> 1.39</b>  | 66.11 $\pm$ 1.36 | 73.47 $\pm$ 2.54                    | 75.95 $\pm$ 1.36                   | 60.69 $\pm$ 4.66                   | 81.39 $\pm$ 2.49                   | 68.80 $\pm$ 1.87  | 74.72 $\pm$ 1.82                   | 28.96 $\pm$ 3.20  | 85.89 $\pm$ 1.27                    |
| Symbols                | <b>93.83 <math>\pm</math> 2.55</b> | 89.89 $\pm$ 7.08                    | 92.52 $\pm$ 1.13                    | 75.94 $\pm$ 0.93 | 85.87 $\pm$ 1.87                    | 83.32 $\pm$ 0.52                   | 34.77 $\pm$ 0.30                   | 85.96 $\pm$ 1.03                   | 61.21 $\pm$ 7.54  | 82.51 $\pm$ 0.70                   | 21.11 $\pm$ 2.93  | 89.33 $\pm$ 0.82                    |
| SyntheticControl       | 95.60 $\pm$ 1.14                   | <b>97.87 <math>\pm</math> 0.61</b>  | 97.60 $\pm$ 0.43                    | 59.07 $\pm$ 3.98 | 94.20 $\pm$ 1.64                    | 90.78 $\pm$ 1.64                   | 59.44 $\pm$ 1.95                   | 79.44 $\pm$ 4.86                   | 68.40 $\pm$ 3.43  | 90.11 $\pm$ 1.35                   | 22.60 $\pm$ 7.62  | 97.33 $\pm$ 0.85                    |
| ToeSegmentation1       | <b>93.25 <math>\pm</math> 1.60</b> | 89.56 $\pm$ 3.33                    | 91.58 $\pm$ 1.68                    | 55.53 $\pm$ 3.14 | 68.95 $\pm$ 9.85                    | 56.73 $\pm$ 5.97                   | 50.29 $\pm$ 6.71                   | 55.85 $\pm$ 3.55                   | 49.39 $\pm$ 2.18  | 55.70 $\pm$ 4.98                   | 50.53 $\pm$ 1.40  | 80.53 $\pm$ 2.75                    |
| ToeSegmentation2       | 81.38 $\pm$ 10.25                  | <b>87.08 <math>\pm</math> 4.53</b>  | <b>85.85 <math>\pm</math> 2.64</b>  | 50.15 $\pm$ 3.62 | 73.38 $\pm$ 4.70                    | 68.21 $\pm$ 10.44                  | 56.67 $\pm$ 7.86                   | 58.46 $\pm$ 6.30                   | 46.77 $\pm$ 3.41  | 61.54 $\pm$ 3.35                   | 43.54 $\pm$ 14.75 | 80.31 $\pm$ 4.70                    |
| Trace                  | 92.20 $\pm$ 4.60                   | <b>100.00 <math>\pm</math> 0.00</b> | <b>95.20 <math>\pm</math> 5.07</b>  | 55.80 $\pm$ 5.72 | 86.20 $\pm$ 10.35                   | 72.67 $\pm$ 1.53                   | 55.67 $\pm$ 4.51                   | 61.33 $\pm$ 4.62                   | 74.80 $\pm$ 3.27  | 78.00 $\pm$ 4.36                   | 46.80 $\pm$ 8.17  | <b>100.00 <math>\pm</math> 0.00</b> |
| TwoLeadECG             | <b>99.86 <math>\pm</math> 0.08</b> | 98.79 $\pm$ 1.25                    | 97.56 $\pm$ 2.37                    | 58.05 $\pm$ 1.68 | 91.89 $\pm$ 5.00                    | 70.85 $\pm$ 3.05                   | 75.62 $\pm$ 1.63                   | 65.26 $\pm$ 1.91                   | 78.86 $\pm$ 1.56  | 78.99 $\pm$ 2.23                   | 54.01 $\pm$ 3.73  | 99.68 $\pm$ 0.24                    |
| TwoPatterns            | 70.89 $\pm$ 4.77                   | 77.56 $\pm$ 2.96                    | <b>78.86 <math>\pm</math> 3.59</b>  | 42.28 $\pm$ 1.08 | 74.73 $\pm$ 6.48                    | 42.55 $\pm$ 1.37                   | 26.22 $\pm$ 0.85                   | 42.35 $\pm$ 1.82                   | 29.82 $\pm$ 3.35  | 37.95 $\pm$ 1.72                   | 25.60 $\pm$ 4.32  | 70.06 $\pm$ 0.70                    |
| UMD                    | 92.78 $\pm$ 3.35                   | 98.33 $\pm$ 1.05                    | <b>98.75 <math>\pm</math> 0.58</b>  | 64.17 $\pm$ 3.05 | 91.53 $\pm$ 4.77                    | 76.62 $\pm$ 2.44                   | 75.93 $\pm$ 7.23                   | 89.58 $\pm$ 4.22                   | 93.06 $\pm$ 1.47  | 87.27 $\pm$ 3.21                   | 47.50 $\pm$ 8.96  | 98.67 $\pm$ 7.42                    |
| UWaveGestureLibraryAll | 48.45 $\pm$ 3.31                   | 48.15 $\pm$ 2.07                    | 49.06 $\pm$ 3.81                    | 80.57 $\pm$ 3.71 | 68.35 $\pm$ 4.39                    | <b>84.19 <math>\pm</math> 2.90</b> | 27.54 $\pm$ 1.68                   | <b>84.63 <math>\pm</math> 3.19</b> | 42.77 $\pm$ 5.45  | 83.32 $\pm$ 3.39                   | 81.20 $\pm$ 2.71  | 52.41 $\pm$ 1.70                    |
| UWaveGestureLibraryX   | 51.98 $\pm$ 1.26                   | 57.08 $\pm$ 2.26                    | 52.99 $\pm$ 2.93                    | 60.99 $\pm$ 2.04 | 57.69 $\pm$ 1.76                    | <b>62.37 <math>\pm</math> 1.85</b> | 17.30 $\pm$ 0.19                   | <b>62.30 <math>\pm</math> 0.55</b> | 36.29 $\pm$ 8.88  | 61.26 $\pm$ 1.95                   | 56.81 $\pm$ 2.90  | 53.50 $\pm$ 1.91                    |
| UWaveGestureLibraryY   | 45.58 $\pm$ 1.39                   | 47.42 $\pm$ 0.67                    | 46.02 $\pm$ 2.85                    | 55.44 $\pm$ 1.29 | 53.52 $\pm$ 3.23                    | <b>56.83 <math>\pm</math> 0.99</b> | 23.04 $\pm$ 3.42                   | <b>56.65 <math>\pm</math> 1.12</b> | 32.16 $\pm$ 4.98  | 55.22 $\pm$ 1.56                   | 46.61 $\pm$ 1.40  | 52.03 $\pm$ 0.97                    |
| UWaveGestureLibraryZ   | 48.34 $\pm$ 2.78                   | 51.60 $\pm$ 2.75                    | 44.60 $\pm$ 1.67                    | 53.18 $\pm$ 1.58 | 53.45 $\pm$ 2.14                    | 54.99 $\pm$ 0.25                   | 20.01 $\pm$ 2.19                   | <b>54.36 <math>\pm</math> 2.47</b> | 34.40 $\pm$ 4.42  | 51.61 $\pm$ 2.72                   | 46.80 $\pm$ 1.81  | <b>60.65 <math>\pm</math> 2.84</b>  |
| Wafer                  | 84.60 $\pm$ 6.19                   | 78.03 $\pm$ 11.28                   | 81.53 $\pm$ 4.78                    | 69.44 $\pm$ 4.07 | <b>84.77 <math>\pm</math> 11.10</b> | <b>82.09 <math>\pm</math> 7.52</b> | 76.30 $\pm$ 12.19                  | 83.57 $\pm$ 7.70                   | 81.81 $\pm$ 10.33 | 81.67 $\pm$ 9.12                   | 78.96 $\pm$ 10.53 | 82.53 $\pm$ 8.58                    |
| Wine                   | <b>68.15 <math>\pm</math> 6.33</b> | 52.59 $\pm$ 5.80                    | 64.07 $\pm$ 8.03                    | 50.74 $\pm$ 1.66 | 50.00 $\pm$ 0.00                    | 58.64 $\pm$ 5.35                   | 50.62 $\pm$ 1.07                   | 50.00 $\pm$ 3.70                   | 50.00 $\pm$ 0.00  | 62.96 $\pm$ 7.41                   | 47.78 $\pm$ 4.97  | 49.26 $\pm$ 2.11                    |
| WordSynonyms           | 35.14 $\pm$ 2.03                   | 30.72 $\pm$ 3.50                    | 35.14 $\pm$ 2.03                    | 35.86 $\pm$ 2.07 | 39.44 $\pm$ 4.12                    | 51.78 $\pm$ 3.11                   | 15.15 $\pm$ 0.48                   | 51.10 $\pm$ 4.21                   | 25.64 $\pm$ 4.61  | 49.01 $\pm$ 4.47                   | 33.92 $\pm$ 1.82  | <b>58.90 <math>\pm</math> 2.04</b>  |
| Worms                  | 50.39 $\pm$ 8.29                   | <b>51.95 <math>\pm</math> 5.35</b>  | 45.71 $\pm$ 7.71                    | 23.38 $\pm$ 5.11 | 39.74 $\pm$ 4.27                    | 23.38 $\pm$ 6.49                   | 26.84 $\pm$ 3.97                   | 27.27 $\pm$ 9.09                   | 25.45 $\pm$ 3.74  | 25.97 $\pm$ 9.09                   | 20.78 $\pm$ 5.35  | 48.57 $\pm$ 4.91                    |
| WormsTwoClass          | <b>64.94 <math>\pm</math> 5.51</b> | 63.12 $\pm$ 4.37                    | 61.30 $\pm$ 7.20                    | 49.09 $\pm$ 4.63 | 47.27 $\pm$ 1.97                    | 50.65 $\pm$ 6.49                   | 48.05 $\pm$ 4.50                   | 48.05 $\pm$ 4.68                   | 48.05 $\pm$ 3.56  | 51.95 $\pm$ 4.50                   | 46.49 $\pm$ 5.15  | 56.36 $\pm$ 3.85                    |
| Yoga                   | 59.91 $\pm$ 2.90                   | <b>59.97 <math>\pm</math> 4.98</b>  | 59.88 $\pm$ 3.11                    | 55.72 $\pm$ 1.71 | 57.07 $\pm$ 1.49                    | 58.79 $\pm$ 3.68                   | 53.03 $\pm$ 3.90                   | 59.02 $\pm$ 1.75                   | 51.43 $\pm$ 2.65  | 59.87 $\pm$ 3.99                   | 51.89 $\pm$ 2.29  | 57.73 $\pm$ 2.05                    |
| Avg. Acc.              | 66.24                              | 64.93                               | <u>67.18</u>                        | 55.43            | 62.72                               | 62.79                              | 46.04                              | 61.94                              | 51.50             | 63.13                              | 46.88             | <b>69.16</b>                        |
| Avg. Rank              | 4.83                               | 5.03                                | <u>4.55</u>                         | 8.00             | 6.23                                | 5.75                               | 9.87                               | 5.73                               | 8.87              | 5.64                               | 9.58              | <b>3.91</b>                         |
| #Best                  | 25                                 | 15                                  | <u>28</u>                           | 6                | 5                                   | 7                                  | 3                                  | 5                                  | 1                 | 8                                  | 1                 | <b>40</b>                           |



Table 7: Key few-shot adaptation settings for DuoTSP.

| Component                       | Setting                                                  |
|---------------------------------|----------------------------------------------------------|
| Base LLM                        | meta-llama/Llama-3.2-1B                                  |
| Vision backbone                 | facebook/dinov2-base                                     |
| Pseudo-image construction       | tivit_sqrt_overlap                                       |
| Patch length / stride           | 16 / 8                                                   |
| Hidden size                     | 128                                                      |
| Default downstream checkpoint   | Stage II (TSQA transfer)                                 |
| Warm-up / joint epochs          | 5 / 55                                                   |
| Batch size / eval batch size    | 8 / 8                                                    |
| Gradient accumulation           | 4                                                        |
| LR (encoder / projector / LoRA) | $2 \times 10^{-4} / 1 \times 10^{-4} / 1 \times 10^{-4}$ |
| Language-side adaptation        | LoRA enabled in Phase II ( $r = 16, \alpha = 32$ )       |
| Tokenizer training mode         | class_rows                                               |
| Prediction rule                 | candidate class-token-plus-eos scoring                   |
| Freeze vision backbone          | true                                                     |
| Data augmentation               | none                                                     |

## B Additional Experimental Details

We summarize the key implementation choices of DuoTSP for the few-shot benchmark in Table 7. These settings are fixed across the main benchmark reported in Section 4.

### B.1 Pretraining Stages and Data Sources

Table 8 summarizes the three training stages discussed in the main paper and clarifies which stage provides the default downstream checkpoint.

The main benchmark therefore avoids UCR TEST leakage at every pretraining stage, but it is not benchmark-disjoint with respect to UCR TRAIN data because Stage I additionally uses raw TRAIN-split sequences from 98 UCR datasets. Downstream few-shot adaptation still uses only the sampled support set of the target dataset.

### B.2 Baseline Training Recipes and Hyperparameters

Table 9 summarizes the training rules used by the reported baseline families. The goal of this table is to make the few-shot comparison protocol explicit: all methods share the same sampled support/query split, but they do not all use the same checkpoint-selection rule or the same epoch schedule.

## C Additional Plot

Figure 2 visualizes the shot-scaling trend over the 128 shared UCR datasets. DuoTSP remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

Table 8: Stage-wise data sources and roles for DuoTSP.

| Stage     | Training data                                                                                                        | Validation / monitoring data                                                                           | Role in the main UCR few-shot benchmark                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage I   | Raw TSQA-train series, raw M4-train series, and official TRAIN-split raw series from a 98-dataset UCR training pool. | Raw TSQA validation/test series and raw M4 validation/test series. No UCR official TEST split is used. | Learns dual-view representations before the LLM interface. This stage introduces benchmark-train overlap with UCR TRAIN data but does not use UCR TEST data. |
| Stage II  | TSQA train split only, with standard language-model training on time-series question answering.                      | TSQA validation/test splits.                                                                           | Provides the default downstream checkpoint used by all main few-shot results in Section 4.                                                                   |
| Stage III | Optional mixture of TSQA, M4 captioning, and synthetic attribute supervision derived from TSQA/M4 raw series.        | Corresponding validation/test splits of the same mixture.                                              | Used only for additional semantic enrichment or caption specialization; not used for the default main benchmark initialization unless explicitly stated.     |

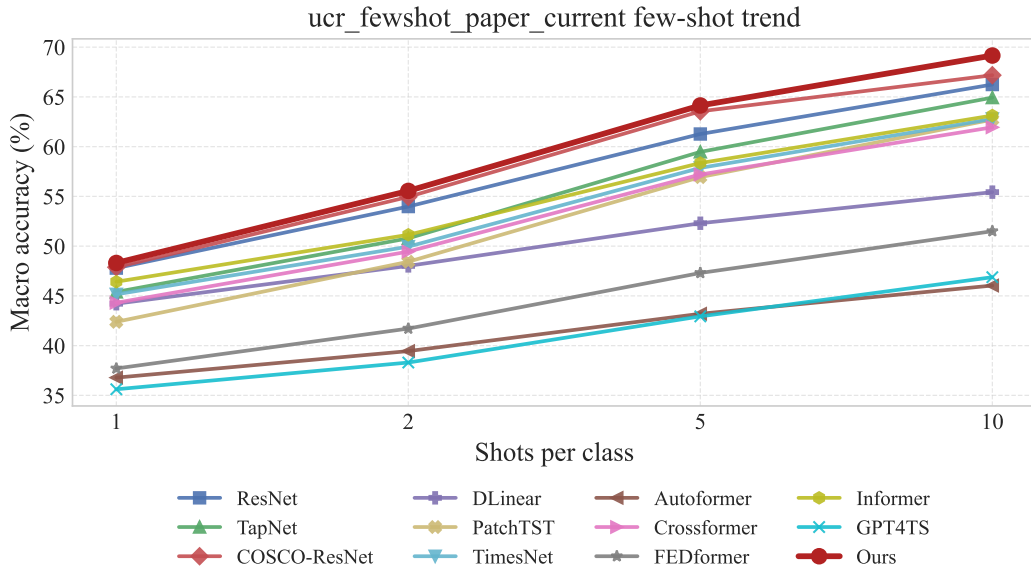


Figure 2: Shot-scaling trend on the 128 shared UCR datasets. Each point reports the mean accuracy over five runs. DuoTSP remains the strongest method across all four shot settings.

Table 9: Few-shot training recipes for DuoTSP and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on the official TEST split restricted to the sampled class set.

| Method / family                                                                         | Optimizer                                       | Epochs                                                                                                | Batch / LR                                                             | Model selection rule                                   | Val | ES | Few-shot notes                                                                                                                              |
|-----------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|-----|----|---------------------------------------------------------------------------------------------------------------------------------------------|
| DuoTSP                                                                                  | AdamW                                           | 5 warm-up + 55 joint                                                                                  | $8 / 2 \times 10^{-4}$ (encoder), $1 \times 10^{-4}$ (projector, LoRA) | Report the last Phase II checkpoint                    | No  | No | Class-token rows remain trainable; language-side adaptation uses LoRA in Phase II.                                                          |
| ResNet, Tap-Net                                                                         | AdamW                                           | Fixed schedule in the few-shot runner (60 in the benchmark configs)                                   | model-specific / benchmark configs use $10^{-4}$                       | Best support-set accuracy, tie-broken by support loss  | No  | No | Trained only on the sampled support set; the support set doubles as the model-selection set.                                                |
| COSCO-ResNet                                                                            | SAM with SGD base optimizer + prototypical loss | Fixed schedule (100 in the benchmark configs)                                                         | 128 / $10^{-2}$ by default                                             | Final model; support centroids computed after training | No  | No | Query predictions are made by distances to support centroids rather than a learned classifier head.                                         |
| PatchTST                                                                                | AdamW + linear warmup                           | Two-phase fixed schedule (head warm-up + joint fine-tuning; 60 total epochs in the benchmark configs) | 16 / separate backbone-head LR                                         | Report the last Phase II checkpoint                    | No  | No | Uses the same support/query split and fixed class subset as the main protocol.                                                              |
| TSLib family (DLINER, AUTO-FORMER, CROSS-FORMER, FED-FORMER, INFORMER, TIMESNET) GPT4TS | RAdam + cosine schedule                         | Fixed model-specific schedule from the TSLib profile or benchmark override                            | usually 16 / $10^{-3}$                                                 | Report the last checkpoint                             | No  | No | Single-stage training; evaluation masks logits to the sampled class subset. TimesNet uses a shorter profile than the other TSLib baselines. |
|                                                                                         | RAdam (default runner setting)                  | Fixed schedule (60 in the benchmark configs)                                                          | 64 / $10^{-4}$ in the benchmark configs                                | Report the last checkpoint                             | No  | No | Implemented with the One-Fits-All classification runner under the same sampled support/query protocol.                                      |

Table 10: Win/Tie/Loss of DuoTSP against selected baselines on 128 shared UCR datasets.

| Baseline     | 1-shot       | 2-shot       | 5-shot       | 10-shot     |
|--------------|--------------|--------------|--------------|-------------|
| COSCO-ResNet | 68 / 0 / 60  | 68 / 1 / 59  | 58 / 0 / 70  | 64 / 0 / 64 |
| PatchTST     | 97 / 0 / 31  | 97 / 0 / 31  | 99 / 0 / 29  | 94 / 2 / 32 |
| TimesNet     | 88 / 0 / 40  | 92 / 0 / 36  | 86 / 0 / 42  | 93 / 1 / 34 |
| GPT4TS       | 106 / 2 / 20 | 110 / 0 / 18 | 114 / 1 / 13 | 121 / 1 / 6 |
| ResNet       | 64 / 1 / 63  | 66 / 1 / 61  | 67 / 0 / 61  | 69 / 0 / 59 |
| TapNet       | 75 / 0 / 53  | 75 / 3 / 50  | 76 / 1 / 51  | 72 / 3 / 53 |

## 526 D Dataset-Level Win/Tie/Loss Summary

527 Table 10 reports head-to-head Win/Tie/Loss counts against representative baselines on the 128 shared  
528 UCR datasets. This table complements the averaged benchmark metrics in Section 4 by showing how  
529 often DuoTSP improves over each baseline at the dataset level.

Table 11: Ablation of aggregate curriculum pretraining on the 128 shared UCR datasets under the same few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full DuoTSP model.

| Variant                    | 1-shot       | 2-shot       | 5-shot       | 10-shot      | Avg          | Drop |
|----------------------------|--------------|--------------|--------------|--------------|--------------|------|
| w/o Curriculum Pretraining | 44.70        | 51.81        | 60.65        | 66.60        | 55.94        | 3.35 |
| DuoTSP                     | <b>48.31</b> | <b>55.55</b> | <b>64.14</b> | <b>69.16</b> | <b>59.29</b> | –    |

## E Ablation Tables

Table 2 in the main paper reports the compact ablation summary. Tables 11–13 expand that summary into three focused views under the same few-shot protocol. Each value is the mean accuracy over five runs, and the final column reports the average accuracy drop relative to the full DuoTSP model.

Table 12: Ablation of dual-view native prompting on the 128 shared UCR datasets under the same few-shot protocol as the main results. These comparisons assess the effect of removing one top-level view at a time. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full DuoTSP model.

| Variant       | 1-shot       | 2-shot       | 5-shot       | 10-shot      | Avg          | Drop |
|---------------|--------------|--------------|--------------|--------------|--------------|------|
| Temporal only | 43.51        | 50.55        | 58.40        | 64.39        | 54.21        | 5.08 |
| Visual only   | 45.88        | 50.02        | 60.86        | 63.58        | 55.09        | 4.20 |
| DuoTSP        | <b>48.31</b> | <b>55.55</b> | <b>64.14</b> | <b>69.16</b> | <b>59.29</b> | –    |

Table 13: Ablation of candidate label scoring on the 128 shared UCR datasets under the same few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full DuoTSP model.

| Variant                     | 1-shot       | 2-shot       | 5-shot       | 10-shot      | Avg          | Drop |
|-----------------------------|--------------|--------------|--------------|--------------|--------------|------|
| w/o Candidate Label Scoring | 44.46        | 50.44        | 60.08        | 65.59        | 55.14        | 4.15 |
| DuoTSP                      | <b>48.31</b> | <b>55.55</b> | <b>64.14</b> | <b>69.16</b> | <b>59.29</b> | –    |

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